outside, but prevented the introduction of any airborne microorganisms, which would get caught in the twists and bends of the flasks' necks. If a life force besides the airborne microorganisms were responsible for microbial growth within the sterilized flasks, it would have access to the broth, whereas the microorganisms would not. He correctly predicted that sterilized broth in his swan-neck flasks would remain sterile as long as the swan necks remained intact. However, should the necks be broken, microorganisms would be introduced, contaminating the flasks and allowing microbial growth within the broth. Pasteur's set of experiments irrefutably disproved the theory of spontaneous generation and also articulated "Omne vivum ex vivo" ("Life only comes from life")

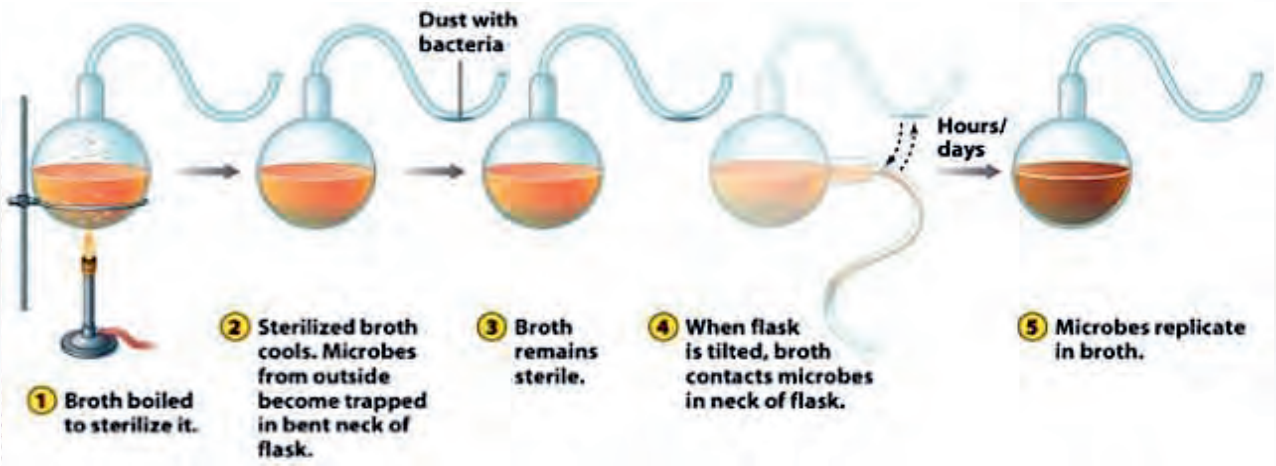


Figure 4.2 Louis Pasteur's experimental set up to disprove spontaneous generation

These two scientists showed that both macro-organisms (Redi) and micro-organisms (Pasteur) could only arise from pre-existing organisms, disproving the theory of spontaneous generation. But what about the first ever cell? Unless we believe that life is eternal, with no beginning and no end, there had to be a first cell. It could not have come from a pre-existing cell because it was the first.

Keyword

Eternity of life theory- claims that the Universe has always existed and that there has always been life in the Universe.

Activity 4.2

Write a report describing carefully how we now know that spontaneous generation is not possible and does not happen.

3. Eternity of life

In this theory of life, there is no beginning and no end to life on Earth and so life neither needs special creation nor does it need to be generated from non-living matter. Supporters of this theory believe that life is an inherent property of the Universe and it has always existed as has the Universe.

At the time when such theories were being propounded, many eminent scientists – including Albert Einstein – believed that the Universe was unchanging. They reasoned that 'if life is found today in an unchanging Universe, then it must always have been there'.

4 Cosmozoan theory, Panspermia or Spore broth theory

Activity 4.3

Make a small group and search for books or other resources (internet) about steady state theory of the Universe? Can eternity of life theory linked to this theory? Report your finding to the class.

According to **Cosmozoan theory** life has reached this planet Earth from other cosmological structures, such as meteorites, in the form of highly resistant spores. This theory was proposed by Richter (1865). According to this theory, 'protoplasm' reached the earth in the form of spores or germs or other simple particles from some unknown part of the universe with the cosmic dust, and subsequently evolved into various forms of life.

Figure 4.3 A meteorite fall to Earth

Keyword

Cosmozoan theory claims that life on Earth originally came from elsewhere in the Universe (possibly from another planet).

Helmholz (1884) speculated that 'protoplasm' in some form reached the earth with falling meteorites.

Activity 4.3

Search from books or other sources about panspermia and pseudo-panspermia theories. How do they explain the origin of life on earth? Try to gather evidence against this theory as living matter cannot survive the extreme cold, dryness and ultra-violet radiation from the sun required to be crossed for reaching the earth. Report your finding to the class.

4. Biochemical origin

The current ideas we have about how life may have evolved on Earth as a result of biochemical reactions (sometimes called abiogenesis) owe much to two biologists working early in the twentieth century.

Keyword

Biochemical theory suggests that life on Earth originated as a result of a number of biochemical reactions producing organic molecules which associated to form cells.

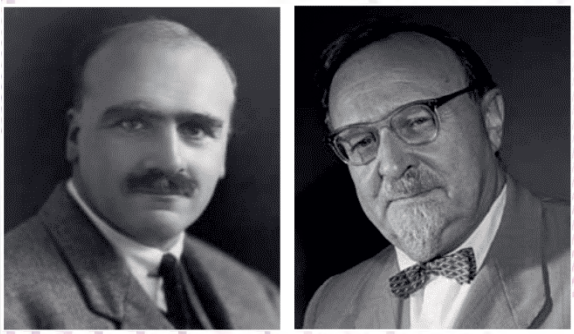


Figure 4.4 J B S Haldane and A I Oparin

Figure 4.5 Clouds of inter-stellar gas have been shown to contain organic molecules

Aleksandr Oparin, a Russian biologist who first put forward his ideas in 1924, and **John Haldane**, an English biologist independently put forward almost identical ideas in 1929 (before Oparin's book had been translated into English). Both believed that organic molecules could be formed from abiogenic materials in the presence of an external energy source such as ultraviolet

radiation and that the primitive atmosphere was reducing (having very low amounts of free oxygen) and contained ammonia and water vapour, among other gases. Both also suspected that the first life-forms appeared in the warm, primitive ocean and were heterotrophic (obtaining preformed nutrients from the compounds in existence on early Earth) rather than autotrophic (generating food and nutrients from sunlight or inorganic materials).

They proposed that common gases in the early Earth atmosphere combined to form simple organic chemicals, and these in turn combined to form more complex molecules. Then, the complex molecules became separated from the surrounding medium, and acquired some of the characters of living organisms.

Figure 4.6 Coacervate droplets – pre-cells

They became able to absorb nutrients, to grow, to divide (reproduce), and so on.

Later Miller had apparently approved the Oparin-Haldane model by mixing the basic elements to produce simple organic compounds, and then combining these to produce the building blocks of proteins andnucleicacid (Figure 4.7).

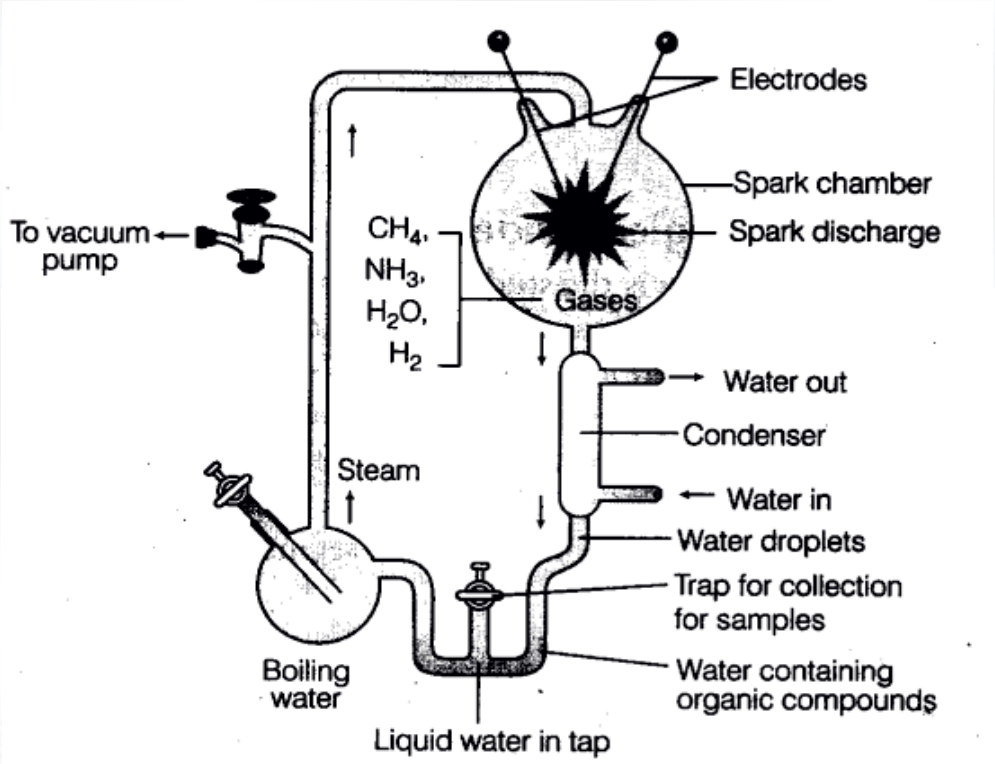


Figure 4.7 Stanley Miller's spark-discharge

Activity 4.4

Refer to different evolution books or the internet and understand the experiment done by Stanley Miller's to support the theory of biochemical origin of life. Explore also some other known scientists how they tried to generate data to support this theory. Present your finding to the class.

Autotrophs

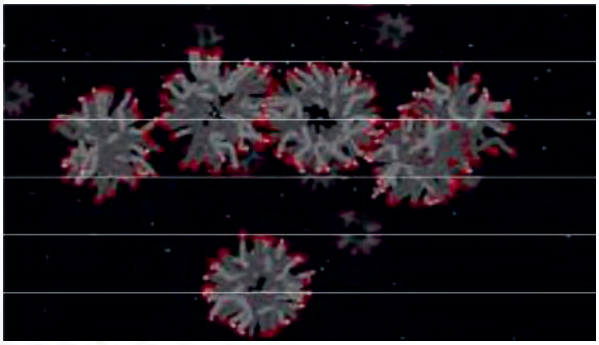
Autotrophs form the basis for all food chains: they are the organisms which create sugars, proteins, lipids, and other materials for life. The first organisms appeared about 4 billion years ago were **prokaryotes**. They had no true nucleus. It seems likely also that they had RNA rather than DNA as their genetic material. It was assumed that they gave rise to three distinct lines of evolution leading to:

- **Archaebacteria** – prokaryotes including thermophilic sulphobacteria, methanobacteria and halophilic bacteria

- **Eubacteria** – prokaryotes; ordinary bacteria and cyanobacteria (blue-green bacteria and sometimes known as blue-green algae).

- **Eukaryotes** – eventually evolving into protoctistans, fungi, plants, animals (nearly all are aerobic).

One great change that affected the evolution of early life forms was the shift from the reducing atmosphere to an atmosphere containing oxygen. This took place about 2.4 billion years ago. The two major types of autotrophs are chemoautotrophs and photoautotroph.



Self-questioning

What are chemoautotrophs? How do they differ from photoautotrophs?

Figure 4.8 Could pre-cells have looked like this when they were dividing?

Keywords

Autotrophs are organisms that produces organic molecules from inorganic material

Heterotrophs are organisms that consume producers or other consumers

Aerobic respiration is a means by which cells release energy from organic molecules using oxygen

Prokaryote is an organism consisting of a prokaryotic cell. All bacteria are prokaryotes

Eukaryote is an organism consisting of one or more eukaryotic cells. All organisms other than bacteria are eukaryotes

Archaebacterium is the first bacteria (and thus the first living organisms) to develop on Earth. They are now only found in extreme conditions

Eubacterium is any bacteria that is not an archaebacterium is a eubacterium

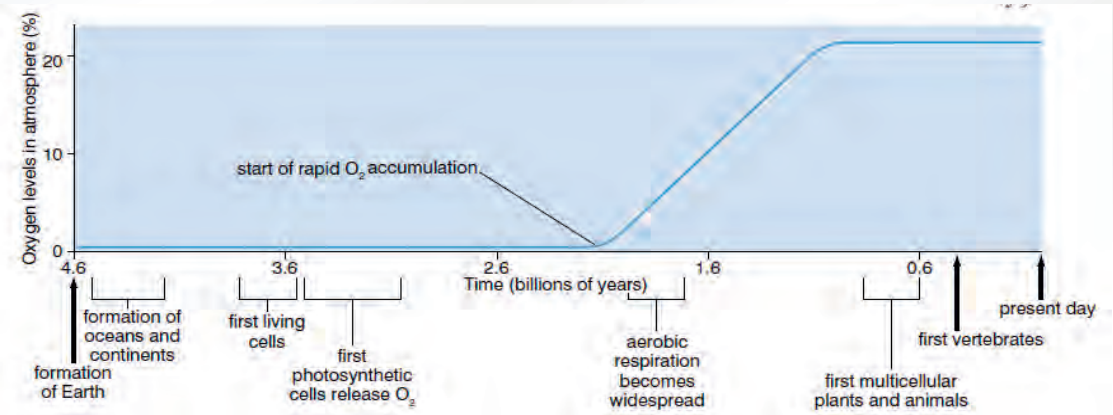


Figure 4.9 Life on Earth has evolved over billions of years

Chemoautotrophs

Chemoautotrophs are organisms that obtain their energy from a chemical reaction (chemotrophs) but their source of carbon is the most oxidized form of carbon, carbon dioxide (CO_2). The best known chemoautotrophs are the chemolithoautotrophs that use inorganic energy sources, such as ferrous iron, hydrogen, hydrogen sulfide, elemental sulfur or ammonia, and CO_2 as their carbon source. All known chemoautotrophs are prokaryotes, belonging to the Archaea or Bacteria domains. They have been isolated in different extreme habitats, associated to deep-sea vents, the deep biosphere or acidic environments. This form of energy conservation is considered one of the oldest on Earth. 0..

Photoautotrophs

Photoautotrophs are organisms that use light energy and inorganic carbon to produce organic materials. Eukaryotic photoautotrophs absorb energy through the chlorophyll molecules in their chloroplasts.

4.1.2. Theories of evolution

By the end of this section, the learner will be able to:

- Compare Lamarck Vs. Darwinian theory

Lamarckism

Jean-Baptiste De Lamarck (1744 - 1829) was a great French naturalist. Lamarck sought a naturalistic explanation for the diversity of modern organisms and the animals seen in

the fossil record. ′ He proposed 'the theory of inheritance of acquired characters' in 1809. He postulated:

A. New Needs: Changes in environment factors like light, temperature, medium, food, air etc or migration leads to origin of new needs in living organisms. To fulfil these new needs, living organisms have to exert special efforts like changes in habits or behavior.

B. Use and disuse of organs: ′ The new habits involve the greater use of certain organs to meet new needs, and the disuse or lesser use of certain other organs which are of no use in new conditions.

C. Inheritance of acquired characters: he believed that the favorable acquired characters are inheritable and are transmitted to the offsprings so that these are born fit to face the changed environmental conditions and the chances of their survival are increased.

D. Speciation: Lamarck believed that in every generation, new characters are acquired and transmitted to next generation, so that new characters accumulate generation after generation. After a number of generations, a new species is formed.

Activity 4.5 Debating the origin of life

Your teacher will assign you to form small groups on the bases of the following theories:

- The creationists
- The spontaneous generationists
- The Eternalists
- The Cosmozoans
- The Abiogenesists

Each group must prepare a 'case' for their theory of the origin of life. Once this is done, each group will, in turn, then start a debate by announcing: This house believes that only (Creationism, for example,) can truly account for the origin of life. The group will have five minutes to put their case to the rest of the class. The group starting the debate will then face five minutes of questions from the other group members.

At the end, those members not involved directly in this debate will vote as to who they think has won the debate – the group proposing the theory or the group questioning the theory.

You must try to vote only on the debate, not on your personal views.

Keyword

Lamarckism is the theory developed by the French biologist Jean-Baptiste Lamarck that claimed that organisms passed on to subsequent generations traits acquired during their lifetime

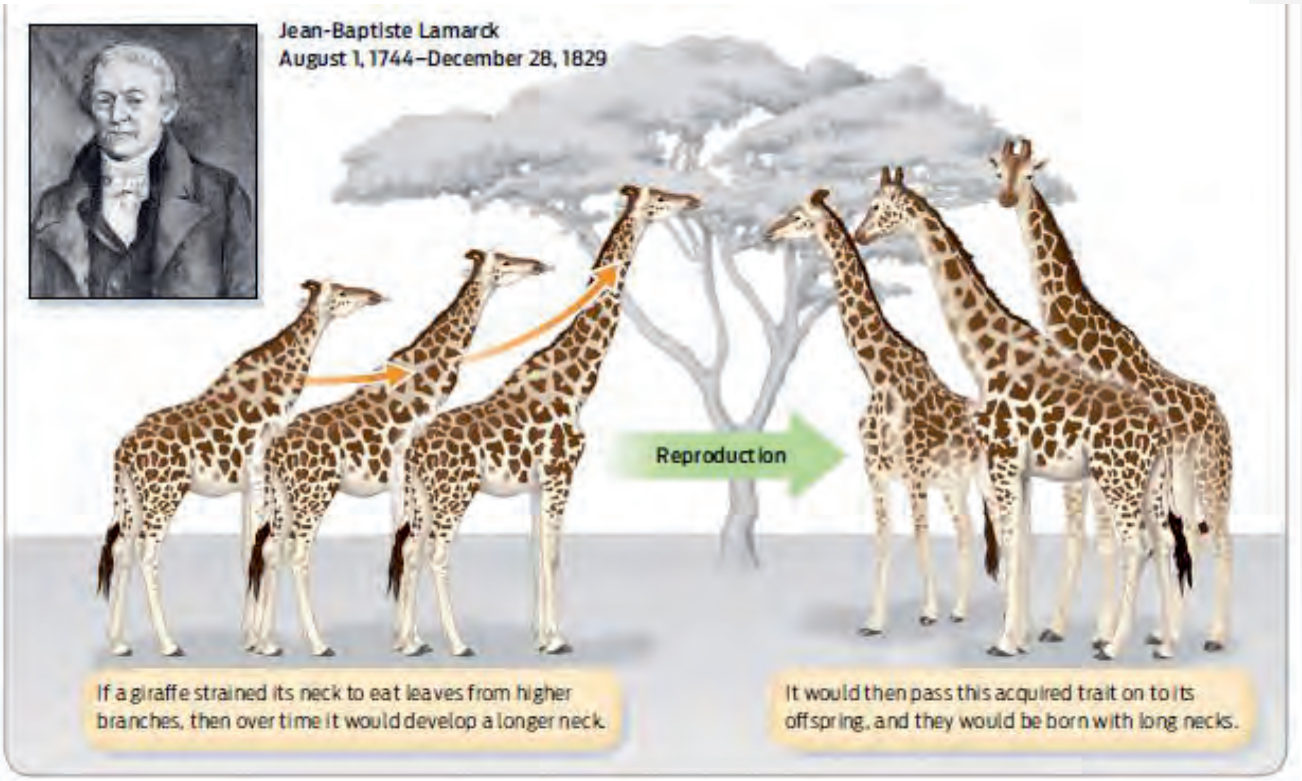


Figure 4.10 Lamarck's ideas of use and disuse and the inheritance of acquired traits

Use and disuse

In this part of his theory, Lamarck suggests that when a structure or process is continually used, that structure or process will become enlarged or more developed. Conversely, any structure or process that is not used or is rarely used will become reduced in size or less developed. The classic example he used to explain the concept of use and disuse is the elongated neck of the giraffe. According to Lamarck, a given giraffe could, over a lifetime of straining to reach high branches, develop an elongated neck. However, Lamarck could not explain how this might happen.

Inheritance of acquired traits

Lamarck believed that traits changed or acquired during an individual's lifetime could be passed on to its offspring. Giraffes that had acquired long necks would have offspring with long necks rather than the short necks their parents were born with. This type of inheritance, sometimes called Lamarckian inheritance, has since been disproved by the discoveries of genetics. However, Lamarck did believe that evolutionary change takes place gradually of evolution and constantly.

1. Significance of Lamarckism

a) It was first comprehensive theory of biological evolution.

b) It nicely explains the existence of vestigial organs in animals due to their continuous disuse. Vestigial organs are organs that are a part of the organism that are no longer in use. That can be used to determine the relatedness of different species. For example, the bone structures of the front flippers of a whale contain bones of limbs that exist in mammals such as cows.

c) It explains the development of strong jaw muscles and claws in the carnivores due to their continued extra use.

d) It stimulated other biologists to look for the mechanism of organic mechanism.

Activity 4.6

Make a small group and search for some of the evidences that were used to support Lamark's theory and the critics forwarded against this theory. Present your work to the class.

Darwinism (Theory of natural selection)

Proposed by Charles Darwin (1809-1882 A.D.), an English naturalist. He went on a voyage on H.M.S Beagle and explored South America, the Galapagos Islands and other islands. He was highly influenced by essay entitled 'On the Tendency of varieties on Depart Indefinitely from the original type'' by Alfred Russell Wallace and another essay

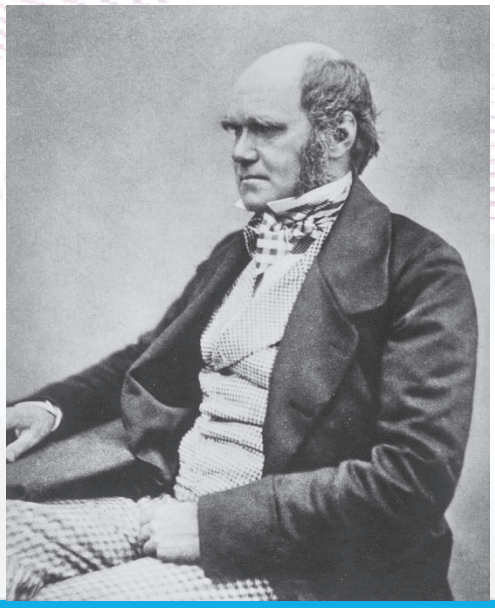


Figure 4.11 Charles Darwin

'Principle of geology'' written by Charles Lyell.

Some of Darwin's evidence came from a visit to the Galapagos Islands. These are a small group of islands in the Pacific Ocean about 600 miles off the coast of Ecuador in South America. Darwin visited five of the Galapagos Islands and made drawings and collected specimens. In particular, Darwin studied the finches found on the different islands and noted that there were many similarities between them, as well as obvious differences. He concluded that an 'ancestral finch' had colonized the islands from the mainland and, in the absence of predators, were able to adapt to the different conditions on the islands and, eventually, evolved into different species (Figure 4.12). Some of the finches had, he suggested, evolved into insect eaters, with pointed beaks. Others had evolved into seed eaters with beaks capable of crushing the seeds.

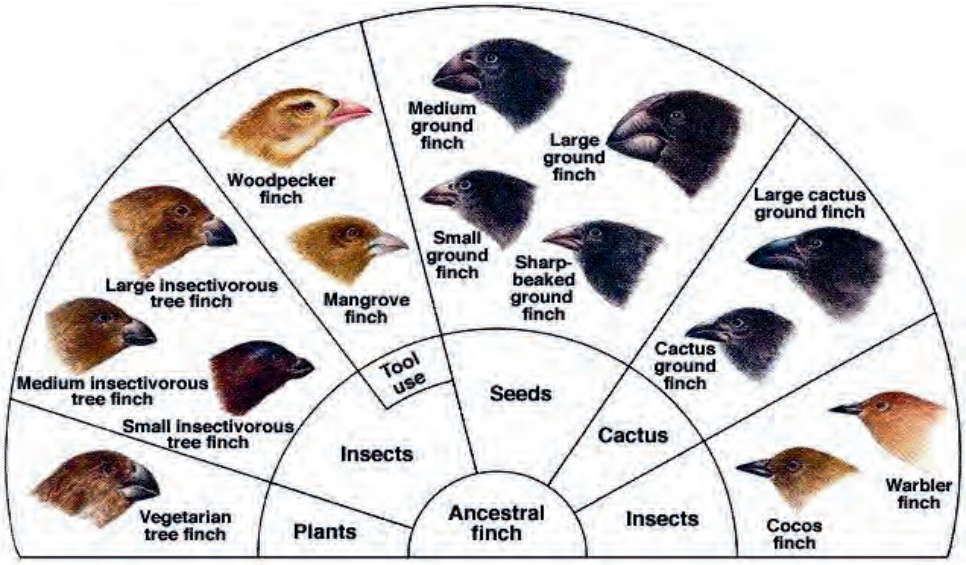


Figure 4.12 Darwin's Theory of Finches on the Galapagos Islands

Basic postulates of Darwinism

Geometric increase: All species tend to produce more offspring than can possibly survive. However, the space and the availability of food supply is limited to support the number of organisms that increase in a geometric ratio.

Struggle for existence: Since the number of individuals produced is far more than the number that can be supported, there is an everlasting competition between organisms at all levels of life.

Variation under nature: No two individuals of a species are exactly similar and they have some differences. These differences are

called variations and without evolution is not possible. Variations give rise to new characters and heredity passes them on to the next generation (inheritance of useful variations).

Natural selection or **survival of the fittest**: Due to struggle for existence and useful heritable variations, only those individuals survive which show high selective value and in the course of time they develop various adaptive modifications to suit the changed conditions of life. Such selection was called natural selection by Darwin.

Origin of species: In the course of long periods of time the best fitted and suitable individuals survived and adjust to the nature. As environment is ever changing, further changes occur and thus new adaptations appear in organisms. The later descendants after several generations become quite distinct from their ancestors. On this way new species appear.

Activity 4.7

1. Make a small group and read from different books about the evidences used to support Darwin's theory and the critics against this theory. Present your work to the class.
2. Why Darwinism theory is said to be descent with modification?
3. Explain the evolution of giraffe in the view of Darwinism and Lamarkism?

Activity 4.8

Form small groups and discuss Lamarck and Darwin theories of evolution. Prepare charts on the main points and elaborate the differences and similarities between the explanations of the two theories.

Neo-Darwinism theory

Charles Darwin knew very little of genetics. Mendel had not carried out his ground-breaking work on inheritance at the time Darwin published his book *On the Origin of Species*. However, we can now incorporate our knowledge of genes and gene action into the theory of natural selection to give a better understanding of what drives evolution.

Keyword

Neo-Darwinism a revised version of Darwin's theory of evolution by means of natural selection. This theory, which is now accepted by most biologists, combines Darwin's original theory, genetic theory and theories about animal behavior.

Genes determine features. But when we think about how a population might evolve into a new species, we need to think not just in terms of the alleles each individual might carry, but also in terms of all the alleles (all the genes) available in the population. We call this the **gene pool** of the population.

Postulates of Neo-Darwinism are:

1. Genetic variability
2. Natural selection
3. Reproductive Isolation

Activity 4.9

Discuss in small group what Neo-Darwinism theory is and the details of the three postulates of Neo-Darwinism. Reflect your opinion to the class.

4.1.3. The evidence for evolution

By the end of this section, the learner will be able to:

- Explain how comparative anatomy embryology, fossil records, and biochemistry provide evidence for evolution

4.1.3.1 Comparative anatomy

Comparative anatomy is one of the strongest forms of evidence for evolution. It looks at the structural similarities of organisms and uses these similarities to determine their possible evolutionary relationships. It assumes that organisms with similar anatomical features are closely related evolutionarily, and that they probably share a common ancestor. Some organisms have anatomical structures that are very similar in form, but very different in

function. We call such structures **homologous structures**. Because they are so similar, they indicate an evolutionary relationship and a common ancestor of the species. Perhaps the best-known example of homologous structures is the forelimb of mammals. When examined closely, the forelimbs of humans, whales, cats and bats are all very similar in structure (Figure 4.13).

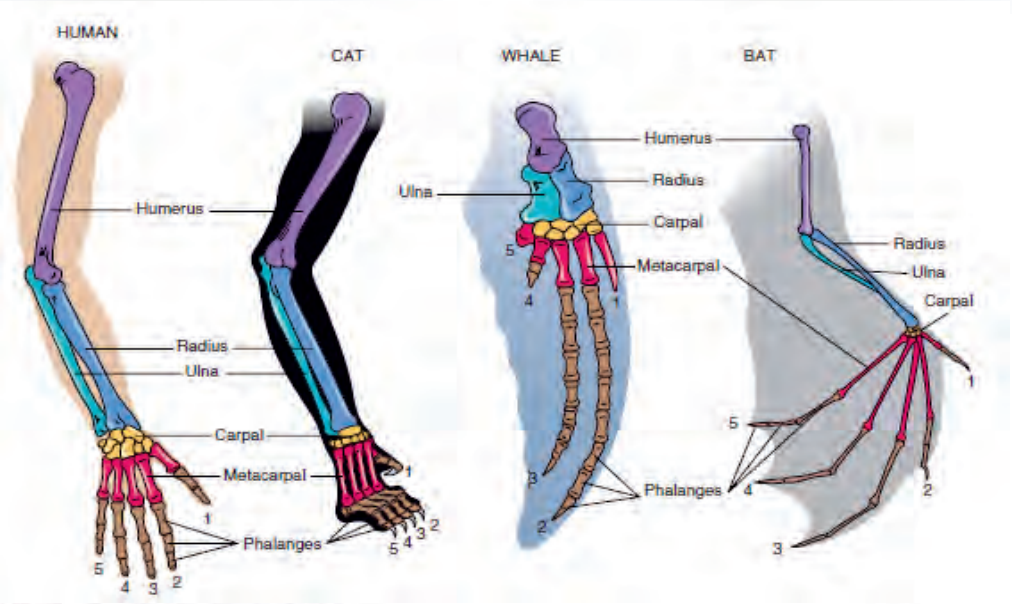


Figure 4.13 The homologous forelimbs of mammals.

Each possesses the same number of bones, arranged in almost the same way while they have different external features that function in different ways as:

- arm for manipulation in humans
- leg for running in cats
- flipper for swimming in whales
- wing for flying in bats

By comparing the anatomy of these limbs, scientists have determined that the basic pattern (called a **pentadactyl limb**) must have evolved just once and that all organisms with this kind of limb were

Keyword

Homologous structures are structures with the same basic anatomy and a common evolutionary origin but having a different function

Keywords

Pentadactyl limb is a limb with five digits

Analagous structures struactures having the same function but different anatomy and different evolutionary origin.

descended from that original type, which they share a common ancestor

However, comparative anatomy needs to be used carefully as evidence for evolution. This is because while sometimes organisms have structures that function in very similar ways, morphologically and developmentally these structures are very different. We call these analogous structures. Because they are so different structurally, even though they have the same function, they cannot indicate that two species share a common ancestor. Although, the wings of a bat, bird and mosquito all serve the same function, yet their anatomies are very different. For example, the bird wing has bones inside and is covered with feathers while the mosquito wing has neither of these (Figure 4.14). They are analogous structures that have evolved separately.

Figure 4.14 Analogy in animals

Activity 4.10

Discuss in small group and explain why homology implies identity and anologus implies similarity. Reflect your opinion to the class.

4.1.3.2 Embryology

Comparative embryology studies the way in which the embryos of vertebrates develop before they hatch or born. This development shows similarities which supports a common ancestry.

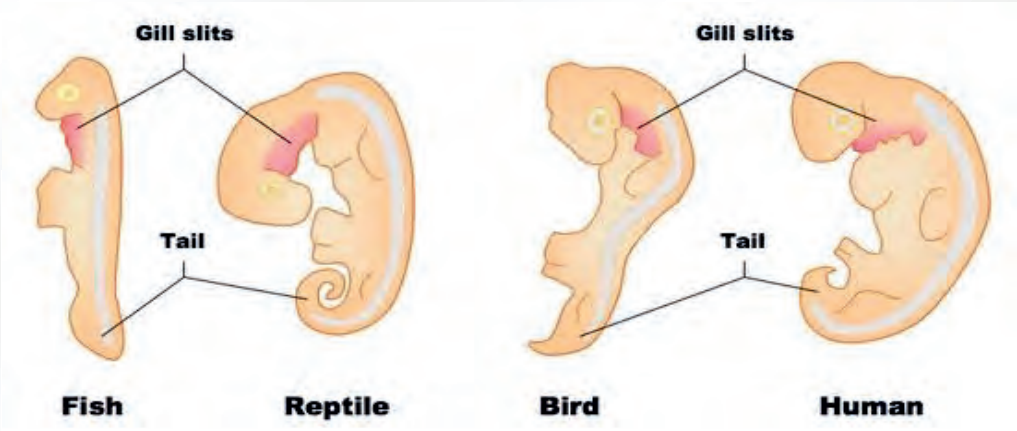


Figure 4.15 Similarities in development of embryos.

For example early in development, all vertebrate embryos have gill slits and tails (Figure 4.15). However, the 'gill slits' are not gills; they connect the throat to the outside, but in many species they disappear later in the embryonic development. However, in fish and larval amphibians they contribute to the development of gills.

Activity 4.10

Form a small group and discuss on the following question:

- Do you think an embryo retraces its evolutionary history? Reflect on how it does to the class.

The embryonic tail does not develop into a tail in all species. In humans, it is reduced during development to the coccyx, or tailbone. The more similar the patterns of embryonic development, the more closely related species are assumed to be. The similarity in the development of vertebrates also suggests a common ancestor.

4.1.3.3 Palaeontology (Paleontology)

The study of fossils is known as **palaeontology**. Fossils are formed when certain remains of organisms or plants get embedded in the soil or water and are preserved for many hundreds of years. They appear either as skeletal remains, footprints, moulds or intact structures as found in the snow. By studying fossils, we are able to establish similarities between the organisms in the present to its ancestor in the past. There can be many similarities that prove the common origins between different closely

related animals and the differences can be studied to establish how they differ now and why. Fossils are very important evidence to prove the theory of evolution and common ancestry. We can group fossils into two categories:

Category 1: The remains of dead animals or plants or the imprint left from the remains, including:

- bones
- teeth
- skin impressions
- hair
- the hardened shell of an ancient invertebrate
- such as a trilobite or an ammonite

an impression of an animal or plant, even if the actual parts are missing

Category 2: Something that was made by the animal while it was living and that it has hardened into stone since then; these are called trace fossils and include:

- footprints
- burrows
- coprolite

(animal faeces)

Self-questioning

How do fossils form?

Type I fossils can be the actual organism or part of an organism, like a piece of bone or hair or feather as it actually was. For example, this spider (Figure 4.17) has been trapped, completely unchanged, inside the amber for millions of years. Amber is fossilised resin from trees. This spider probably became stuck inside the sticky resin and could not escape. As the amber .became fossilised, the spider was protected from micro-organisms and the air which would have led to its decomposition. In many fossils like this, the soft parts of the body have been lost, but the exoskeleton is perfectly preserved. In some cases, however, the entire body remains.

Figure 4.16 A spider preserved in amber.

Dating fossils

Sedimentary rocks are laid down in layers (strata) which help to deduce how the organisms have changed over time. This is called **stratigraphy**. The oldest strata and the oldest fossils are found in the lowest layers and more recent rocks and fossils in

the layers above them nearest to the surface (Figure 4.17). The thickness of each stratum (shown in the diagram) is a measure of the time period during which that stratum was formed.

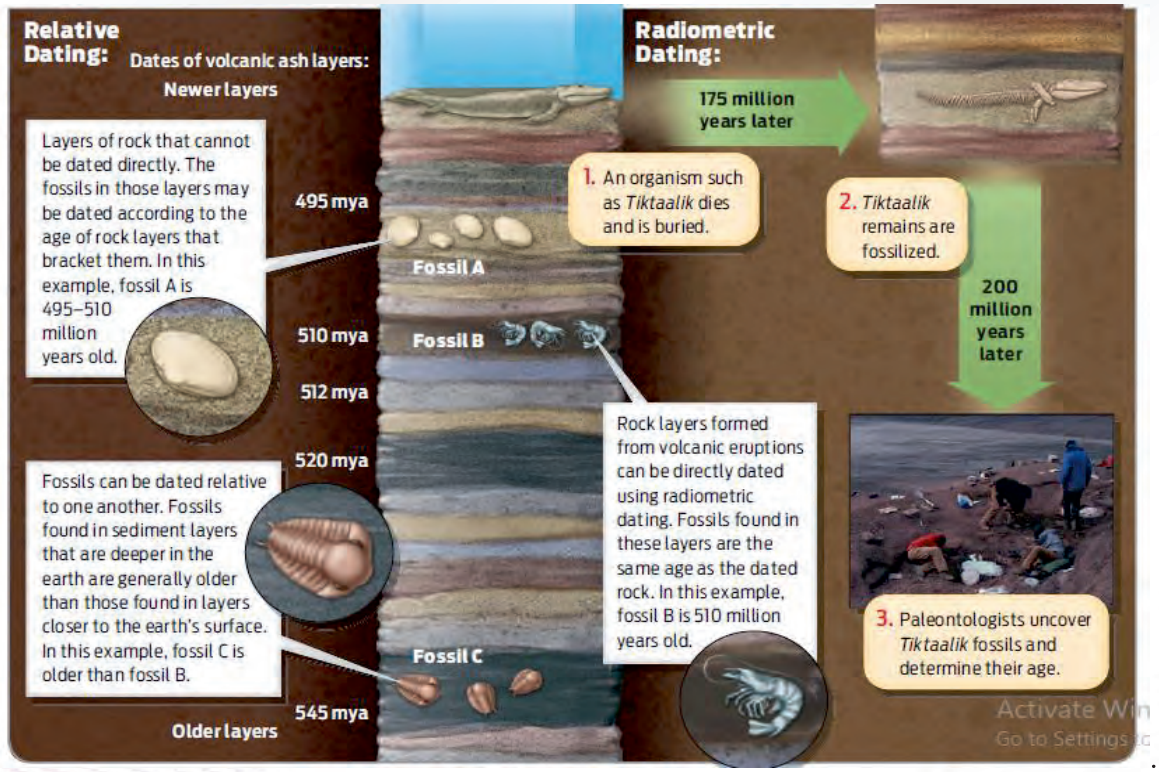


Figure 4.17 Stratigraphy and the how the relative age of fossils dated.

Activity 4.11

Refer from books and other sources how scientists actually date rocks? How do they find out how old each layer is? What does half-life time is? Reflect your answer to the class.

Some minerals in rocks and organic matter (e.g., wood, bones, and shells) can contain radioactive isotopes. The abundances of parent and daughter isotopes in a sample can be measured and used to determine their age. This method is known as **radiometric dating**. The amount of time it takes for half of the parent isotopes to decay into daughter isotopes is known as the **half-life** of the radioactive isotope (Figures 4.18). Fossil age can be determined using two ways; absolute dating which determines the number of years that have elapsed since an event occurred or the specific time when that event occurred. On the other hand, relative dating

determines the age by analyzing rocks and structures placed into chronological order, establishing the age of one thing as older as or younger than another.

Keyword

Half-life: the time needed for half the atoms of a radioactive substance to decay. After two half-lives, three quarters of the atoms will have decayed, and so on.

Fossils: the preserved remains or impressions of once-living organisms.

Fossil record: An assemblage of fossils arranged in order of age, providing evidence of changes inspecies over time.

Paleontologist: A scientist who studies ancient life by means of the fossil record

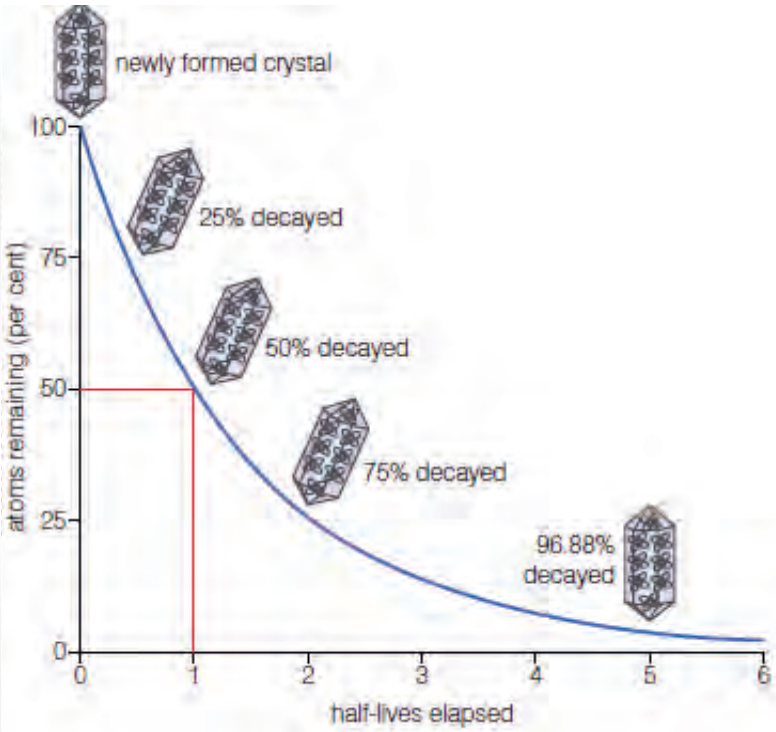


Figure 4.18 Half-life of a radioactive element

Activity 4.12

Discuss about fossils found in Ethiopia and explain why these fossils are thought to be so important? Report your finding to the class.

At time zero, the sample is composed entirely of the radioisotope (Figure 4.19) and the radioactive clock begins ticking. After one half-life, only 50% of the original radioisotope remains. During each succeeding half-life, half of the remaining radioisotope is converted to decay product (s).

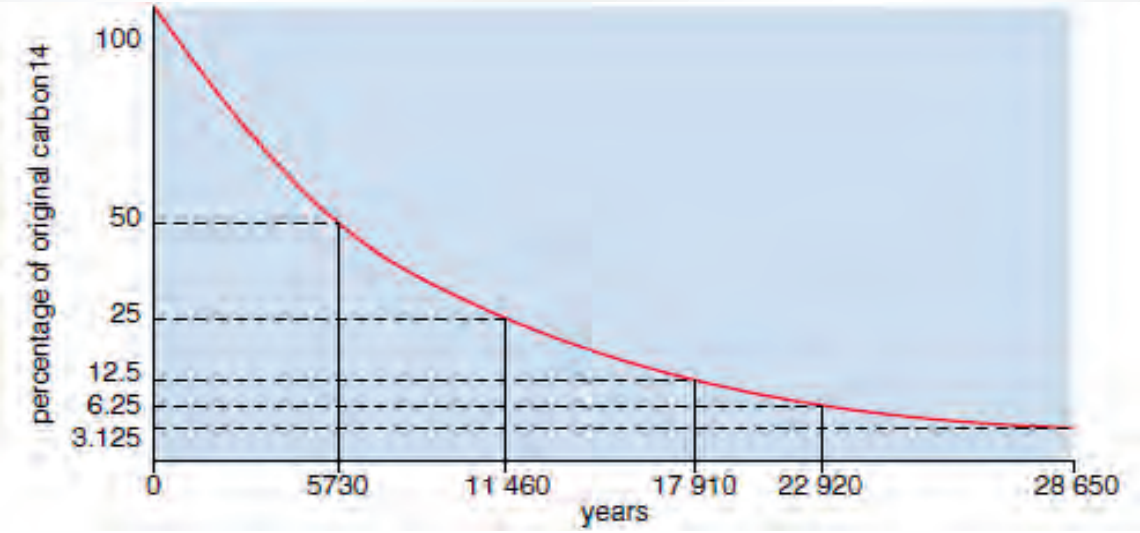


Figure 4.19 Converting the percentage of carbon 14 in a fossil to an age.

How do scientists actually date the rocks? How do they find out how old each layer is? To do this, we scientists use one of two techniques:

- Radiocarbon dating, or
- Potassium–argon dating.

Both these techniques rely on the principle that radioactive atoms decay into other atoms over time. Radioactive carbon atoms (C14) decay into non-radioactive nitrogen atoms (N). Radioactive potassium atoms (K40) decay into argon atoms (A40). Each has what is known as a **half-life**. During this period, half of the radioactive atoms decay. So, starting with a certain number of radioactive potassium atoms, after one half-life, 50% will still be radioactive. After a second half-life, 50% of this will have decayed and 25% of the original number will still be radioactive

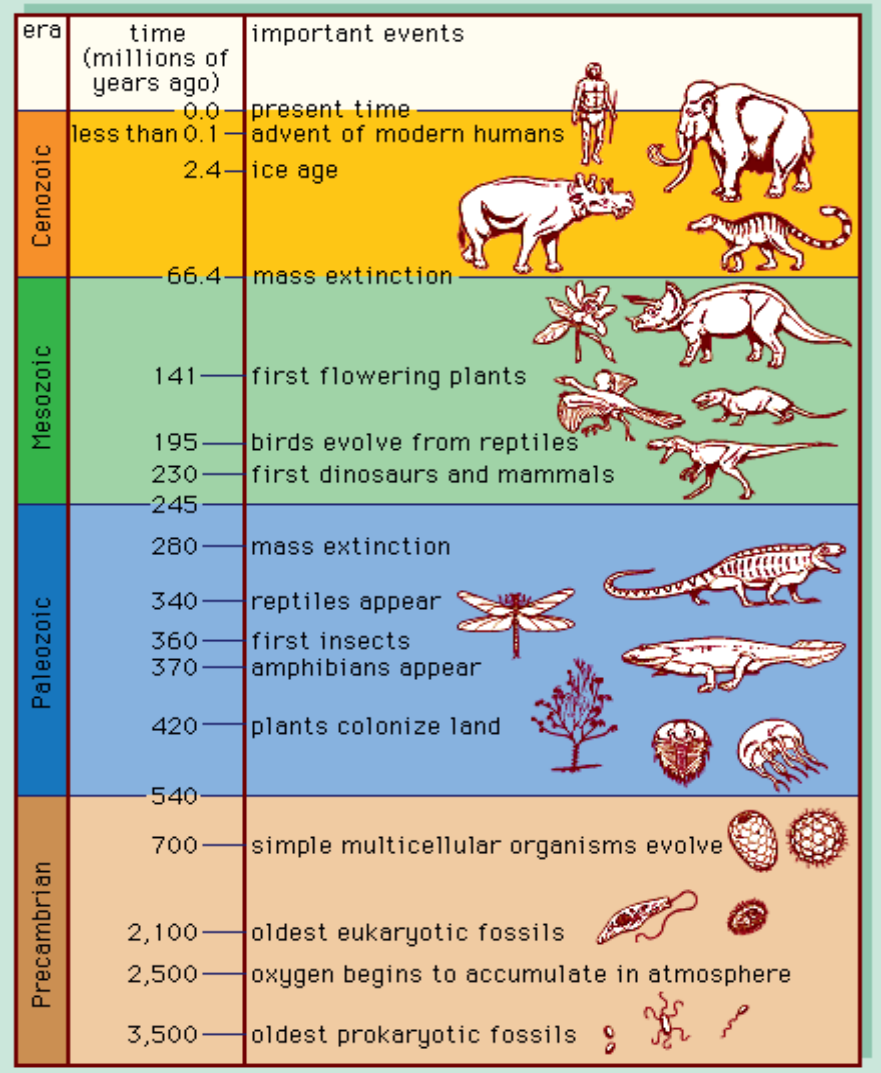


Figure 4.20 Key events in the fossil record of animal evolution

4.1.3.4 Comparative biochemistry

Organisms that share very similar molecules and biochemical pathways are closely related evolutionarily. Chemicals that have been used in such analysis include DNA protein.

Species that are closely related are believed to have the most similar DNA and proteins; those that are distantly related are assumed to share fewer similarities. For example, a comparison of DNA sequences shows that 98% of our DNA is the same as chimpanzees which confirms that chimpanzees are the closest relatives of humans (Figure 4.21).

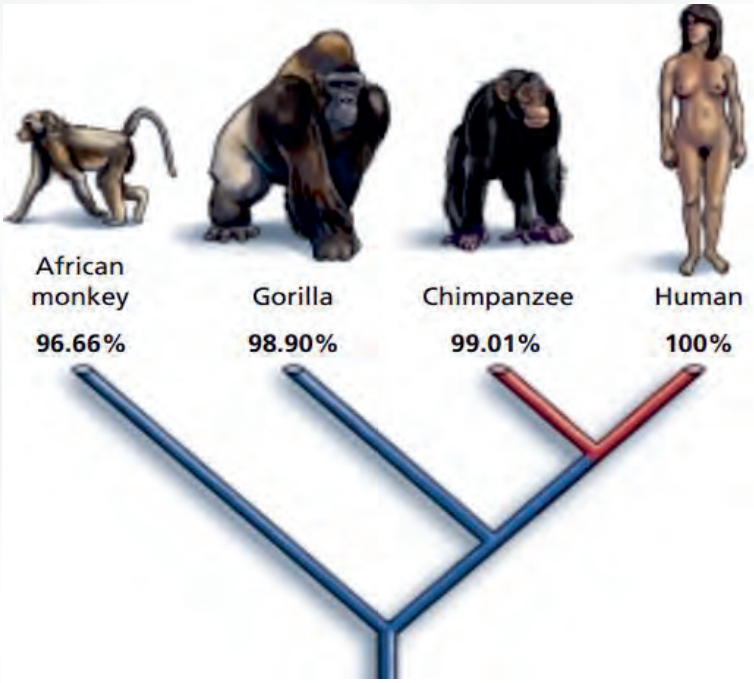


Figure 4.21 A phylogenetic (evolutionary) tree of some animals based on differences in DNA

The **haemoglobin** molecule is similar in all animals that possess it, but there are differences. For example, the haemoglobin of the lamprey (a primitive fish-like animal) has only one polypeptide chain, not four. Most animals have haemoglobin with four chains, but the chains do vary. Figure 4.22 shows the differences in the amino acid sequences of the α chains of haemoglobin of the human and several other animals. The diagram (Figure 4.22) is presented to show how different animals may have diverged from the evolutionary line that led to humans.

Keyword

Hemoglobin is the molecule found in red blood cells that carries oxygen to where it is needed

Table 4.1 Similarities and differences between the polypeptide chains of hemoglobin in four primate species

Species	Alpha-hemoglobin (141 amino acids)	Polypeptides chains Beta-hemoglobin (146 amino acids)	Y-hemoglobin
Human	+	+	+
Chimpanzee	+	+	+
Gorilla	1	1	1
Gibbon	3	3	2

Key: Hemoglobin is composed of four peptide chains, made up of alpha, beta, and gamma (Y) polypeptides.

+ indicates no differences in amino acids sequence from that of human, figures indicates number of amino acid differences.

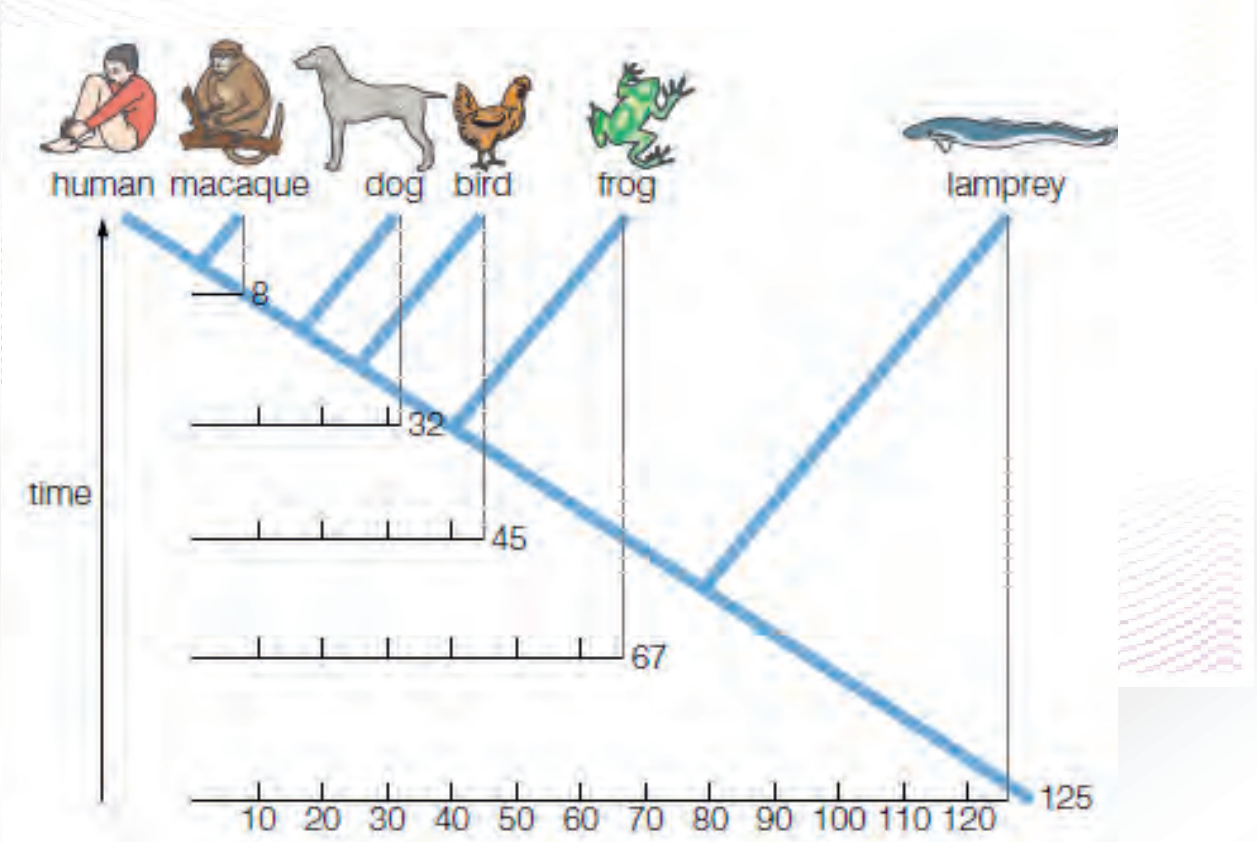


Figure 4.22 The evolutionary relationships of some animals shown by differences in haemoglobin

The molecules that are used to show evolutionary relationships are those that are common to large numbers of organisms. But, clearly, haemoglobin analysis cannot be used to include plants and algae in any phylogenetic tree.

Activity 4.13: The evidence for evolution

Your Teacher will divide the class into groups with the following professions

- The Palaeontologists
- The Anatomists
- The Embryologists
- The Biochemists

Each group will:

- Focus on its own particular area of expertise and try to find four or five pieces of evidence from books in library/ the internet about the pieces of evidence that supports the theory of evolution
- Make a poster of their findings for display

4.1.4. Natural selection: Definition, Types & Examples

By the end of this section, the learner will be able to:

- Define natural selection
- Describe the types of natural selection with examples
- Apply the theory of natural selection in their day-to-day life (survival of the fittest)
- Appreciate the struggle among organisms for survival

Natural selection is the 'driving force' behind evolution. It is the process that brings about changes (over time) in populations that can, eventually, lead to different populations of the same species to become different species. Those members of a species which are best adapted to their environment will survive and reproduce in greater numbers than others that are less well adapted. They will pass on their advantageous alleles to their offspring and, in successive generations, the frequency of these alleles will increase in their gene pool. The advantageous types will, therefore, increase in frequency in successive generations.

Activity 4:14

1. Why natural selection is a key force for ADAPTATION?
2. How natural selection operates on the phenotype of population?
3. Give practical examples for the different types of selections

To appreciate how natural selection can eventually lead to **speciation** (the formation of new species), we must be clear what do we mean by the term species. Obviously humans are different species from chimpanzees. But, the different races of humans are all members of the same species. Why?

Our current definition of species is therefore, is a group of similar organisms with a similar biochemistry, physiology and evolutionary history that can interbreed to produce fertile offspring.

This explains why all humans are members of the same species, but belongs to a different species from the chimpanzee. So how can there be different types of natural selection? All types of natural selection work in the same manner, but their influence on a population is different. The different types of

natural selection include:

Activity 4:15

Read about natural selection and discuss in detail the following points:

a. Directional selection?

b. Stabilising selection

c. Disruptive selection

- directional selection
- stabilising selection and
- disruptive selection

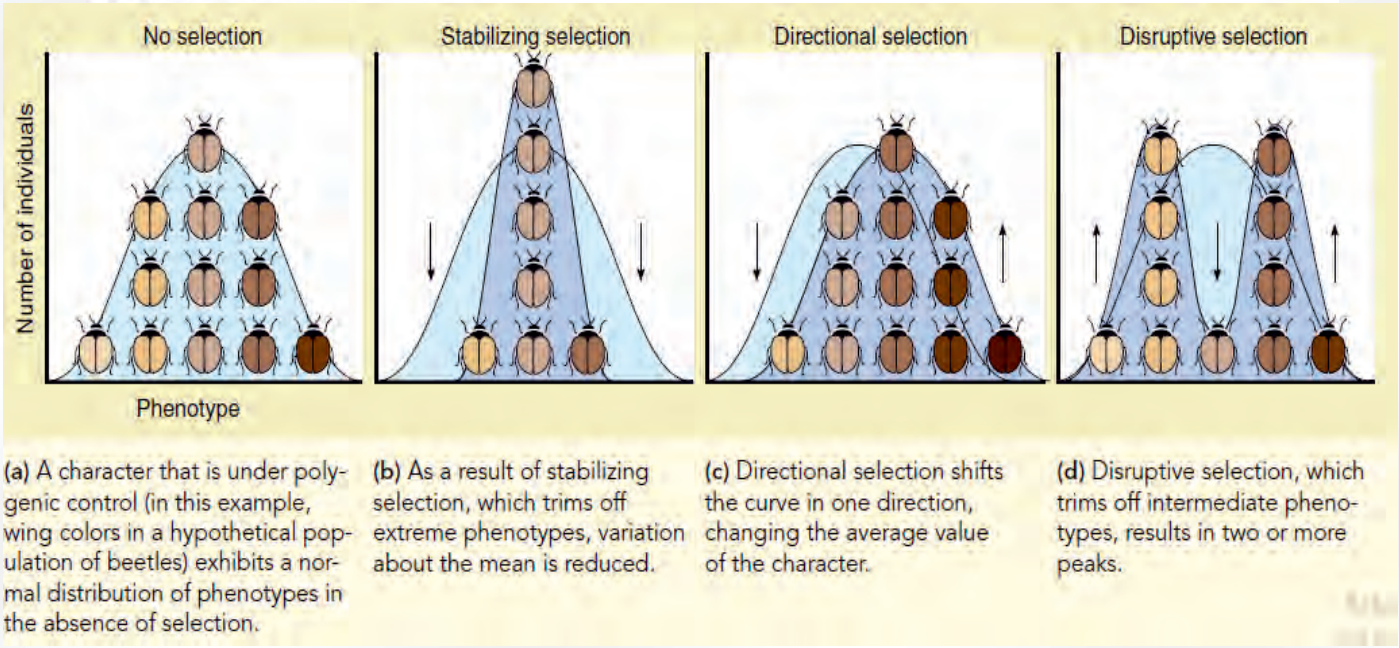


Figure 4.23 A summary of the different types of natural selection

Activity 4.16

Your teacher will facilitate to visit nearby natural forests or grasslands. You are expected to plan and inquire questions and observe what happens with the plants and animals and come up with concluding remarks on concepts like natural selection and

Antibiotic resistance – a modern example of natural selection in action

The spread of antibiotic resistance in pathogenic bacteria is a simple and elegant example of evolutionary adaptation by natural selection. Bacteria can become resistant to antibiotics through mutations that alter the cellular targets of antibiotics. The acquisition of resistance is a very rare event; for example, resistance mutations usually occur in less than 1 in a million bacteria. However, resistant bacteria can continue to grow and reproduce under antibiotic treatments that effectively paralyse or kill their antibiotic susceptible neighbours (Figure 4.24) – this is Darwinian natural selection in its simplest form.

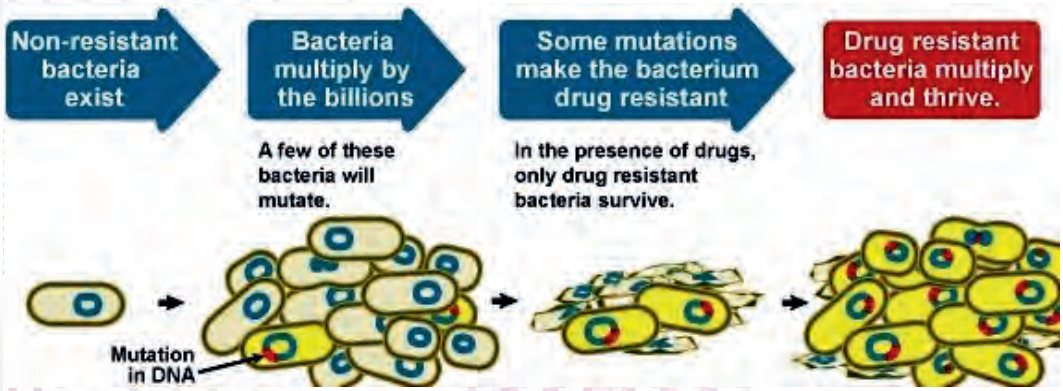


Figure 4.24 Drug resistance in bacteria is often the result of a genetic mutation

Pattern of evolution

Divergent, convergent, and parallel evolution Evolution over time can follow several different patterns. Factors such as environment and predation pressures can have different effects on the ways in which species exposed to evolve. There are three main types of evolution: divergent, convergent, and parallel evolution (Figure 4.25).

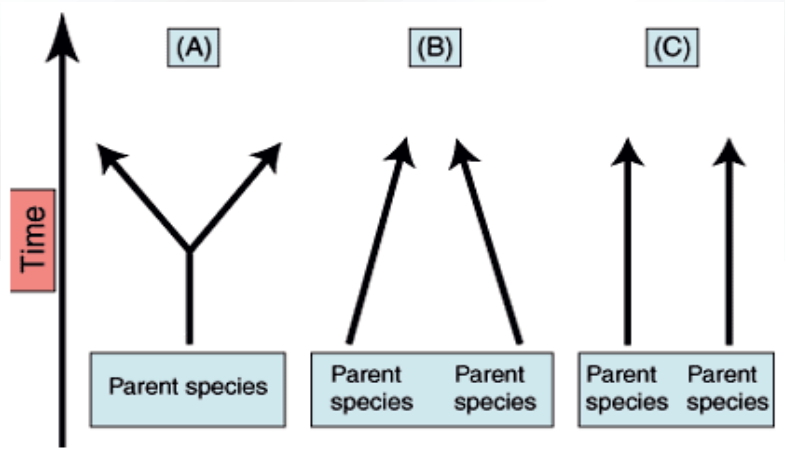


Figure 4.25 Types of evolution; A) Divergent B) Convergent C) Parallel

Divergent Evolution

When people hear the word "evolution," they most commonly think of divergent evolution, the evolutionary pattern in which two species gradually become increasingly different. This type of evolution often occurs when closely related species diversify to new habitats. On a large scale, divergent evolution is responsible for the creation of the current diversity of life on earth from the first living cells. On a smaller scale, it is responsible for the evolution of humans and apes from a common primate ancestor.

Convergent Evolution

Convergent evolution takes place when species of different ancestry begin to share **analogous traits** because of a shared environment or other selection pressure. For example, whales and fish have some similar characteristics since both had to evolve methods of moving through the same medium: water.

Parallel Evolution

Parallel evolution occurs when two species evolve independently of each other, maintaining the same level of similarity. Parallel evolution usually occurs between unrelated species that do not occupy the same or similar niches in a given habitat.

Types of speciation

Natural selection provides a mechanism by which new populations of a species can arise. But, at what point can these populations be considered as distinct species?.

Activity 4:17

1. What is a species?

2.What is the difference between reproduction and speciation?

3.How evolution plays part in the process of speciation?

4.Explain the difference between allopatric and sympatric speciation?What key elements are involved in allopatric and sympatric speciation process?

If two populations become so different, individuals from these different populations cannot interbreed to produce fertile offspring, then we must think of them as different species. There are a number of ways in which this can occur. The two main ways are:

- Allopatric speciation and
- Sympatric speciation.

Keywords

Allotropic speciation occurs when a population from an existing species becomes geographically isolated and the isolated population develops into a new species

Sympatric speciation occurs when a population from an existing species develops into a new species without becoming geographically isolated from other members of the original species.

As long as two populations are able to interbreed, they are unlikely to evolve into distinct species. They must somehow go through a period when they are prevented from interbreeding. Both allopatric and sympatric speciation involves isolating mechanisms that prevent different populations from interbreeding for a period of time (Figure 4.26). During this period, mutations that arise in one population cannot be passed to the other. As a result of this, and the different selection pressures in different environments, genetic differences between the two populations increase. Eventually, the two populations will become so different that they will be unable to interbreed or they are 'reproductively isolated'.

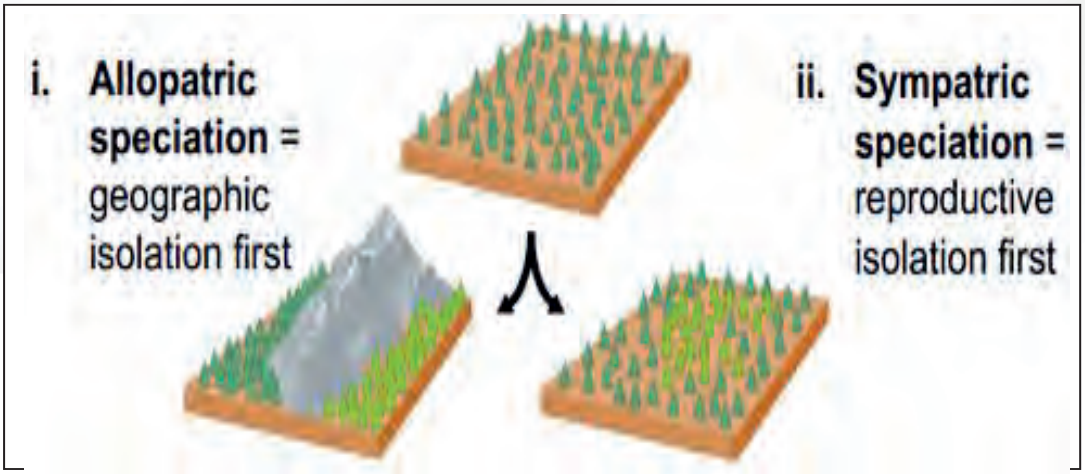


Figure 4.26. The difference between Allopatric and Sympatric speciation

Activity 4.18

- Search from books or other sources about allopatric and sympatric speciation. Write a report and present to the class.

Keywords

Polyploidy occurs when an organism has more than two sets of homologous chromosomes

Tetraploid: a tetraploid organism has four sets of homologous chromosomes

What is polyploidy and why is it important in plant evolution?

Poly- means many. **Polyploid** cells have many sets of chromosomes per cell – sometimes four sets, sometimes eight or more. Some human liver cells have 92 chromosomes per cell – they are **tetraploid** and have four sets of chromosomes per cell.

Polyploidy has been important in plant evolution because it has allowed infertile hybrids to become fertile. When different species form hybrids, very often the hybrid cannot produce offspring because all the chromosomes cannot form bivalents (homologous pairs) in meiosis. So, they cannot form sex cells and cannot reproduce. If the chromosome number were to double, then all chromosomes would be able to form homologous pairs. Meiosis and sex-cell formation can take place and the hybrid is now fertile.

Keywords

Divergent evolution –the evolution from one species of animals or plants of a number of different forms

Convergent evolution—the development of superficially similar structures in unrelated organismds, usually because the organisms live in the same kind of environment

Parallel evolution-the development of related along similar evolutionary passes due to strong selection pressures acting on all of them in the same way

A new species forms when one portion of a population can no longer breed and produce fertile offspring with the rest of the population. That is, the two separate groups no longer share a gene pool, and each begins to follow its own, independent evolutionary path. Two parts of a population can become reproductively isolated in many ways, because successful reproduction requires so many complex events. Any interruption in courtship, fertilization, embryo formation, or offspring development can be a reproductive barrier. Reproductive isolating mechanisms fall into two major categories: **prezygotic reproductive barriers** act *before* fertilization to prevent individuals of different species or populations from mating, whereas **postzygotic reproductive barriers** act *after* fertilization to prevent the development of viable offspring, or to reduce the offsprings' fertility.

Reproductive Isolation

Reproductive Isolating Mechanisms

Prezygotic barriers prevent fertilizatio

Mechanisms of **prezygotic reproductive isolation** affect the ability of two species to combine gametes and form a zygote. These reproductive barriers include the following

Ecological (or habitat) isolation: A difference in habitat preference can separate two populations in the same geographic area. For example, one species of ladybird beetle eats one type of plant, while a closely related species eats a different plant. The two species never occur on the same host plant, although they interbreed freely in the laboratory. The different habitat preferences are the reproductive barrier that keeps the gene pools of the two species separate.

Temporal isolation: Two species that share a habitat will not mate if they are active at different times of day or reach reproductive maturity at different times of year

Behavioral isolation: Behavioral differences may prevent two closely related species from mating. The males of two species of tree frogs, for instance, use distinct calls to attract mates. Female frogs choose males of their own species based on the unique calls. Likewise, sexual selection in many birds is based on intricate mating dances

Mechanical isolation: In many animal species, male and female parts fit together almost like a key in a lock. Any change in the shape of the gamete-delivering or –receiving structures may prevent groups from interbreeding. In plants, males and females do not copulate, but mechanical barriers still apply

Gametic isolation: If a sperm cannot fertilize an egg cell, then no reproduction will occur. For example, many marine organisms, such as sea urchins, simply release sperm and egg cells into the water. These gametes display unique surface molecules that enable an egg to recognize sperm of

Keywords

Prezygotic Barriers - Prevent fertilization

Temporal isolation - Similar species reproduce at different times

Habitat isolation - Similar species reproduce in different habitats

Behavioral isolation - Similar species have distinctive courtship behaviors Mechanical isolation - Similar species have structural differences in their reproductive organs

Gametic isolation - Gametes of similar species are chemically incompatible

Postzygotic Barriers- Reduce viability or fertility of hybrid

Hybrid inviability - Interspecific hybrid dies at early stage of embryonic development

Hybrid sterility - Interspecific hybrid survives to adulthood but is unable to reproduce successfully

Hybrid breakdown - Offspring of interspecific hybrid are unable to reproduce successfully

the same species. In the absence of a "match," fertilization will not occur, and the gene pools will remain separate.

Activity 4:19

Make a small group to discuss and reflect the following question

What difference have you seen between

A. temporal and habitat isolation

B. Gametic and mechanical isolation

Postzygotic Barriers Prevent Viable or Fertile Offspring

Individuals of two different species may produce a hybrid zygote.Even then, **postzygotic reproductive isolation** may keep the species separate by selecting against the hybrid offspring, effectively preventing genetic exchange between the populations. Collectively, these postzygotic barriers are sometimes called hybrid incompatibility.

Postzygotic reproductive barriers include the following:

Hybrid inviability: A hybrid embryo may die before reaching reproductive maturity, typically because the genes of its parents are incompatible. Since the hybrid offspring cannot reproduce, the gene pools of the parent species remain isolated from one another.

Hybrid infertility (sterility): Some hybrids are infertile. A familiar example is the mule, a hybrid offspring of a female horse and a male donkey. Mules are infertile because a horse's egg has one more chromosome than a donkey's sperm cell. The animal can grow and develop, but meiosis does not occur in the mule's germ cells because the chromosomes are not homologous

Activity 4.20

Make a small group and discuss

- What would happen if a muel interbreed with horse and/or donkey and produce a viable offspring? Reflect your discussion to the class.

Hybrid breakdown: Some species produce hybrid offspring that are fertile. When the hybrids reproduce, however, their offspring may have abnormalities that reduce their fitness. Some second-generation hybrid offspring of the mosquito species *Aedes aegypti* and *Aedes mascarensis*, for example, have abnormal genitalia that make mating difficult. The strong selective pressure

against hybrid offspring limits gene flow between the two mosquito species.

Activity 4.21: Apes to humans

Make a small group and discuss a topic from the list below:

- Tools and tool use by early hominids before Homo sapiens
- Who's who on the *Homo sapiens* evolutionary tree
- The preservation of non-human primates in the wild
- The common ancestor of humans and chimpanzees

Activity 4.22 Making a fossil

You will need:

- light plastic container to make the fossil
- a sponge; natural and synthetic are both fine, but sponges with more holes work better
- some fine sand, enough to half fill your container
- magnesium sulphate
- a saucer or small tray

Carry out the following:

- Create a shape for your fossil by cutting it out from the sponge – it could be a leaf, a shell, a bone or a whole animal.
- Cut two or three small holes in the bottom of your container. Place it on the saucer/ tray.
- Put sand in the container to a depth of 1 cm.
- Place your sponge shape on top and cover with another 2 cm of sand.
- Mix four spoons of bath magnesium sulphate and warm water and pour into your container.
- Allow the mix to sink through the sand then leave it somewhere safe and warm (for example, a window ledge).
- Add more of the water and salt mix at least once a day for at least five days. The longer you leave it the more 'fossilized' it becomes! (The holes in the sponge trap the salts, mineralising the sponge; as they dry out they solidify to create a fossil. How crumbly this fossil is depends on the consistency of your local sand)
- After this time, leave the sand to dry out for two days before removing the 'fossil' sponge. If it is still a little wet, leave the fossil for a few days before

4.1.5. Human evolution

By the end of this section, the learner will be able to:

- Outline human evolution.
- State the role of paleontological discoveries in Ethiopia in explaining human evolution e.g. Lucy (*Australopithecus afarensis*).

Who are we and where have we come from?

Activity 4.23

Make a small group, and use internet, pictures, diagrams, and video material on human evolution. Reflect your findings to the class.

There is often a lot of very loose language used in describing human evolution. You will hear people say 'we evolved from monkeys' or 'we evolved from apes' or 'we evolved from chimpanzees'. None of these statements are accurate. There has been a 'line of evolution' for millions of years that has given rise to old world monkeys, new world monkeys, the great apes and the different species of humans that have lived (Figure 4.27) But, we are *Homo sapiens* and we are the latest of several humans to live on the planet.

We have two features in particular that distinguish us from other primates. These are:

- A very large brain, and
- Bipedalism – the ability to truly walk on just two legs.

There was a lot of debate amongst biologists as to which of these came first and also about exactly how this 'evolutionary tree' (Figure 4.27) has given rise to the various groups although they may disagree.

Keyword

Homo sapiens is the species that all humans alive today belong to.

Activity 4.24

Form small groups and discuss differences among the human races.

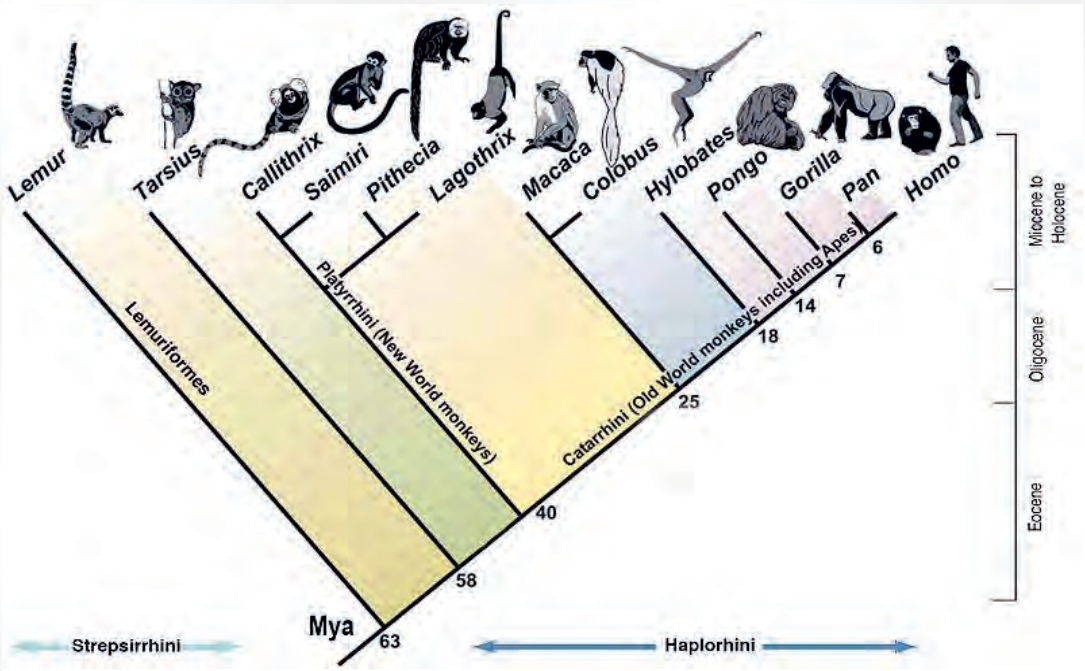


Figure 4.27 The evolutionary tree for modern primates

Over the details, they all agreed about the idea – a line evolution that has branched to give the different groups of primates (including apes and humans) that exist today has existed in not too distant past. Figure 4.28 shows the part of the evolutionary tree of humans and the living great apes in more detail.

Keywords

Genus (plural **genera**) a group of closely related species

Hominin any member of the genus Homo. This includes modern humans (Homo sapiens), neanderthals and Homo erectus

Hominid a group of species that includes all the species belonging to the genus Homo along with other species such as Ardipethecus ramidus and Australopithecus afarensis (Lucy).

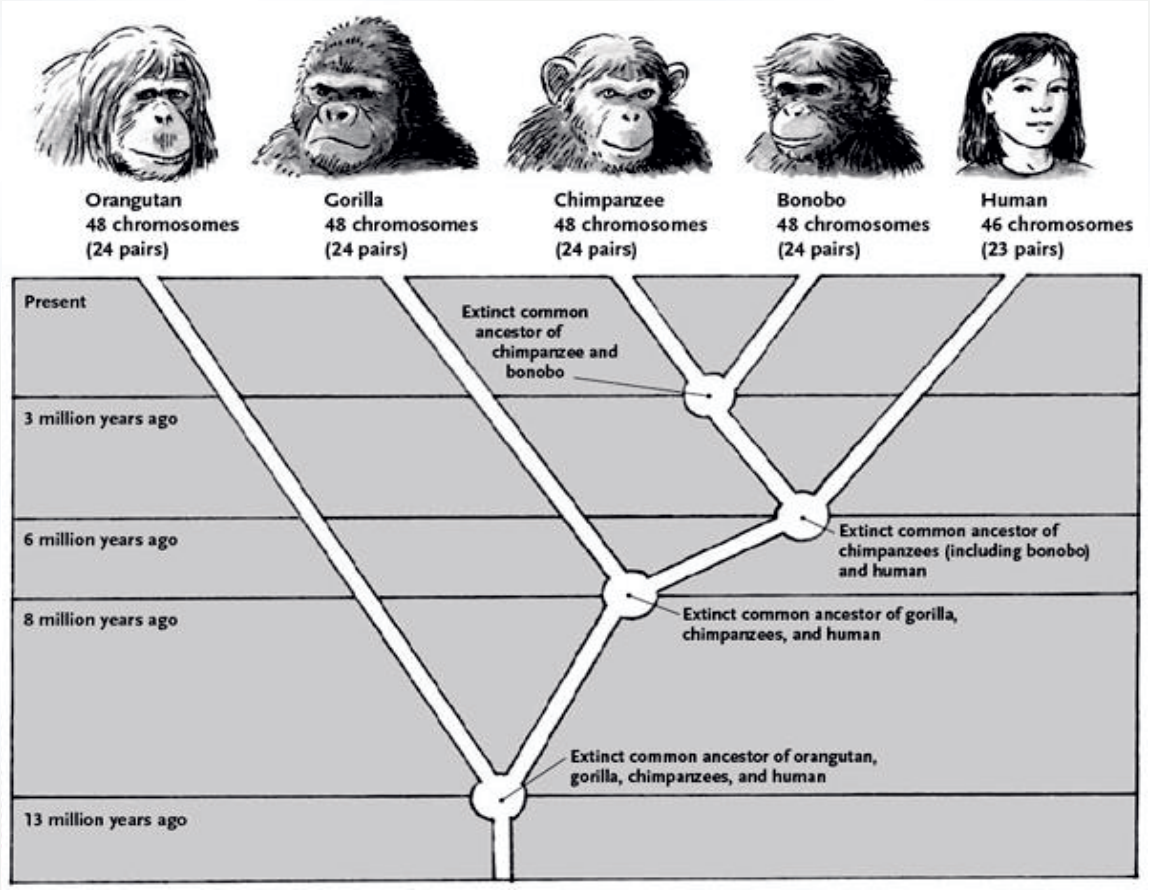


Figure 4.28 The evolutionary tree of humans and the great apes

Branching points in the evolutionary tree represent ancestors (Figure 4.28). At these points it is assumed that an ancestral type became divided into at least two populations which subsequently evolved along different lines. For example, humans and chimpanzees both evolved from a common ancestor that lived about 6 million years ago (Figure 4.28).

So far, we have talked about 'humans' rather than the one specific type of human (ourselves – *Homo sapiens*) that now inhabits the planet. There were other humans before us and, before them, what we might call 'pre-humans'. However, all humans belong to the **genus** *Homo*.

Figure 4.29 shows a timeline for the major **hominin** and **hominid** species based on the currently available fossil evidence. Looking carefully at Figure 4.30, you begin to see the evolution of humans. Fossils of many of the species along the early part of the timeline were found in Ethiopia. The country is therefore, the 'cradle of mankind'.

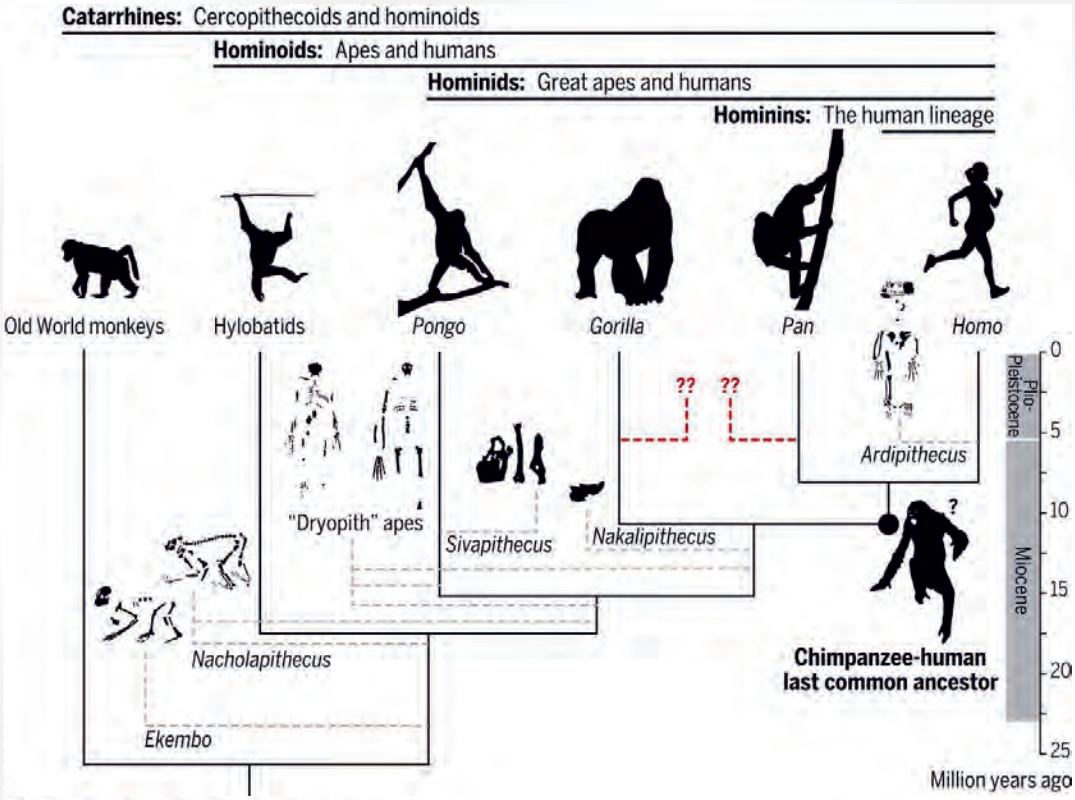


Figure 4.29 A timeline for the major hominin and hominid species

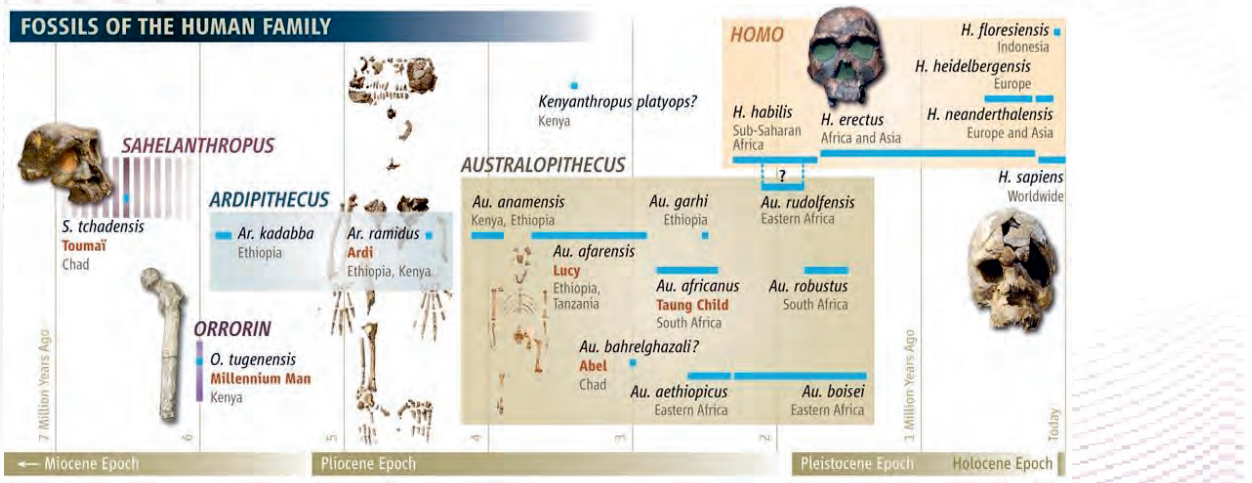


Figure 4.30 A timeline for the fossils of human family.

Lucy and Ardy

Both Lucy and Ardy are important fossils in explaining the evolution of modern humans and chimpanzees from a common ancestor. Lucy was discovered by Donald Johanson and Tom Gray in 1974 at Hadar, Ethiopia. Lucy is a fossil dated at about 3.2 million years. She was an adult female of about 25 years and belonged to the species *Australopithecus afarensis*. Her skeleton was about 40% complete, an unusually high proportion for a fossil skeleton. Her pelvis, femur (the upper leg bone) and tibia show that she was bipedal (could walk upright on two legs) (Figure 4.31).

However, there is also evidence that Lucy was partly arboreal (tree-dwelling). She was about 107 cm (42″) tall and about 28 kg (62 lbs) in weight. At the time she was discovered, Lucy represented one of the oldest fossil hominins. The proportions of her humerus and femur were mid-way between those of modern humans and chimpanzees.

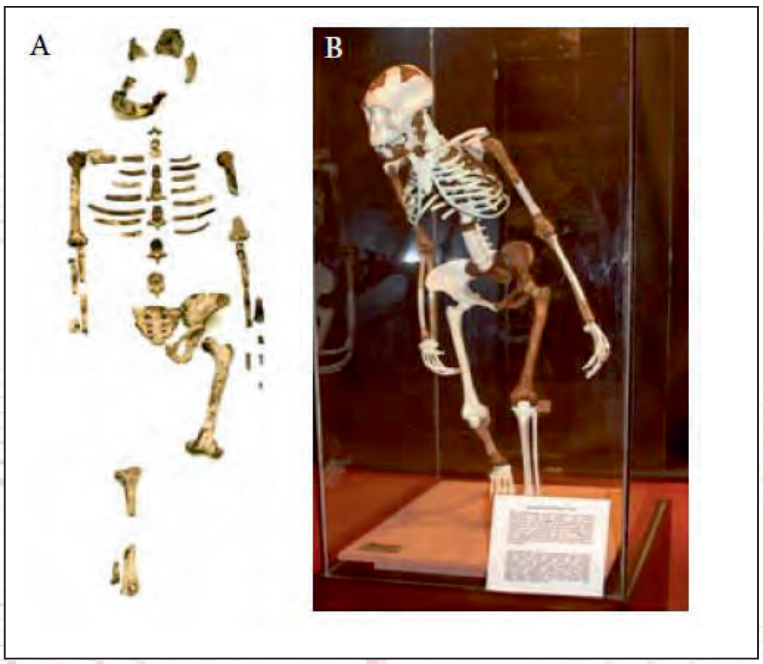


Figure 4.31 A – The original Lucy fossil; B – The Lucy display including reconstructed parts

Lucy had a brain about the same size as that of a chimpanzee, so her discovery was able to settle a debate amongst biologists at the time– which came first, large brain or bipedalism? Clearly bipedalism came before big brains.

The Ardi fossil (together with many other similar fossils) was first discovered in 1992, in the Afar dessert in Ethiopia, but it was only in 2009 after many years' analysis, that

research papers were finally published that gave Ardi a unique position in the human evolution.

Ardi was 1.2 million years older than Lucy, was also female who belonged to the species *Ardipithecus ramidus*. One significant feature about Ardi was that she was also bipedal. At 4.4 million years old, Ardi is the nearest fossil to the 'common ancestor' of humans and chimpanzees that has so far been found. This finding finally proved that the common ancestor of humans and chimpanzees could not have resembled a chimpanzee, as chimpanzees are not truly bipedal. However, there was signs of being adapted for both bipedal walking and arboreal life (Figure 4.32).

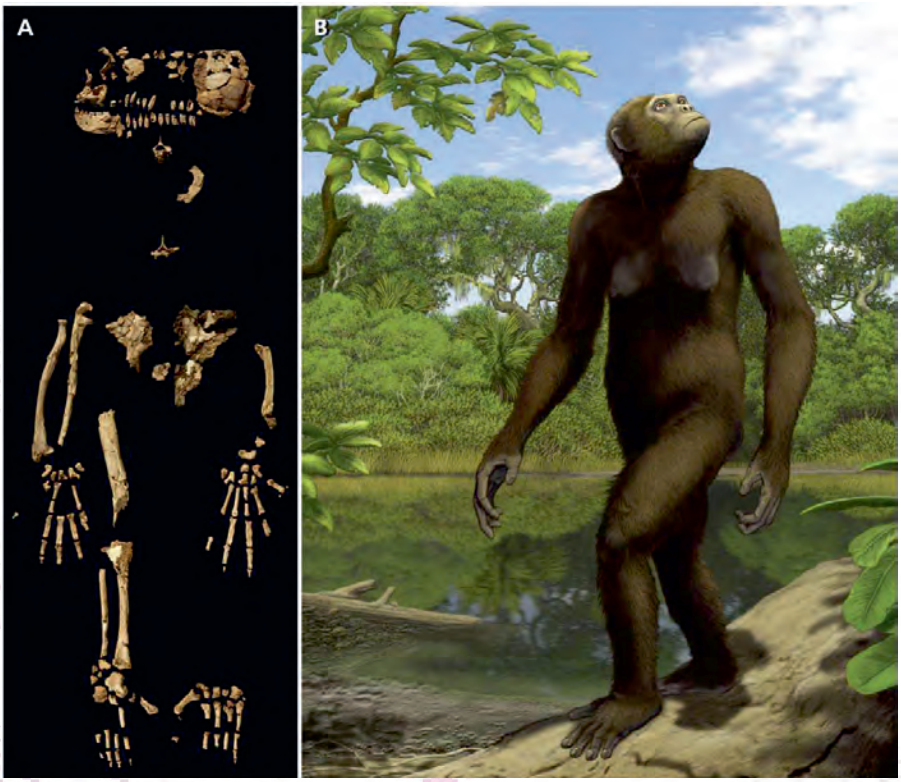


Figure 4.32 A relatively complete skeleton of *Ardipithecus*, which lived 4.4 million years ago. B: *Ardipithecus* shows signs of being adapted for both bipedal walking and arboreal life

How brain size changed during human evolution?

During the course of human evolution, the brain has got bigger. Studies on comparative anatomy of fossils revealed that the cranial capacity has increased with each new hominid species evolved (Figure 4.33). However, the brain has increased in size as a proportion of body mass. Species of *Australopithecus* have a brain that is between 0.7% and 1.0% of their body mass, whereas modern humans have a brain size between 1.8% and 2.3% of their body mass. The brain of *Homo sapiens* uses 25% of the resting energy requirement, compared with 8% in the great apes.

A larger brain allows humans to:

- run faster and in a more upright posture
- plan in advance to avoid attack
- develop and use tools and weapons

Activity 4.25

Read about emergence of *Homo sapiens* and present to the class.

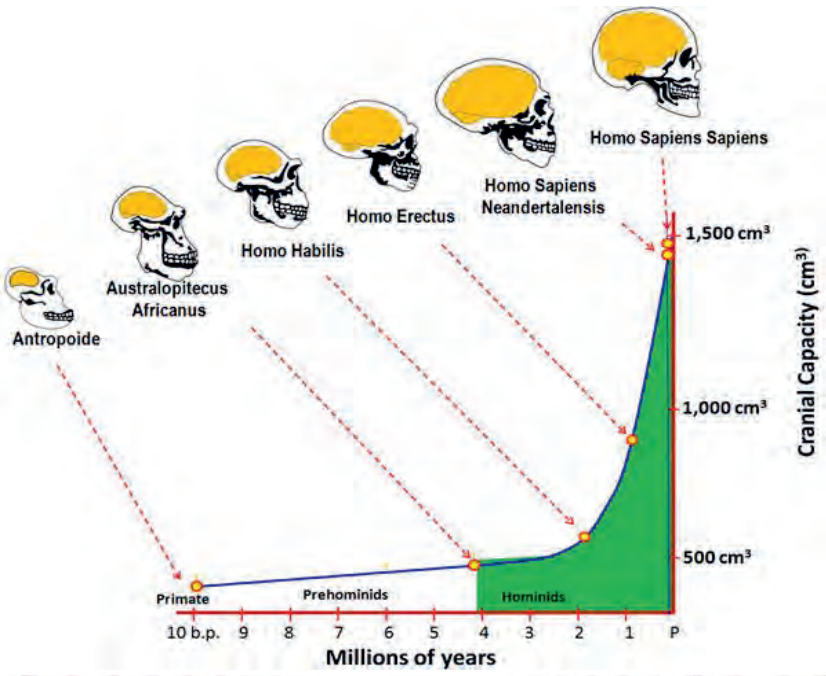


Figure 4.33 Brain size in different hominids

These abilities clearly depend on other physical adaptations such as longer legs, more nimble fingers and a straighter spine, but, without the larger brain to co-ordinate the activities; the physical changes would not confer the same advantage.

Are we still evolving?

Homo sapiens (modern humans) first appeared in Africa and have since migrated to all other parts of the world. Figure 4.34 shows these migratory patterns together with the time (thousands of years ago) when they took place.

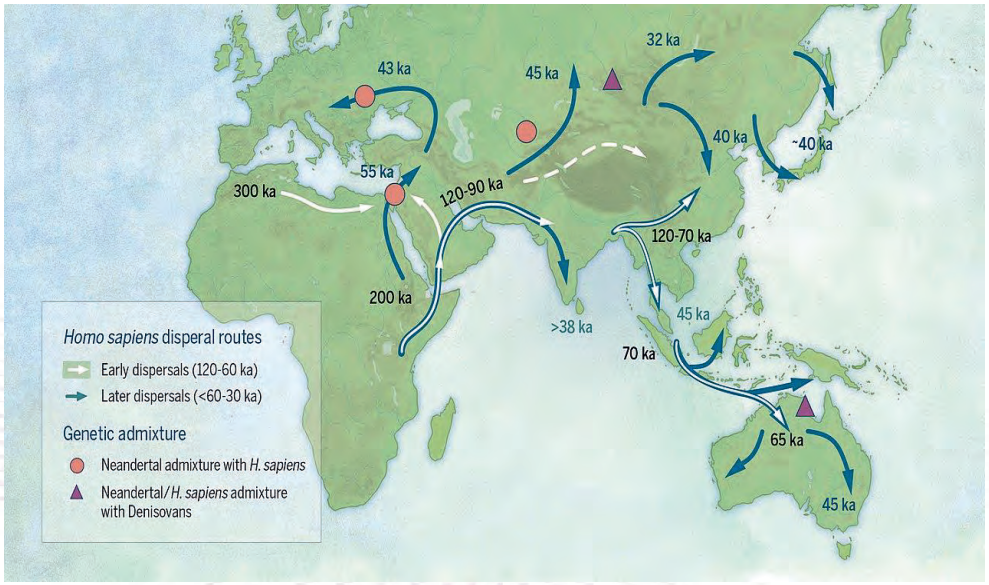


Figure 4.34 The migration of modern humans out of Africa – it all begins in East Africa. Numbers indicate the time (in years) since each stage of the migration.

As humans moved from Africa into different areas of the world, they encountered different environments. Different selection pressures in the different environments resulted in the different human populations evolving along different lines.

For example, as humans encountered colder climates, body features that gave a survival advantage to conserve heat were selected for. These included:

- a shorter, squatter body shape; this reduces the surface-area-to-volume ratio and so reduces the rate of heat loss by radiation
- an increased layer of adipose tissue under the skin to act as insulator.

- increased hairiness; this reduces heat loss by convection

Humans have been evolving into different 'races' for thousands of years. The classification of these races is difficult and there is some disagreement about their exact nature.

Activity 4.26

Search from books and other sources about different types of human races and the disagreement on the classification of the races. Report your finding to the class.

Activity 4.27

Ethiopia plays a central role in the story of human evolution. Using this text book, other books and internet, read as much as you can on the discovery and importance of Lucy, Ardi and Selam in the story of human evolution. Reflect your findings to the class.

4.1.6. Mutation

By the end of this section, the learner will be able to:

- Define mutation
- Examine the different types of mutations with examples
- Describe the causes of mutation
- Appreciate the effects of mutations on living things

Keyword

Mutation is a random change in genetic information

A **mutation** can be caused by several factors but is divided into two parts. If the agent that caused the mutation cannot be identified then it is known as a **spontaneous mutation**. If the mutation can be identified then it is called an **induced mutation**. Substances that cause mutations are radiation, x-ray, ultra-violet radiation, nuclear radiation and certain chemical substances. These agents can also be called **mutagenic agents** or **mutagens**. There can be large

structural changes involving the whole chromosomes or parts of chromosomes, or changes that involve only a single base. The changes involving only a single base are called **point mutations**.

Point mutation

There are several types of point mutation, in which one of the bases in the DNA sequence of a gene is altered, usually by being copied wrongly when the DNA replicates. The different point mutations are:

- substitution
- addition
- deletions

Activity 4.28

Make a small group and use the internet, books, and other relevant sources to discuss and describe the causes and effects of mutation on living things. Reflect your findings to the class.

These mutations occur quite randomly when the DNA is replicating and each involves a change to just one base, but the change to the gene can be dramatic and the result can be that the protein the gene should code for is not made at all or a different protein is made (Figure 4.35)

A point mutation is a change in a single nucleotide in DNA. This type of mutation is usually less serious than a chromosomal alteration. An example of a point mutation is a mutation that changes the codon UUU to the codon UCU.

Table 4.2 Description of point mutation

Type	Description	Example	Effect
Silent	mutated codon codes for the same amino acid	CAA (glutamine) → CAG (glutamine)	none
Missense	mutated codon codes for a different amino acid	CAA (glutamine) → CCA (proline)	variable
Nonsense	mutated codon is a premature stop codon	CAA (glutamine) → UAA (stop) usually	serious

Substitution

Guanine replaces thymine in this substitution. The triplet ATT has been changed to ATG (no other triplet is affected). The original triplet, ATT, codes for the amino acid isoleucine.

However, the new triplet, ATG, codes for methionine (see Figure 4.35). As a result, a different protein will be synthesised, which may or may not be significantly different from the original. One different amino acid in a protein does not always make a functional change.

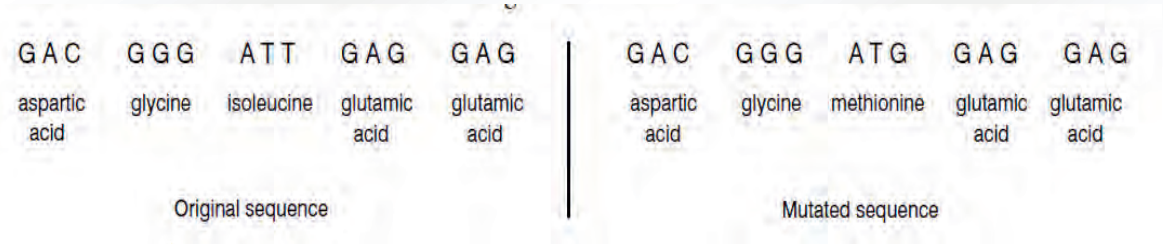


Figure 4.35 A substitution mutation

If a substitution of just one base in the sixth triplet of the gene coding for one of the four polypeptides in the haemoglobin molecule alters the triplet from GAG to GTG.

This results in the amino acid valine replacing glutamate in the polypeptide chain. The different haemoglobin molecule formed results in the condition known as **sickle-cell anaemia** (Figure 4.36)**.**

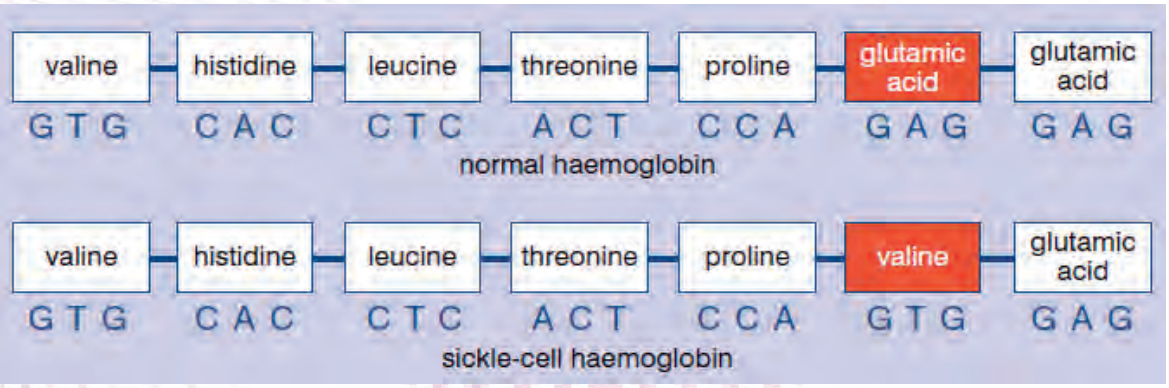


Figure 4.36 Sickle-cell anaemia

Addition and deletion

In a **deletion mutation,** a base is 'missed out' during replication, whereas in additions, an extra base is added **(Figure 4.37).** Both deletion and additions are more significant mutations than substitutions. The reason for this is that they do not just alter the triplet in which the mutation occurs. Because there is one fewer or one extra base, the whole sequence after the

Keyword

Deletion mutation a mutation caused by one DNA nucleotide being omitted from the sequence.

point of the mutation is altered. We say that there has been a frameshift and these are frameshift mutations. A totally different mRNA is produced (if one is produced at all) and a non-functional protein or no protein at all. Sometimes, a whole triplet is missed out or inserted. This will result in either one extra or one fewer codon in the mRNA. In turn, this will lead to one extra or one fewer amino acid in the polypeptide chain.

Another way of thinking about frameshifts

Look of this sequence of letters: THEMANWASHOTANDRANF ORHISHAT

If we give this a 'reading frame' of three letters, it becomes:

THE MAN WAS HOT AND RAN FOR HIS HAT

and it makes sense. But if we take out the S at the end of WAS (a deletion mutation), it becomes:

THE MAN WAH OTA NDR ANF ORH ISH AT

In other words it no longer makes sense. In genetic terms it is **missense** mutation.

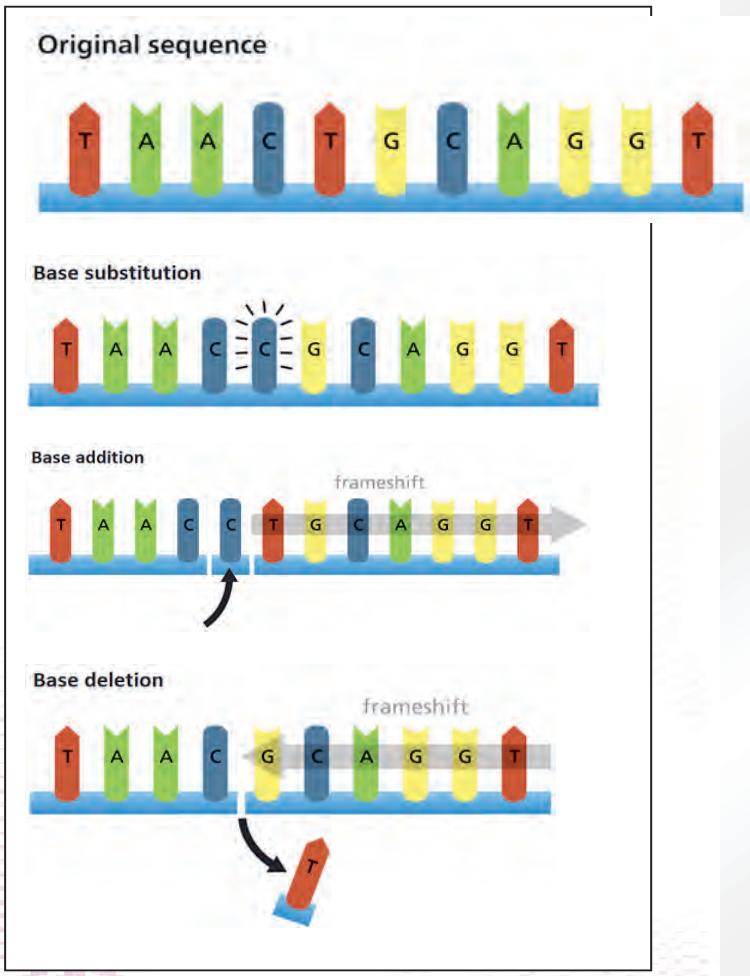


Figure 4.37 Types of point mutation

What causes point mutations and what are the consequences of gene mutations?

Activity 4.29

Search different types of sources and prepare a report to the class the following points

a. Possible causes of mutation.

b. Consequences of gene mutations

Chromosomal mutation

A **chromosomal mutation** is a mutation involving a long segment of DNA. Chromosomal mutation occurs when there is any change in the arrangement or structure of the chromosomes. Chromosomal mutation occur most often during meiosis at crossing over in prophase I. There are several different mutation types that result in a change in the structure of a chromosome such as duplication, deletion, inversion and translocation (Figure 4.38). They are much bigger events than point mutations and usually result in the death of a cell. They may also affect the whole organism. For example, if essential parts of the DNA are affected by chromosomal mutations, a foetus may be aborted. There are different types of chromosome mutations.

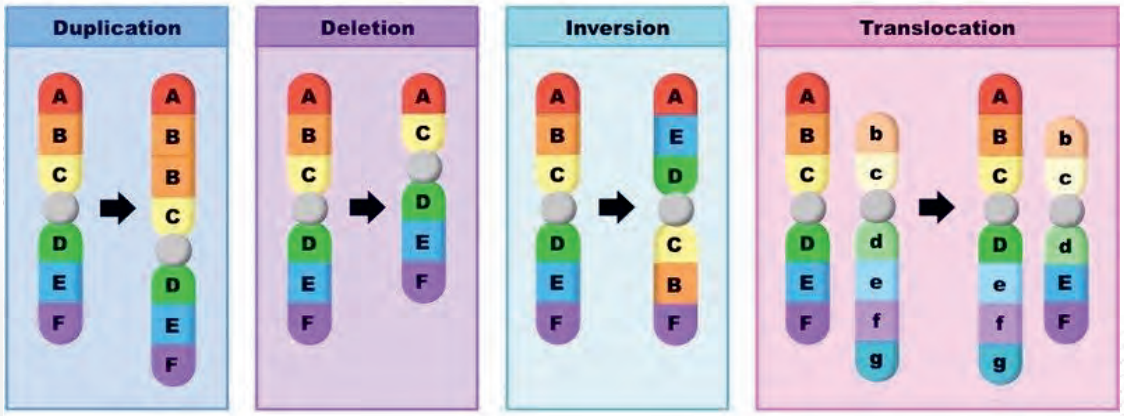


Figure 4.38. Types of Chromosomal mutations

Types of Chromosomal Mutations

Most chromosomal mutations are either Chromosomal Mutations I or Chromosomal mutations II.

- Chromosomal Mutations I involve the processes of inversion, deletion, duplication/amplification or translocation

- Chromosomal Mutations II involves aneuploidy and polyploidy.

A. Chromosomal mutations I

Chromosomal mutations I mutations alter the structure of the chromosome since they tend to break and these pieces often form sticky ends.

Inversion

Inversion chromosomal mutation is a kind of chromosomal mutations I and its segments are reversed from end to end. A piece of the chromosome is removed then reattached but in the opposite direction than it was originally. When this does not include the center or the centromere of the chromosome, it is called **paracentric inversion**. When the inversion does include the centromere, it is **pericentric**.

Deletion

Also known as partial monosomies, these occur when a piece of a chromosome accidentally gets removed or deleted. There can be cases with one piece deleted at the end (terminal deletion), two deletions – one deleted within the chromosome, and one on the end – (interstitial deletion). Microdeletions also occur when the deletions in the chromosomes are too small to be detected.

Duplication/Amplifications

As the name implies, an extra copy of a segment or the entire chromosome is present in the nucleus. These are also known as partial trisomies. Often an organism that usually has two copies of a chromosome will have three in the case of duplication. This can happen anywhere along the chromosome whether in the middle or on the ends.

Translocation

Translocation chromosomal mutation follows the route of biological translocation. This happens when a segment of a chromosome breaks off and then relocates to a completely different chromosome. This creates fusion chromosomes as one type of chromosome fuses with another.

Reciprocal translocation occurs when pieces of chromosomes "swap" places. **Robertsonian translocation** occurs when a segment of a chromosome is attached to another chromosome, causing an elongation of it. These can be balanced or unbalanced, where the chromosome is fully functional with no missing genetic information (balanced) or with important missing pieces and cannot function as normal (unbalanced). The translocation of chromosome 21 onto the 14th chromosome

causes the common genetic mutation of **down syndrome**.

B. Chromosomal Mutations II

The other type of chromosomal mutation categories Chromosomal mutations II than consists of aneuploidy and polyploidy. The general category of these is called heteroploidy since they cause a mutation by changing the number of chromosomes present in the cell.

Aneuploidy

This mutation either causes the loss or addition of a chromosome by the contortion of the chromosome
set. **Nondisjunction** during meiosis or mitosis usually results in this mutation.

Polyploidy

This mutation causes the creation of two sets of genomes within an organism. It is not usually common naturally however, it can be observed in some plants. It usually causes effects like giantism and the reduction of fertility.

3. The Advantages of Chromosomal Mutations

In some cases, chromosomal mutations can benefit the organism. Some mutations can help the organisms to survive better than others. Lactose tolerance became an advantage to have over others when numerous populations depended on cows and goats as sources of food. On the other hand, chromosomal mutations can be dangerous and even detrimental to the life of living organisms. Some of these can cause numerous problems within animals, plants, and humans.

Genetic Disorder

These are specific disorders or disabilities caused by mutations within the organism's DNA. These can be small genetic issues that may barely affect the individual or larger issues that may bring major concerns to the individual. A chromosomal disorders list can be seen below:

- Trisomy 21: Down Syndrome
- Trisomy 18
- Trisomy 13
- Klinefelter Syndrome
- XYY Syndrome
- Turner Syndrome
- Triple X Syndrome

Activity 4.30

Search from different books and internet about different types of chromosome mutations (inversion, deletion, insertion, duplication etc). Report your finding to the class.

4.1.7. Genetic Drift

By the end of this section, the learner will be able to:

Discuss Genetic drift.

What is genetic drift?

Keyword

Genetic drift is a mechanism of evolution in which allele frequencies of a population change over generations due to chance events. Genetic drift is change due to "sampling error" in selecting the alleles for the next generation from the gene pool of the current generation.

Allele frequencies can change due to chance alone. This is called genetic drift. Drift is a binomial sampling error of the gene pool. What this means is, the alleles that form the next generation's gene pool are a sample of the alleles from the current generation. When sampled from a population, the frequency of alleles differs slightly due to chance alone.

Alleles can increase or decrease in frequency due to drift. The average expected change in allele frequency is zero, since increasing or decreasing in frequency is equally probable. A small percentage of alleles may continually change frequency in a single direction for several generations just as flipping a fair coin may, on occasion, result in a string of heads or tails. A very few new mutant alleles can drift to fixation in this manner.

The variance in the rate of change of allele frequencies is greater in small populations than in large populations. Let's make the idea of drift more concrete by looking at an example. As shown in Figure 4.39, we have a very small rabbit population that's made up of 8 brown individuals (genotype BB or Bb) and 2 white individuals (genotype *bb*). Initially, the frequencies of the *B* and *b* alleles are equal. What if, purely by chance, only the 5 circled individuals in the rabbit population reproduce? (Maybe the other rabbits died for reasons unrelated to their coat color, e.g., they happened to get caught in a hunter's snares.) In the surviving group, the frequency of the *B* allele is 0.7 and the frequency of the *b* allele is 0.3.

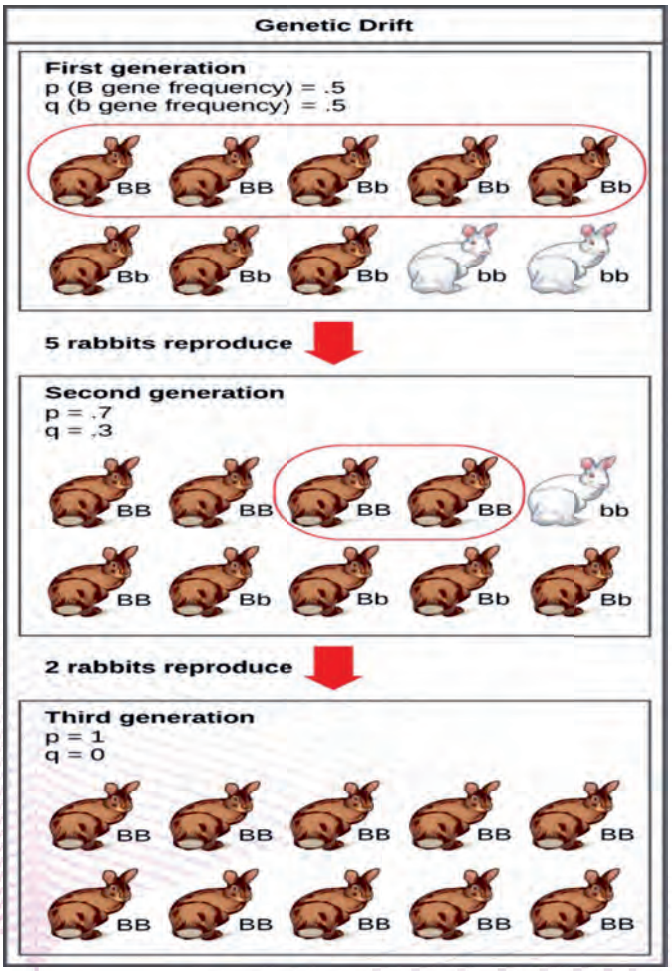


Figure 4.39 Example of genetic drift in a rabbit population

The allele frequencies of the five lucky rabbits are perfectly represented in the second generation. Because the 5-rabbit "sample" in the previous generation had different allele frequencies than the population as a whole, the frequencies of *B* and *b* in the population have shifted to 0.7 and 0.3 respectively. From this second generation, what if only two of the *BB* offspring survive and reproduce to yield the third generation? In this series of events, by the third generation, the *b* allele is completely lost from the population.

However, the overall rate of genetic drift (measured in substitutions per generation) is independent of population size. If the mutation rate is constant, large and small populations lose alleles to drift at the same rate. This is because large populations will have more alleles in the gene pool, but they will lose them more slowly. Smaller

populations will have fewer alleles, but these will quickly cycle through. This assumes that mutation is constantly adding new alleles to the gene pool and selection is not operating on any of these alleles.

Hardy–Weinberg Equilibrium

At Hardy–Weinberg Equilibrium, Allele Frequencies Do Not Change

The study of population genetics relies on the intimate relationship between allele frequencies and genotype frequencies. Each genotype's frequency is the number of individuals with that genotype, divided by the total size of the population. For example, if 64 of the 100 individuals in a population are homozygous recessive, then the frequency of that genotype is 64/100, or 0.64.

Hardy–Weinberg equilibrium is the highly unlikely situation in which allele frequencies and genotype frequencies do not change from one generation to the next. It occurs only in populations that meet the following assumptions:

(1) Natural selection does not occur;

(2) Mutations do not occur, so no new alleles arise;

(3) The population is infinitely large, or at least large enough to eliminate random changes in allele frequencies;

(4) Individuals mate at random; and

(5) Individuals do not migrate into or out of the population.

Hardy–Weinberg equilibrium is named after mathematician

Godfrey H. Hardy and physician Wilhelm Weinberg. They independently developed two simple equations that represent the relationship between allele frequencies and genotype frequencies. To understand their logic, begin by assuming that a gene has only two possible alleles, with frequencies p and q.

The first equation represents the frequencies of both alleles in the population:

$$p + q = 1$$

The two frequencies add up to 1 because the two alleles represent all the possibilities in the population. For example, the frequency of the dark fur allele (D) is 0.6; the frequency of the alternative allele d, which confers tan fur, is 0.4. (Tally the D and d alleles in the picture of the ferrets to verify these numbers.)

At Hardy–Weinberg equilibrium, we can use allele frequencies to calculate genotype

frequencies, according to the second equation in figure 12.12

p2 + 2pq + q2 = 1

In this equation, the proportion of the population with genotype DD equals p2 (0.36 for our ferrets) and the proportion with genotype dd equals q2 (0.16). To calculate the frequency of the heterozygous class, multiply pq by 2 (0.48). Since the homozygotes and the heterozygotes account for all possible genotypes, the sum of their frequencies must add up to 1.

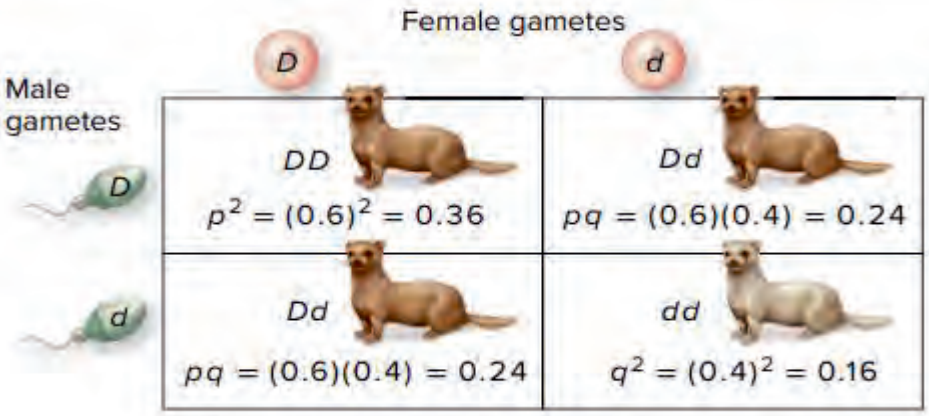
Allele frequencies: $p + q = 1$

Definition/equation	Example
p = frequency of dominant allele	p = frequency of *D* (dark fur) = 0.6
q = frequency of recessive allele	q = frequency of *d* (tan fur) = 0.4
$p + q = 1$	0.6 + 0.4 = 1

Genotype frequencies: $p^2 + 2pq + q^2 = 1$

Definition/equation	Example
p^2 = frequency of *DD* genotype	0.6 x 0.6 = 0.36
$2pq$ = frequency of *Dd* genotype	2 x 0.6 x 0.4 = 0.48
q^2 = frequency of *dd* genotype	0.4 x 0.4 = 0.16
$p^2 + 2pq + q^2 = 1$	$(0.6)^2 + (2 \times 0.6 \times 0.4) + (0.4)^2 = 1$

Reproduction (random mating)



Hardy–Weinberg Equilibrium. At Hardy–Weinberg equilibrium, allele frequencies remain constant from one generation to the next; evolution does not occur."

Table 4.3 Conditions for Hardy-Weinberg Equilibrium

Condition	Consequence if Condition Does Not Hold
1. No mutations	The gene pool is modified if mutations occur or if entire genes are deleted or duplicated.
2. Random mating	If individuals mate within a subset of the population, such as near neighbors or close relatives (inbreeding), random mixing of gametes does not occur and genotype frequencies change.
3. No natural selection	Allele frequencies change when individuals with different genotypes show consistent differences in their survival or reproductive success.
4. Extremely large population size	In small populations, allele frequencies fluctuate by chance over time (a process called genetic drift).
5. No gene flow	By moving alleles into or out of populations, gene flow can alter allele frequencies.

The genetic bottleneck effect

The **bottleneck effect** is an extreme example of genetic drift that happens when the size of a population is severely reduced. Events like natural disasters (earthquakes, floods, fires) can decimate a population, killing most individuals and leaving behind a small, random assortment of survivors.

The allele frequencies prior to the natural disasters may be very different from those of the population after the event, and some alleles may be missing entirely. The smaller population will also be more susceptible to the effects of genetic drift for generations (until its numbers return to normal), potentially causing even more alleles to be lost.

Imagine a bottle filled with marbles, which, represent individuals in a population. If a bottleneck event occurs, a small, random assortment of individuals survive the event and pass through the bottleneck (and into the cup), while the vast majority of the population is killed off (remains in the bottle). The genetic composition of the random survivors (Figure 4.40) is now the genetic composition of the entire population. A population bottleneck yields a limited and random assortment of individuals. This small population will now be under the influence of genetic drift for several generations.

Activity 4. 31

Discuss the following question in small group and reflect your opinion to the class.

How can a bottleneck event reduce genetic diversity?

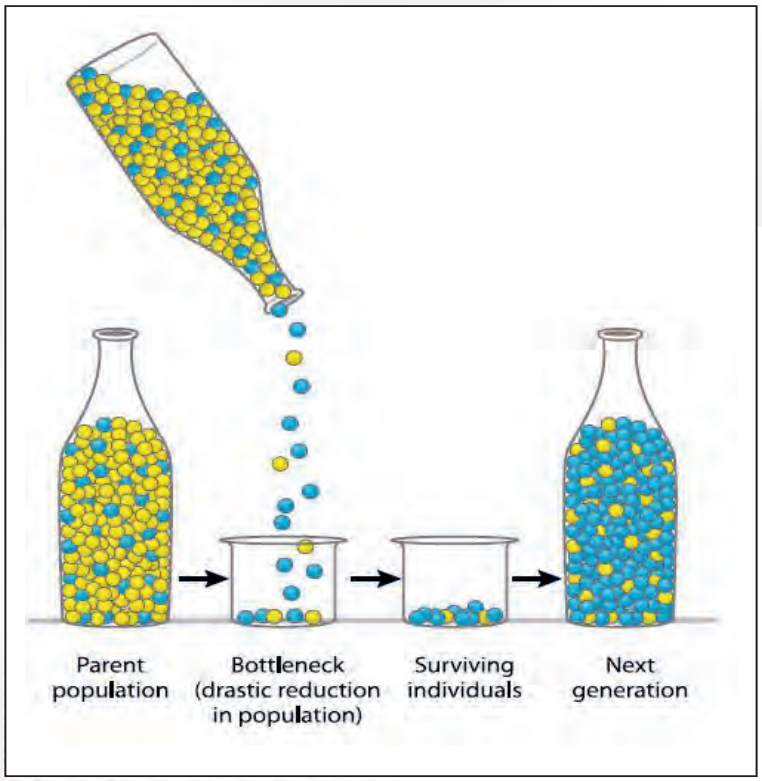


Figure 4.40. Bottleneck effect eliminates genes from a population

The founder effect

The **founder effect** is another extreme example of genetic drift that occurs when a small group of individuals breaks off from a larger population to establish a colony (Figure 4.41).The new colony is isolated from the original population, and the founding individuals may not represent the full genetic diversity of the original population. That is, alleles in the founding population may be present at different frequencies than in the original population, and some alleles may be missing altogether. The small size of the new colonies means that they will experience strong genetic drift for generations. The founder effect is similar in concept to the bottleneck effect, but it occurs via a different mechanism (colonization rather than catastrophe).

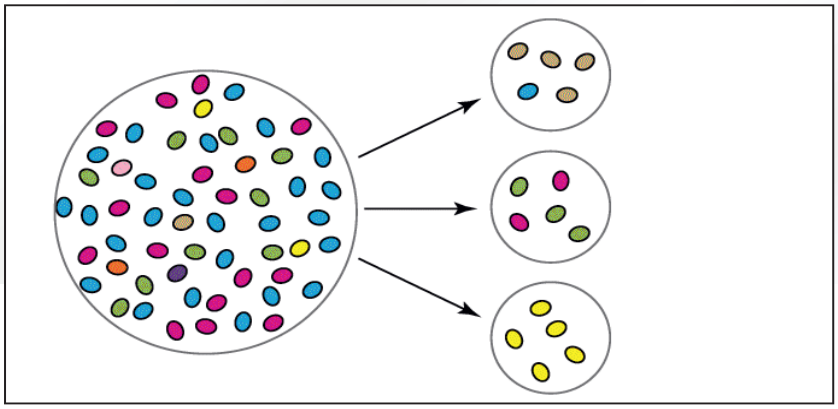


Figure 4.41 Founder effect eliminates genes from a population

Activity 4.32

Search from different sources (books, internet) about Ellis-Van Creveld syndrome (whose symptoms include polydactyly, or extra fingers, and other physical abnormalities). With the information you have about genetic drift and write a report of your findings.

4.1.8. Gene flow (immigration and emigration)

By the end of this section, the learner will be able to:

- Discuss Gene flow
- Describe the roles of immigration and emigration on gene flow and speciation.

Gene flow also called migration- is any movement of individuals, and/or the genetic material they carry from one population to another. Gene flow includes lots of different kinds of events, such as pollen being blown to a new destination or people moving to new cities or countries. If gene versions are carried to a population where those gene versions previously did not exist, gene flow can be a very important source of genetic variation. In Figure 4.42 below, the gene ersion for brown coloration moves from one population to another.

Keywords

Gene flow *is* any movement of individuals, and/or the genetic material they carry, from one population to another

Migration: The permanent movement of genes into or out of a population, causing a change in allele frequency

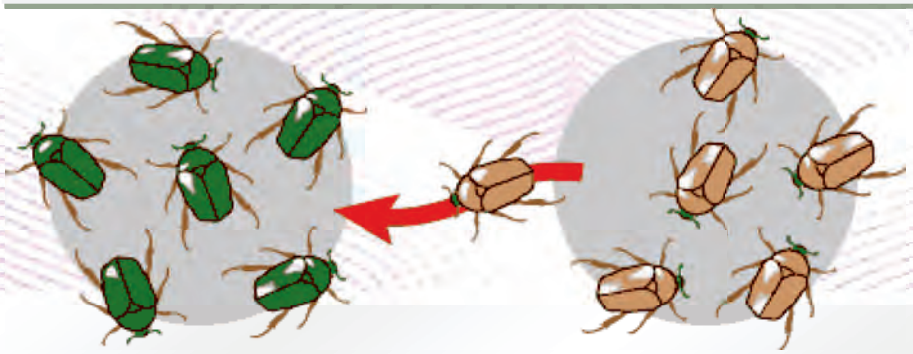


Figure 4.42 Gene flow (immigration)

Immigration is when new organisms join a population, changing allele frequencies. Emigration is when members of a population leave, taking with them their genes. These phenomena change the overall balance of the gene pool of the populations. Gene transfer is the flow of alleles from one species to another. Horizontal gene transfer is especially common in bacteria.

4.1.9. Causes of species extinction

By the end of this section, the learner will be able to:

- Explain causes of species extinction

Evidence suggests that anthropogenic effects and natural disaster played significant role for the direct and indirect causes of species extinction.

Activity 4.33

Form a small team and explore drivers of extinction like climate change, human activities, greenhouse gases, and natural disasters from various sources (books, internet etc). You are expected to seek solutions to mitigate habitat loss and prevent extinction. Prepare a report and discuss your findings to the class.

4.2. Renowned Anthropologists in Ethiopia

By the end of this section, the learner will be able to:

Appreciate the works of renowned anthropologists in Ethiopia.

Activity 4.34

Search from different sources (books, internet) about gene flow. Write a summary of your findings and present your findings to the class.

Activity: 4.35

Your teacher may wish you to know the work of renowned Ethiopian anthropologists

4.3. Renowned Evolutionists in Ethiopia

By the end of this section, the learner will be able to:

- Appreciate the works of renowned evolutionists in Ethiopia

Activity: 4.36

Make a small group and explore from the internet and other sources about human evolution discoveries from Ethiopia. Also search for the work of Ethiopian scientists involved in the discovery of human evolution in Ethiopia. Prepare a report and

Unit summary

In this unit,

You have learnt about evolution which can be defined as the change in genetic composition of a population over successive generations. You have been introduced with the major theories that explain the origin of life on Earth, which include:

- Special creation theory: a 'supreme being' is believed to have created life or directs its creation and evolution.
- Spontaneous generation theory: life is believed to arise from non-living matter; this was finally disproved by the experiments of Francesco Redi and Louis Pasteur
- Eternity of life theory: life is believed to have existed forever and will continue to exist forever. Hence, no origin is required.
- Cosmozoan theory: either life forms or the organic molecules needed for the origin of life are believed to have been brought to Earth by meteorites and comets.
- Biochemical origin theory:life is believed to have originated as a result of biochemical reactions creating first the necessary organic molecules, which then

became assimilated into 'pre-cells', which eventually evolved into cells.

You also studied that Miller's 'spark discharge' experiment showed that the organic molecules essential for life could be synthesized on Earth 4.5 billion years ago. The oldest photo-autotrophs are the cyanobacteria and they were largely responsible for the increase in free oxygen in the atmosphere. In 1809, Lamarck proposed a two-part theory to explain evolution based on:

- use and disuse
- inheritance of acquired characteristics

- In 1859 Darwin proposed the theory of natural selection based on:

- a struggle for existence
- natural variation in the offspring

- Particularly, you have learnt about Darwin's theory of natural selection, which stated that 'those members of a species which are best adapted to their environment will survive and reproduce in greater numbers than others less well adapted'.

- Apart from theory of natural selection, a detailed account of concepts regarding have been provided. Based on this, it has been stated that Neo-Darwinism takes into account our knowledge of genetics, biochemistry and ethology to modify Darwin's original theory to include the effect of selection on allele frequency and frequency of behavior patterns.

- On a similar vein, other insights on the origin of life have been provided. Evidence supporting the theory of evolution comes from many areas, including:

- palaeontology (the fossil record)
- comparative anatomy
- comparative embryology
- comparative biochemistry

- Another important point you have studied in this unit was about the use of fossils to determine the origin of life. In this case, fossils can be dated using:

- Stratigraphy –analyzing the sequence and thickness of different layers (strata) of rocks
- radioactive carbon (C14) dating – measuring the ratio of radioactive carbon to normal carbon – is suitable for fossils up to 60 000 years old

This unit was also devoted to illustrating the following scientific concepts, facts and evidence related to evolution. These are:

- Homologous structures are evidence of a common origin and divergent evolution.
- Analogous structures are evidence of a different origin and convergent evolution.
- Similar patterns of embryological development in vertebrates suggest a common origin.
- The extent of differences in molecules common to many species (for example, DNA, cytochrome c, hemoglobin) is a measure of their relatedness.

A species can be defined as 'a group of similar organisms with a similar biochemistry, physiology and evolutionary history that can interbreed to produce fertile offspring.

The gene pool is the sum of all the alleles of all the genes in a population. The gene pool is constantly changing as a result of mutations introducing new genes into the population and disadvantageous alleles being lost through natural selection. In natural selection:

- individuals with an advantageous allele survive to reproduce in greater numbers
- the process repeats over many generations in which the frequency of the advantageous allele increases in each generation

In directional selection, one extreme of a range of values for a feature has a survival advantage; the range of values for the population shifts towards the extreme with the selective advantage. In stabilizing selection, the two extremes are at a selective disadvantage compared to those showing the mean values for a particular feature; the range is compressed around the mean. In disruptive selection, both extremes have a selective advantage compared with the mean; two distinct types begin to emerge showing the extreme values of the original population. If two populations of the same species are isolated for sufficient time, they may become so different genetically as to evolve into separate species. Speciation involving geographical separation is called allopatric speciation. Speciation involving separation within one area which is a result of different breeding strategies is called sympatric speciation; the different strategies can involve. Divergent evolution involves adaptive radiation and is the evolution of one basic 'type' into several different 'types' as a result of different selection pressures. Examples include

- the divergent evolution of the pentadactyl limb into flippers, legs, wings, etc.

- the divergent evolution of the beaks (and other features) of Darwin's finches on the Galapagos Islands

Convergent evolution is the evolution of similar 'types' with similar adaptations from several different original 'types'. Examples include:

- the elongated 'snouts' (and other features) of the different anteaters of the world
- the wings of birds, insects, pterodactyls, etc.

Modern humans and other primates have evolved from a common primate ancestor that lived before the dinosaurs became extinct. Modern humans and chimpanzees have evolved from a common ancestor that lived about 6 million years ago. Two distinctive features of modern humans are:

- large brains
- true bipedalism

The fossil Lucy was significant because it showed that bipedalism evolved before large brains. The fossil Ardi was significant because it showed that the common ancestor of humans and chimpanzees cannot have resembled a chimpanzee.

Unit review questions

Choose the best answer

1. Which of the following is *not* a part of Darwin's theory of natural selection?

   A. Individuals of a population vary

   B. Organisms tend to over-reproduce themselves

   C. There are limited resources for which individuals compete

   D. Modifications an organism acquires during its lifetime can be passed to its offspring

   E. Variations possessed by individuals of a population are heritable

Direction: Answer question number 2 based on the following diagram

2. Why did Pasteur bend the neck at curve

   a. To make the flask air tight

   b. To trap dust & microbes

   c. To enable the steam – easily comes out

   d. To make dust & microbes pan in & out

3. In the Miller and Urey's simulation experiment to study organic molecules formation in primeval atmosphere of the earth, all the following gases were used except which?

   A. Ammonia

   B. Nitrogen

   C. Methane

   D. Oxygen

   E. Hydrogen

4. If the amount of C-14 relative to C-12 in a sample is one-quarter of that in living organisms at present, what is the age of the fossil at 5,730 half-life years?

   A. 11,460 years

   B. 17,190 years

   C. 22,920 years

   D. 50,000 years

5. Which one of the following includes all the others?

   Natural selection

   Disruptive selection

   Directional selection

   Stabilizing selection

6. Birth mass in humans is an example of which of the following selections?

   A. Disruptive selection

   B. Directional selection

   C. Stabilizing selection

   D. Diversifying selection

7. When the human and other primate DNA compared, human's DNA are found distantly related to

   Rhesus monkey

   Chimpanzees

   House mouse

   Gibbon

8. The many species of Galapagos Finches are adapted to eat different foods. This is the result of

   A. Gene pool

   B. Genetic Drift

   C. Survival of the fittest

D. Adaptive radiation

9. A bell shaped curve in a variable indicates which kind selection?

A. Stabilizing selection C. Divergent selection

B. Unidirectional selection D. Disruptive selection

10. According to the theory of evolution proposed by Charles Darwin, which of the following is the most important driving force of evolution?

A. Over reproduction C. Mutation

B. Natural selection D. Variation

11. Theory of inheritance of acquired characters was given by

A. Wallace B. Darwin C. Lamarck D. Hugo De Vries E. Mendel

12. During which of the following periods did the extinction of dinosaurs happen?

A. Cambrian B. Ordovician C. Jurassic D. Cretaceous E. Permian

13. The Study of fossils is

A. Herpetology B. Serology C. Palaeontology D. Homology

14. Which of the following is the direct ancestor of Homo sapiens?

A. *Australopithecus* C. *Homo erectus*

B. H. sapiens Neanderthals D. Homo sapiens fossils

15. Species occurring in different geographical area are called as

A. Sympatric C. Sibling

B. Allopatric D. Neopatric.

16. Which of the following are homologous organs?

A. Wings of bird and hands of human

B. Wings of bird and wings of insect

C. Nails of human being and claws in animals

D. Wings of bat and wings of cockroach

17. Similarities in organism with different genotype indicates

A. Microevolution C. Convergent evolution

B. Macroevolution D. Divergent evolution

18. Which of the following is most important for speciation?

A. Seasonal isolation C. Behavioral isolation

B. Reproductive isolation D. Tropical isolation

19. Adaptive radiation refers to

A. Evolution of different species from a common ancestor

B. Migration of members of a species to different geographical areas

C. Power of adaptation in an individual to a variety of environments

D. Adaptations due to geographical isolation.

20. Which theory of the origin of life states that a supreme being Somehow involved in the creation of life

A. Eternity life C. Special creations

B. Cosmozoan theory D. Spontaneous generation

21. On which of the following evolutionary ideas do Lamarck & Darwin fully agree?

A. Acquired traits are inherited

B. Natural selection leads to evolution

C. Life is a result of natural processes

D. Evolution occurs by the use & disuse of body parts

22. The scientist who tested the biochemical hypothesis of the origin of life

A. Oparin B. Miller C. Pasture D. Fredi

Provide the correct answer for the following questions

1. List at least one point in favour and one point against the theories on the origin of life listed here under. You may have to imagine yourself as a 'person of the times' in some cases.

a. Special creation

b. Spontaneous generation

c. Eternity of life

d. Cosmozoan (panspermia)

e. Biochemical (abiogenesis)

2. (a) Explain what is meant by each of the following terms:

i. Evolution

ii. convergent evolution

iii. divergent evolution

(b) How does the fossil record provide evidence for evolution?

(c). Explain how neo-Darwinism has modified Darwin's original theory of natural selection.

3. (a) In each of the following examples of natural selection, identify:

- the selection pressure (feature of the environment that is selecting for some types and against others), and the type within the population that is best adapted.

i. wildebeest hunted by lions

ii. Bacteria in a hospital where penicillin is widely used

iii. Nettle plants with different-sized leaves in a shaded woodland area.

(b) Allopatric speciation and sympatric speciation are two processes by which new species can evolve. Explain:

- one similarity between the two processes

- one difference between the two processes

4. (a). Explain what is meant by the term 'species'.

(b) King cheetahs have a different pattern of spots from ordinary cheetahs. At first it was thought that they might be a different species. Suggest how:

i. the difference in spot pattern might have arisen

ii. biologists have been able to show that king cheetahs are members of the same species as other cheetahs

5. The amino acid sequences of one of the polypeptide chains of haemoglobin from nine animals were determined. The results are shown in the table.

Type of haemoglobin	Number of amino acids different from human haemoglobin
Human	0
Gorilla	1
Gibbon	2
Rhesus monkey	8
Horse	25
Chicken	45
Frog	67
Sea slug	127

(a) Use the information to draw a phylogenetic tree of the organisms.

(c) It is possible to use DNA hybridisation to suggest relationships between species. Explain why

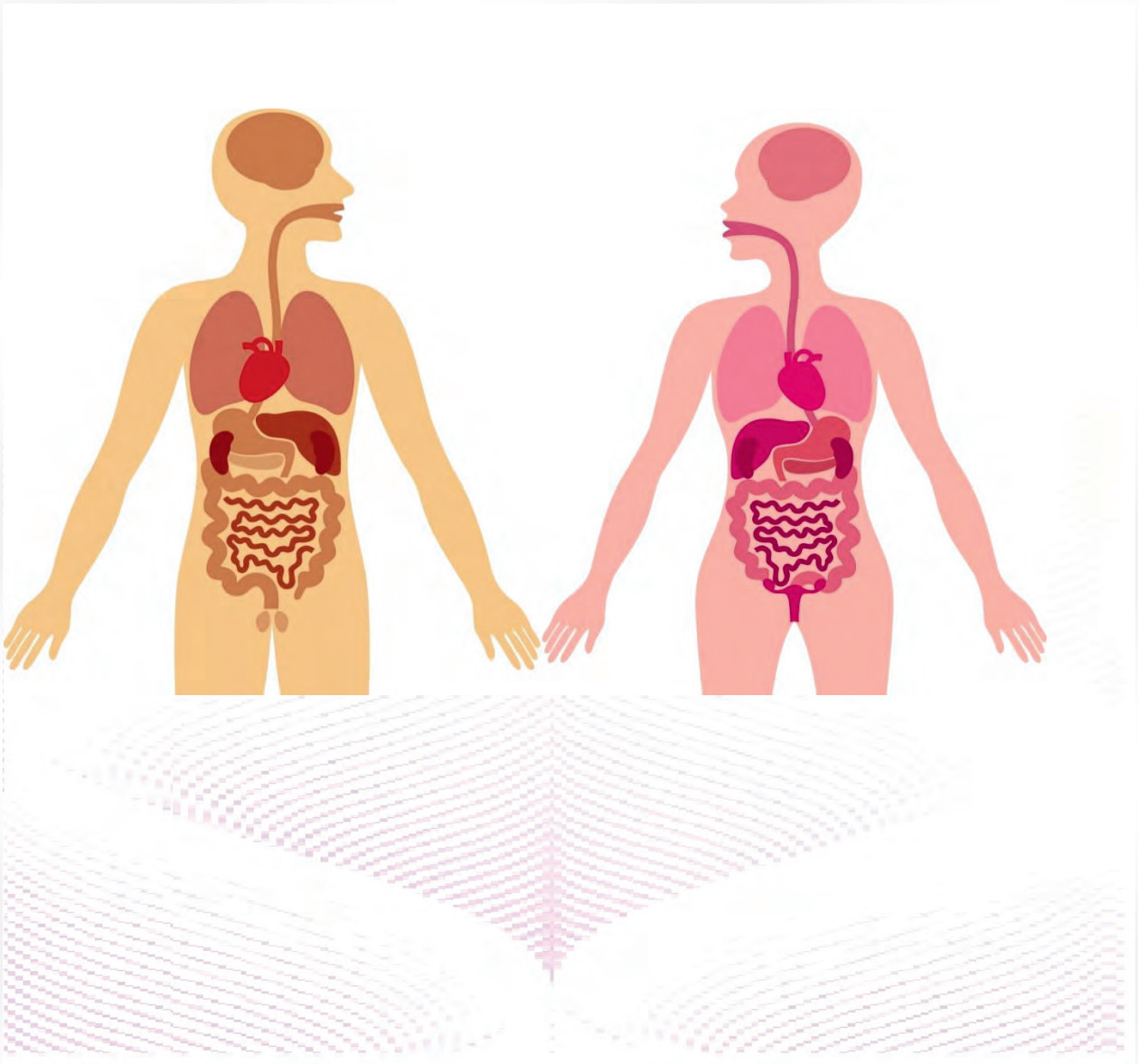
6. Explain the importance of each of the following in speciation:
   - Isolation of different populations
   - Mutation
   - Selection pressures
   - Reproductive isolation
7. Describe how the experiments of Redi and Pasteur were able to disprove the theory of spontaneous generation.
8. Describe (a) the Oparin/Haldane theory of abiogenesis (the biochemical origin of life).

(b) Three pieces of evidence that support this theory.

9. Write a short essay on human evolution. Include the following aspects of human evolution in the essay:

- The idea of a common ancestor with chimpanzees
- Some of the early humans that have existed
- The importance of bipedalism and large brain size
- The significance of the Lucy, Ardi and Selam fossils.

Unit 5: Human body system



After successful completion of this unit, the learner will be able to:

- Explain the structures and functions of the nervous system
- Explain how the organs of each organ system work together
- Explain what homeostasis is and how the human nervous and endocrine system helps maintain homeostasis
- Discuss how the nervous and endocrine system works
- Describe the effects of drug abuse on the normal functioning of the nervous and endocrine system

Self-questioning

Which parts your body systems get involved for coordination and rapid responses to the changes in the environment?

Body Coordination

There are two main forms of coordination in animals as a whole. These are nervous and hormonal coordination. Thus, the nervous system and the endocrine system are the two important systems that are responsible for body coordination. The nervous system uses nerve cells to pass electrical impulses along their length. They stimulate their target cells by secreting chemicals, known as neurotransmitters, directly on to them. On the other hand, the endocrine system produces chemicals (hormones) that are transported in the blood plasma to their target cells. The target cells have specific receptors on their cell-surface membranes and the change in the concentration of hormones stimulates them.

Table 5.1. Nervous system versus endocrine system

	Nervous system	Endocrine system
Speed of action	Seconds	Minutes to hours (even days)
Duration of action	Seconds to minutes	Minutes to days
Method of transmitting messages	Electrical	Chemical
Transport method	Neurones	Hormones

5.1. The Nervous System

After successful completion of this unit, the learner will be able to:

- Define nervous system.
- Describe the structure of nervous system.
- Describe the structure of neurons
- Explain the functions of different types of neurons
- Define reflex action
- Show the root of reflex arc with example
- Identify types of the nervous system
- Describe the mechanism of the conduction of nerve impules.
- Illustrate the effects of drug abuse on the nervous systems with local and international example

Activity 5.1

Make a small group and discuss on the following point and reflect your idea to the class

- Desecibe the differences and similarities of nervous system and endocrine system

The nervous system

Self-questioning

- What is nervous system?
- What are the constituents of central nervous system and peripheral

The **nervous system** is responsible for the reception and processing of sensory information from both the external and internal environments. The nervous system has two major divisions (Figure 5.1). The **central nervous system** (**CNS**) consists of the brain and spinal cord. The **peripheral nervous system** (**PNS**) consists of nerves outside the CNS.

The nervous system has three specifc functions:

1. The nervous system receives sensory input. Sensory receptors in skin and other organs respond to external and internal stimuli by generating nerve signals that travel by way of the PNS to the CNS.

2. The CNS performs information processing and integration, summing up the input it receives from all over the body.

3. The CNS generates motor output. Nerve signals from the CNS go by way of the PNS to the muscles, glands, and organs.

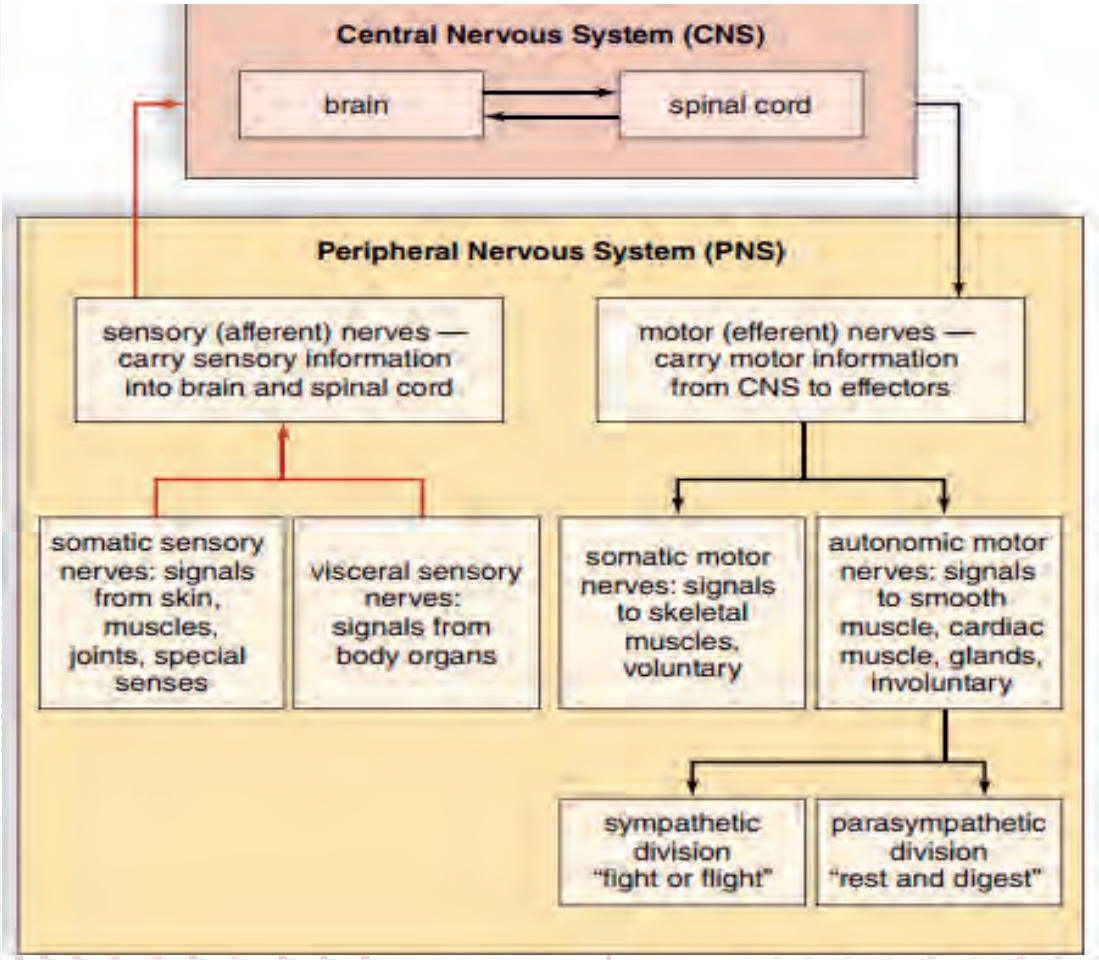


Figure 5.1 Nervous system with their brief functions

5.1.1. Neurons and their functions

By the end of this section, the learner will be able to:

- Describe the structure of neurons.
- Mention the three types of neurons.
- Describe the function of neurons.

Self-questioning

- What are the functional difference between neurons and glial cells?

The nervous system is composed of only two principal types of cells - neurons and supporting cells (glial cells). **Neurons** are the basic structural and functional units of the nervous system. They are specialized to respond to physical and chemical stimuli, conduct electrochemical impulses, and release chemical regulators. Through these activities, neurons enable the perception of sensory stimuli, learning, memory, and the control of muscles and glands.

Supporting cells aid the functions of neurons and are about five times more abundant than neurons. In the CNS, supporting cells are collectively called **neuroglia,** or simply **glial cells** (from the Middle Greek *glia* = glue). **Neuroglia** serve as supporting cells, providing support and nourishment to the neurons.

The basic unit of communication in the nervous system is the nerve cell (**neuron**). All neurons consist of : (1) a cell body, (2) dendrites, and (3) an axon (Figure 5.2). Dendrites and axons can be referred to generically as processes, or extensions from the cell body.

1). **Cell body:** The cell body contains the nucleus, as well as other organelles. Also known as a soma, the cell body is the neuron's core. The cell body consists of nucleus that carries genetic information, maintains the neuron's structure, and provides energy to drive activities. Most of the neurone cell bodies are located inside the central nervous system and form the grey matter. The Cell bodies that are located in the peripheral nervous system are called **ganglia.**

2). **Axon:** Each neurone has only one axon that conducts information away from the cell body. The axon will also branch at its terminal into many axon terminals. The axon delivers the impulse to another neurone or a gland or a muscle. Individual axons are termed nerve fibers, and collectively they form a **nerve.** Many axons are insulated with a fatty substance called **myelin sheath.**

3). **Dendrites:** Dendrites are short extensions that receive signals from sensory receptors or other neurons. Incoming signals from dendrites can result in nerve signals that are then conducted by an axon. Neurons can have more than one set of dendrites, known as dendritic trees. The number of dendritic tree neurons have generally depends on their role.

Myelin sheath: Myelin is a fatty substance whose purpose is to protect the neurone and to electrically insulate it, speeding up impulse transmission. The axons of peripheral nerves and long or large axons are covered by a myelin sheath. Within the peripheral nervous system Schwann cells wrapped in layers around the neurone form the myelin sheath. The gaps where there is no myelin sheath are called **nodes of Ranvier**. Some nerve fibres are unmyelinated, and this makes nerve impulse transmission significantly slower.

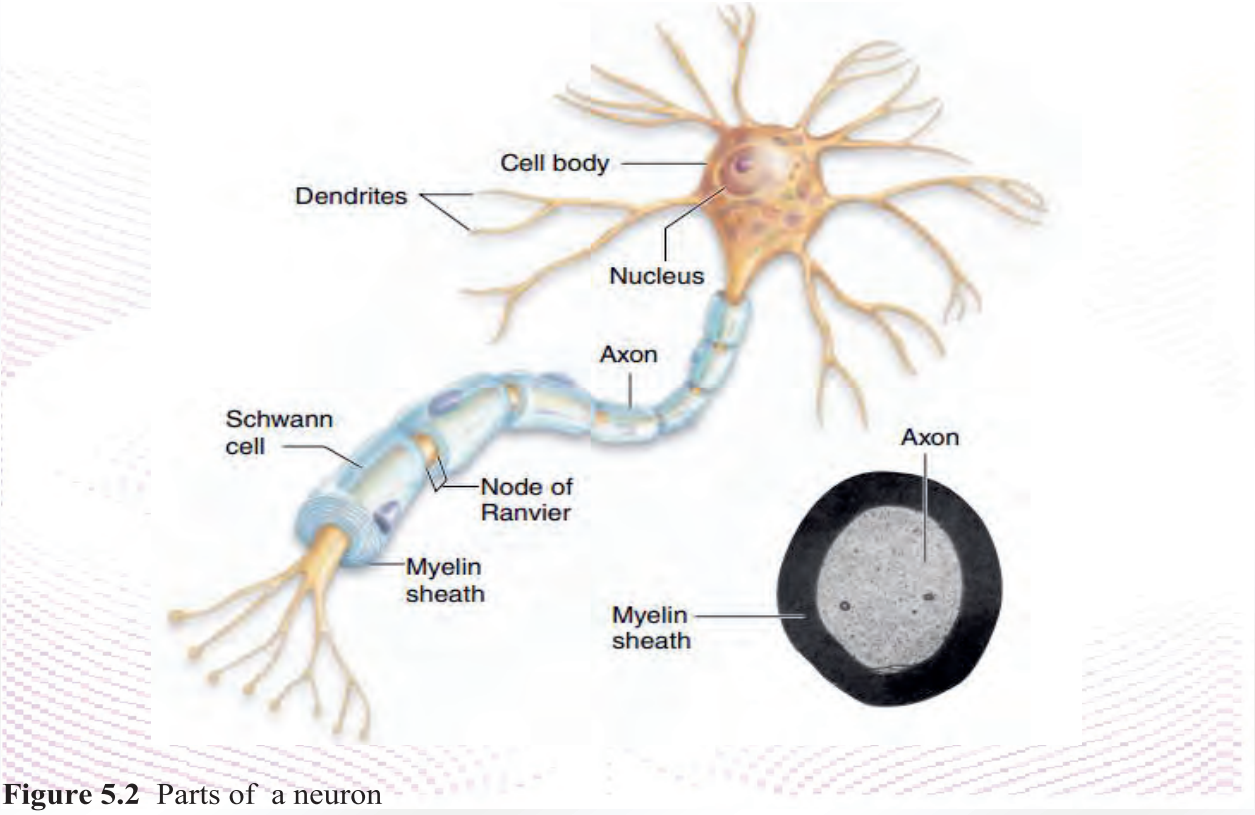


Figure 5.2 Parts of a neuron

Keywords

Dendrites: tree-like branches from nerve cell bodies that receive signals from other nerve cells at synapses

Axon: long process from nerve cell that carries the nerve impulse

Cellbody: the enlarged portion or rod granule that contains the nucleus.

Myelin sheath: fatty insulating sheath that grows around many nerves.

Node of Raniver: periodic gap in the insulating sheath (myelin) on the axon of certain neurons that serves to facilitate the rapid conduction of nerve impulses

Activity 5.2

Make a small group and discuss on the following point and reflect your idea to the class

- What is the difference between a neuron and a nerve?
- Do you think neurons are continuous like a 'wire'?

Types of neurons

Neurons vary in structure and function. In terms of function, scientists classify neurons into three broad types: **sensory**, **motor**, and **interneurons**.

1. **Sensory (affector/ afferent) neurons:** Sensory neurons carry messages from sensory receptors to the central nervous system. They have typically a long dendrite and short axon. Sensory neurons are triggered by the physical and chemical inputs from environment (stimuli) such as sound, touch, heat, and light.

2. Motor (effector/ efferent) neurons: Motor neurons transmit messages from the central nervous system to the effectors (muscles or glands). Motor neurons have a long axon and short dendrites. Motor neurons play a role in voluntary and involuntary movements. These neurons allow the brain and the spinal cord to communicate with muscles, organs, and glands all over the body.

3. Interneurons (association neurons): Interneurons are found entirely within the central nervous system. They pass signals

from sensory to motor neurons, or to integrate these functions.

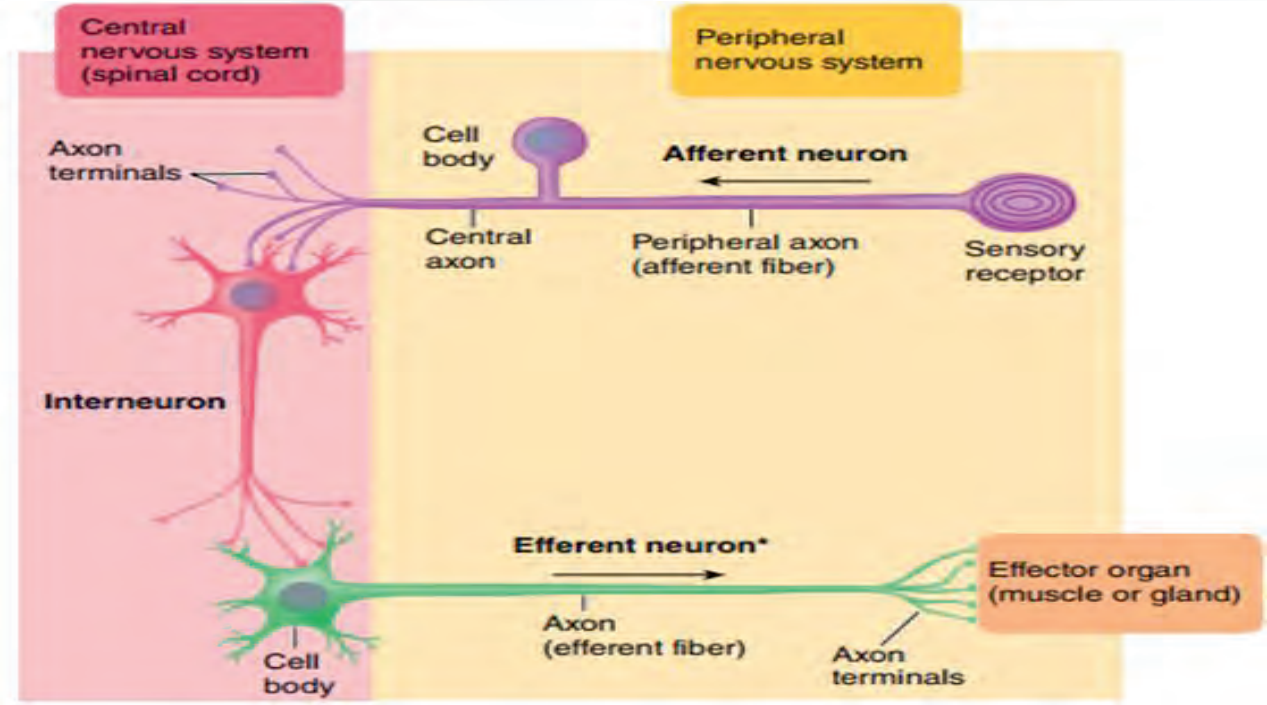


Figure 5.3 Types of neurones

The nervous system takes in information from our receptors. Once a stimulus is picked up by a receptor, the information is passed along special nerve cells, affector or afferent (sensory) neurons, to the central nervous system (CNS). In the CNS interneurons (associated neurons) pass signals from sensory neurons to motor neurons and other interneurons. Once the information has been processed in the CNS, instructions are sent out to the body along motot neurons called effector or efferent (motor) neurons. The summary of information flows through the nervous system in the following sequence:

Keywords

Affector (afferent) neuron that sends impulses from organs to the spinal cord and brain

Effector (efferent) neuron that sends impulses from the brain and spinal cord to organs

Interneurons form connections between afferent and efferent neurons

Stimulus → Reception by sensory receptor → transmission by afferent neuron →

integration by interneurons in CNS ⟶ transmission by efferent neuron

→ action by effectors

Nerves and nerve tracts

Activity 5.3

Make a small group and discuss on the following point and reflect your idea to the class

- Distinguish between Stimulus and response
- How sensory neurons, interneurons and motor neurons work.
- How CNS interpret and analyze information.
- How response of the effector works.

A **nerve** is a group of neurons (bundles of neurons) with blood vessels and connective tissue.

Sensory (afferent) nerves are made only of sensory neurons. The optic nerves for vision and olfactory nerves for smell are examples of nerves with a purely sensory function.

Motor (efferent) nerves are made only of motor neurons; autonomic nerves are motor nerves.

A **mixed nerve** contains both sensory and motor neurons. Most of our peripheral nerves, such as the sciatic nerves in the legs, are mixed nerves.

The term **nerve tract** refers to groups of neurons within the central nervous system. All the neurons in a nerve tract are concerned with either sensory or motor activity. These tracts are often referred to as white matter; the myelin sheaths of the neurons give them a white color.

Activity 5.4

Can you draw a typical nerve cell and label the parts, and explain how it functions to the class?

- The sodium channel and potassium channels on the membrane of neuron are closed.

- This state of resting neuron is called Polarized state and it is electro-negatively charged.

Depolarization or Action Potential

A stimulus above the threshold which can be initiated by mechanical, chemical and physical stimulation cause the initiation of impulse

5.1.2. The Nerve Impulse and transmission

By the end of this section, the learner will be able to;

Explain nerve impulse and transmission.

Self-questioning

Why is the presence of synapse important for neurotransmission?

What is action potential ?

How does it passes from one neuron to the other?

How does impulse passes from one

The nervous system relies on nerve impulses travelling along a neuron. Each nerve impulse is a minute electrical event that works as a result of charge differences across the membrane of the axon. The wave of positive charge inside the axon when the neuron is stimulated is known as the **action potential**.

Resting Potential

When an axon is resting, its membrane is polarized; that is, the outside is positive compared to the inside (which is negative). When the axon is not conducting an impulse, the voltmeter records a membrane potential equal to about -65 mV (millivolts), indicating that the inside of the neuron is more negative than the outside (Figure 5.4). This is called the resting potential because the axon is not conducting an impulse.

Keywords

Action potential a short- term change in the electrical potential on the surface of a cell when it is stimulated

Synapse the junction between two neurons (axon-to-dendrite) or between a neuron and a muscle

Threshold potential the voltage at which depolarization of a cell leads to generation of an action potential.

Hyperpolarization increasing neuronal membrane potential to more than its usual resting potential (making it harder to induce the cell to produce an action potential)

Refractory period a brief period following the generation of an action potential, during which a neuron is hard to re-excite

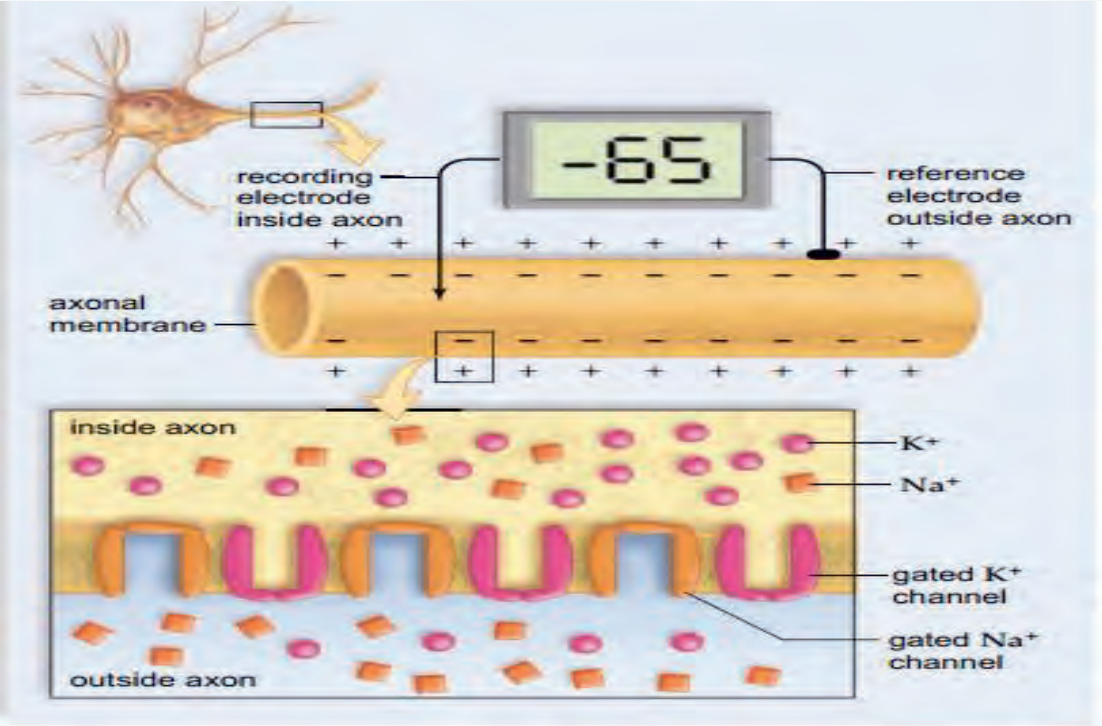


Figure 5.4 Resting potential

Polarization

- In the resting state the inside of membrane has negative electrical potential compared to outside. This difference in potential is called resting potential (about -40mv to -90mv).
- The interstitial fluid has high concentration of Na+ ion which is about 16 times higher outside the neuron than inside neuron. Similarly, the axoplasm has high concentration of K+ ion which is about 25 times higher inside than in outer interstitial fluids.
- Due to difference in Na, K concentration, the Na+ ions diffuse inside the exoplasm and K+ ions tends to diffuse outside of it.
- The membrane of neuron at resting is more permeable to K+ ion than Na+ ion. So, K+ leaves the neuron faster than Na+ enter the neuron.
- Na+and K+ are transported across the membrane against their concentration gradient by carrier protein, which is called Na-K pump and energy is used through ATP.
- The difference in permeability results in accumulation of high concentration of cation (+ve charged ion) outside the neuron compared to the concentration of cation inside.
- .Sodium channels are opened but potassium channels are closed and

Na+ions flood in through cell membrane and create a positive charge of +40mv.

- It is a very short periods that change in potential and last for 3 milliseconds.
- When an action potential occurs, the axon is said to be depolarized.
- The depolarization of the membrane stimulates the adjacent voltage channel, so the action potential passes as a wave along the length of neuron.

Depolarization: This process is started by a nerve impulse. Acetylcholine released by the axon terminal makes the sarcolemma very permeable to Na+ ions, which enter the cell and cause a reversal of charges to (-ve) outside and (+ve) inside. The depolarization spreads along the entire sarcolemma and initiates the contraction process. Folds of the sarcolemma called T tubules carry the depolarization into the interior of the muscle cell. As soon as depolarization takes place, the neuron membrane becomes very permeable to K+ ions, which rush out of the cell. This restores the positive charge outside and the negative charge inside, and is called repolarization. Then the sodium and potassium pumps return Na+ ions outside and K+ ions inside, and the neuron is ready to respond to another stimulus and transmit another impulse. An action potential in response to a stimulus takes place very rapidly and is measured in milliseconds. An individual neuron is capable of transmitting hundreds of action potentials (impulses) each second

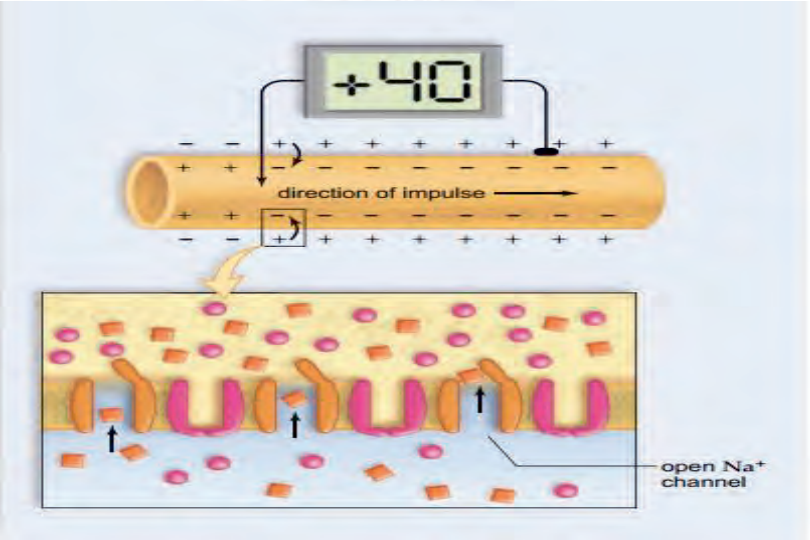


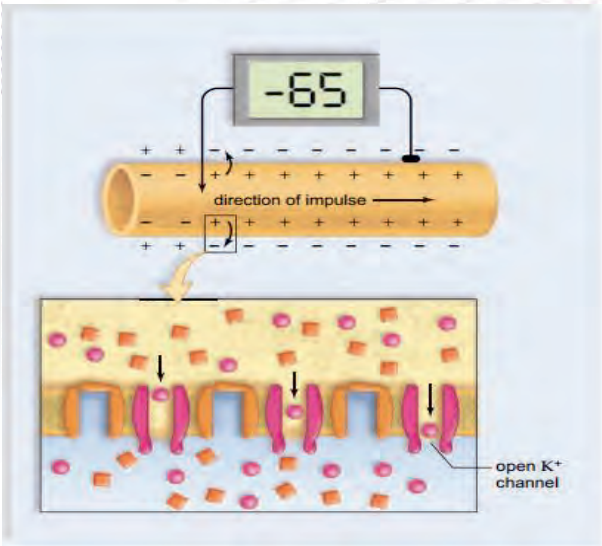
Figure 5.5 Action potential begins: depolarization occurs

Repolarization

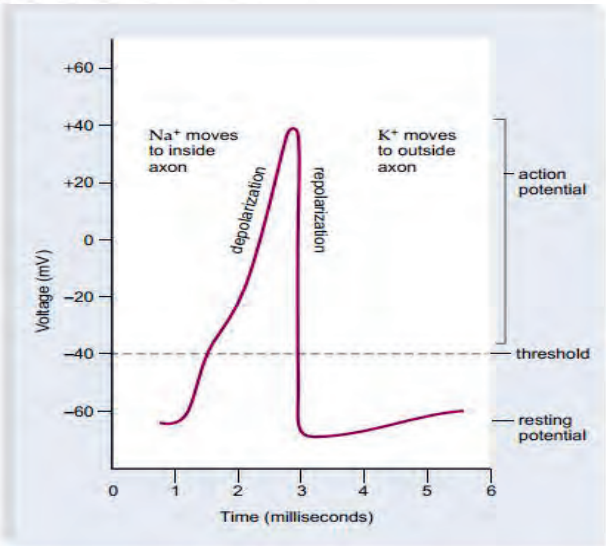
- Sodium channels are closed potassium channel are opened and K^+ions diffuse out along their concentration gradient. This starts repolarization and resting potential going to reestablish.
- At the same time nerve become less permeable for Na+than K+. So many K+ flow out and inside charge become more negative than that it was originally.
- Na-k pump starts and normal concentrations of Na and K ions are reestablished. Each pump actively transports two K+ions into the cell to every three Na+ ions transported out.
- The membrane is once again at its resting potential.

Action Potential: An action potential is a rapid change in polarity across a portion of an axonal membrane as the nerve impulse occurs. An action potential uses two types of gated ion channels in the axonal membrane.

- When the nerve fiber is conducting a nerve impulse (action potential), a change in polarity occurs across the axon's membrane. First, the inside of an axon becomes positive compared to the outside (this is called depolarization), and then the inside becomes negative again (this is called repolarization). An action potential requires two types of channels in the membrane: One channel can allow Na ions to pass through the membrane, and the other can allow K ions to pass through the membrane. During depolarization, Na ions move to the inside of the axon, and during repolarization, K ions move to the outside.



A) Action potential ends



B) Action potential graph

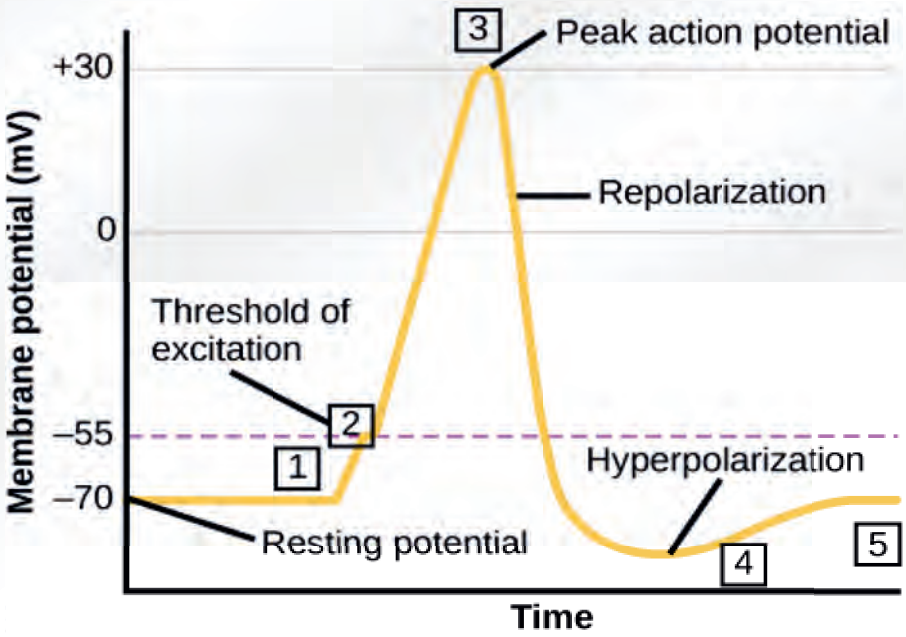


Figure 5.6 Conduction of Action Potentials

Syanapse

By the end of this section, the learner will be able to;

- Explain how impulse crosses the synaptic cleft
- Describe the function of acetylecholinestrase

Self-questioning

- How do synapses work?

Whenever one neuron ends and another begins there is a gap known as a s**ynapse**. Neuron-neuron synapses usually involve a connection between the axon of one neuron and the dendrites, cell body, or axon of a second neuron. The axon of the presynaptic neuron ends in slight swellings, the axon terminals, which hold the **synaptic vesicles** that contain neurotransmitter molecules.

The axon terminal of the **presynaptic neuron,** which conducts its action potentials *toward* the synapse, ends in a slight swelling, the **synaptic knob.** The synaptic knob contains **synaptic vesicles,** which store a specifc chemical messenger, a

neurotransmitter that has been synthesized and packaged by the presynaptic neuron. The synaptic knob comes close to, but does not touch, the **postsynaptic neuron,** whose action potentials are propagated away from the synapse. The space between the presynaptic and postsynaptic neurons is called the **synaptic cleft.**

The process by which the impulse in the presynaptic neuron signals the postsynaptic cell is called **synaptic transmission.** As a result of synaptic transmission, the presynaptic neuron stimulates or inhibits a postsynaptic cell (Figure 5.7). Synaptic transmission is a one-way process carried out by *neurotransmitters*. An impulse travels along the axon of the presynaptic neuron to the axon terminal. Most axons have several rounded synaptic knobs at their terminals, which dendrites do not ave. These knobs have arrays of membranous sacs, called synaptic vesicles, that contain neurotransmitter molecules. When an impulse reaches a synaptic knob, voltage-sensitive calcium channels open and calcium diffuses inward from the extracellular fluid. The increased calcium concentration inside the cell initiates a series of events that fuses the synaptic vesicles with the cell membrane, where they release their neurotransmitter by exocytosis.

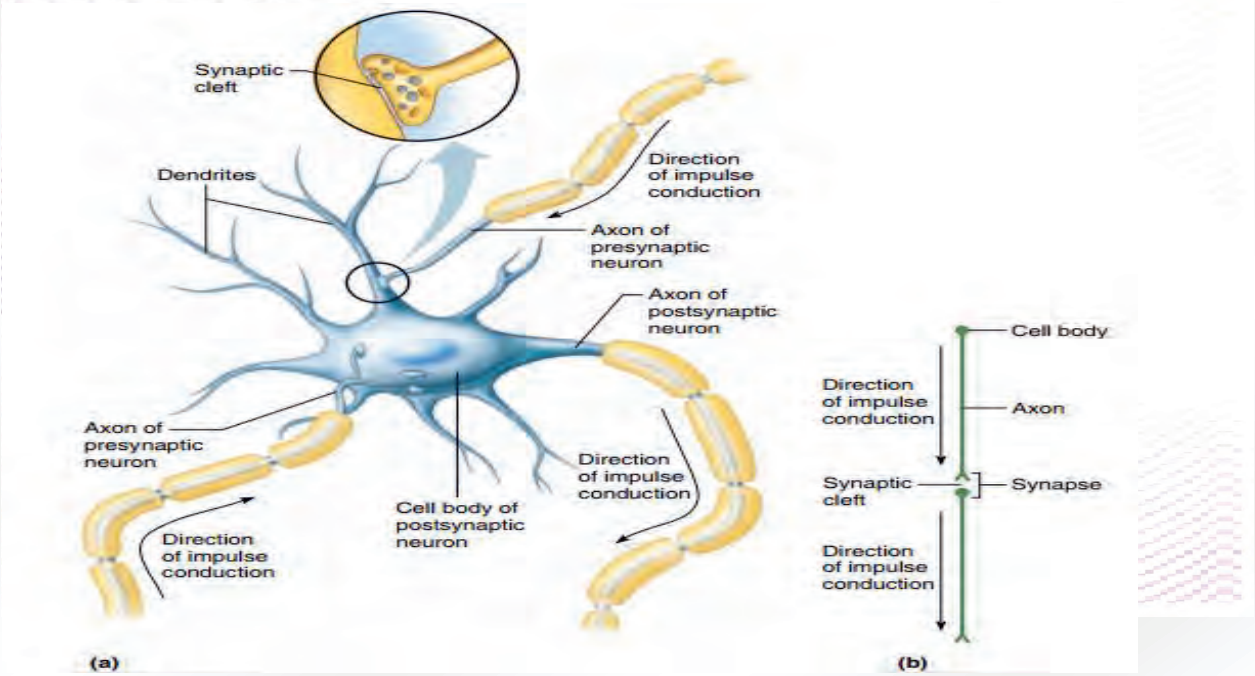


Figure 5.7 Synapsis occurs from axon terminal one neuron to the denderities or cell body another neuron

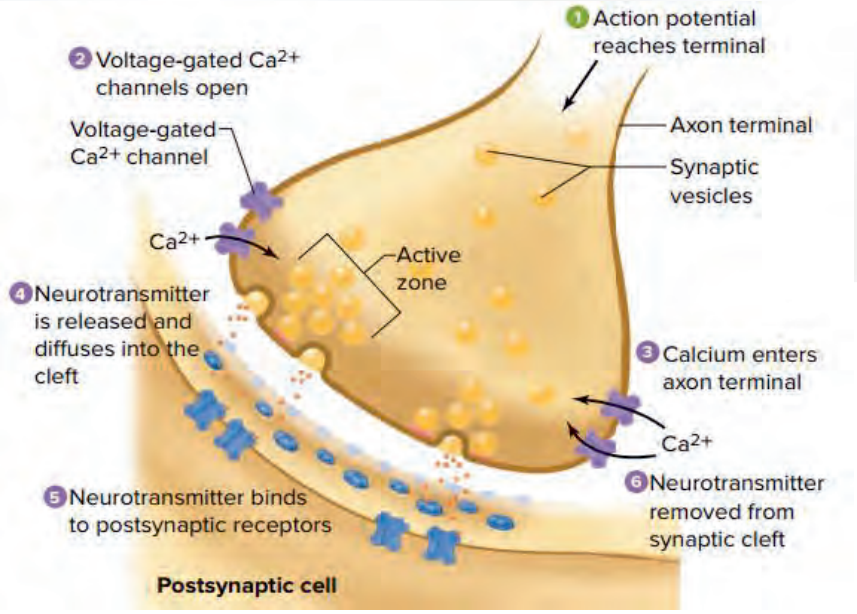


Figure 5.8 Signal transmission through a synapse.

5.1.3. Neurotransmitters

At the end of this section, the learner will be able to:

- Describe the role of neurotransmitters.

Neurotransmitters are often referred to as **the body's chemical messengers**. They are the molecules used by the nervous system to transmit messages between neurons, or from neurons to muscles. Acetylcholine (ACh) is **an abundant neurotransmitter in the human body**. It is found in both the central nervous system (CNS) and the peripheral nervous system (PNS).

An **excitatory transmitter** promotes the generation of an action potential in the receiving neuron, whereas an **inhibitory transmitter** prevents it. Whether a neurotransmitter is excitatory or inhibitory depends on the receptor it binds to.

Keywords

Excitatory transmitter promotes the generation of an electrical signal (action potential) in the receiving neuron

Inhibitory transmitter decrease the likelihood that the neuron generate action potential (action potential).

Activity 5.5

Make a small group and search from different sources about synapses and special types of synapses and answer the following questions.

a. Do we have special types of synapses?

b. What are neuromuscular junctions? What is their role? Where do we find them? Reflect your findings to the class.

5.1.4. Types of the nervous system

Activity 5.6

Search from different sources (books and the internet) about the various types of neurotransmitters and their role in the nervous system. Reflect your finding to the class.

By the end of this section, thestudent will be able to:

- Identify types of the nervous syste

Human nervous system has two main parts. These are the central nervous system (CNS), which is made up of **brain** and **spinal cord,** and the **peripheral nervous system** which consists of the nerves that branch out from the brain and spinal cord.

Keywords

Spinal cord a column of nerve tissue within the backbone

Neurons nerve cells

Sensory receptors nerve endings that can sense stimuli, e.g. pressure, pain, temperature, and start a nerve impulse

A. The Central Nervous System

The Central Nervous System (CNS) consists of two main parts: – the **brain** and the **spinal cord**.

The Brain

Self-questioning

Which parts your body systems get involved for coordination and rapid responses to the changes in the environment?

Central nervous tissue is delicate. Because of this characteristic, and because damaged nerve cells cannot be replaced, this fragile, irreplaceable tissue must be well protected. Four major features help protect the CNS from injury:

1. It is enclosed by hard, bony structures. The *cranium (skull)* encases the brain, and the *vertebral column* surrounds the spinal cord.

2. Three protective and nourishing membranes, the *meninges*, lie between the bony covering and the nervous tissue.

The meninges consist of three connective tissue layers:

• **Dura mater** – this layer lies closest to the bone of the skull and is a double layer of tough, fibrous, connective tissue. The outer layer is called the periosteal layer (the spinal cord lacks this layer), and the meningeal layer lies closest to the brain.

• **Arachnoid mater** – between the dura mater and the arachnoid mater there is a space called the subdural space. The arachnoid mater is a delicate serous membrane. The subarachnoid space is below the arachnoid mater and above the pia mater. The subarachnoid space contains CSF and is also home to some of the larger blood vessels serving the brain.

• **Pia mater** – this is a delicate connective tissue layer that clings tightly to the brain. It contains many tiny blood vessels that serve the brain.

The brain "floats" in a special cushioning fluid, the *cerebrospinal fluid (CSF)*. It is a thin fluid similar to plasma and has several important functions:

• it acts as a cushion, supporting the weight of the brain and protecting it from damage;
• it helps to maintain a uniform pressure around the brain and spinal cord;
• there is a limited exchange of nutrients and waste products between neurones and CSF.

Protected within the skull, the brain is composed of the **cerebrum**, **cerebellum**, and **brainstem (Medula)** (Figure 5.3).

Cerebrum

The **cerebrum,** also called the telencephalon, is the largest portion of the brain in humans. **Cerebrum:** is the largest part of the brain and is composed of **right** and **left** hemispheres. It performs higher functions like interpreting touch, vision and hearing, as well as speech, reasoning, emotions, learning, and fine control of movement.

The **hypothalamus** ("under the thalamus") is the center for homeostatic control of the internal environment. It receives signals about the state of the body and regulates thirst, appetite, and body temperature. It also controls sex drive and is an endocrine gland that interacts with the adjacent pituitary gland.

Thalamus

The thalamus is superior to the hypothalamus andinferior to the cerebrum. The third ventricle is a narrow cavity that passes through both the thalamus andhypothalamus. Many of the functions of the thalamusare concerned with sensation. The **midbrain** acts as a relay station for tracts passing between the cerebrum and the spinal cord or cerebellum. The tracts cross in the brain stem so that the right side of the body is controlled by the left portion of the brain, and the left portion of the body is controlled by the right portion of the brain.The brain stem also has reflex centers for visual, auditory, and tactile responses.

The **cerebellum** lies under the occipital lobe

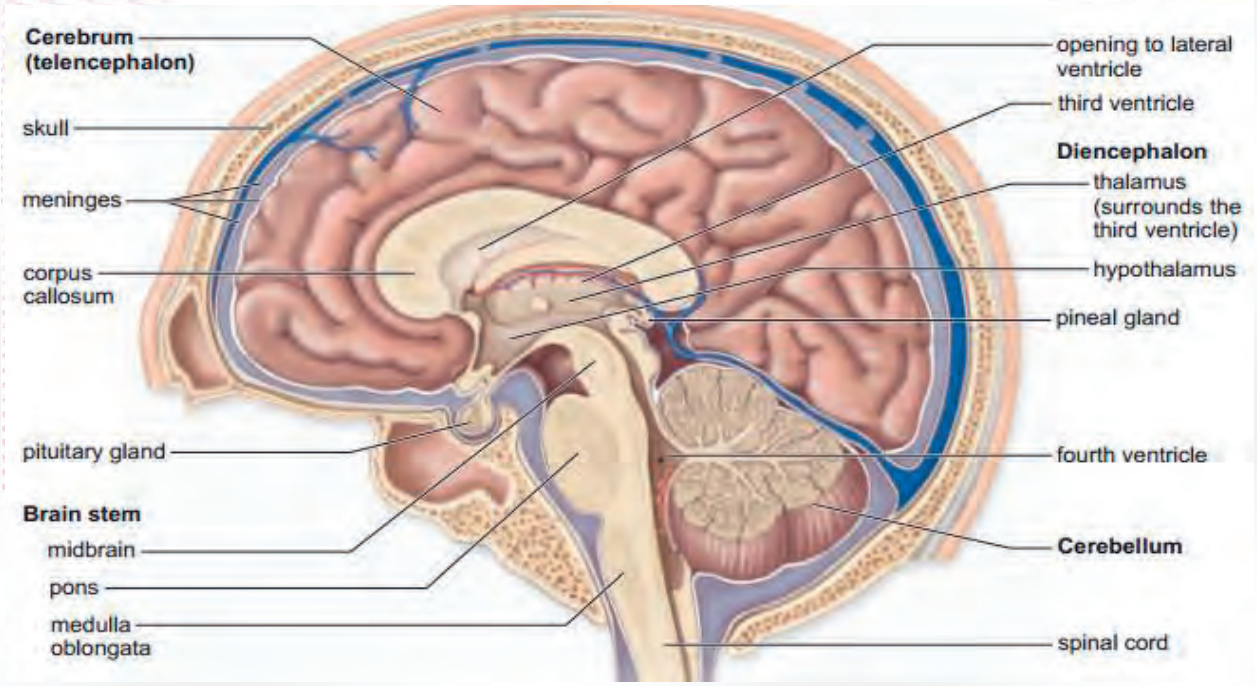


Figure 5.9 Parts of human brain

of the cerebrum and is separated from the brain stem by the fourth ventricle. It is the largest part of the hindbrain. The cerebellum receives sensory input from the eyes, ears, joints, and muscles about the present position of body parts, and it also receives motor output from the cerebral cortex about where these parts should be located. After integrating this information, the cerebellum sends motor impulses by way of the brain stem to the skeletal muscles. In this way, the cerebellum maintains posture and balance.

It also ensures that all of the muscles work together to produce smooth, coordinated voluntary movements. The cerebellum assists the learning of new motor skills such as playing the piano or hitting a baseball. New evidence indicates that the cerebellum is important in judging the passage of time.

Medulla (Brainstem)

The medulla oblongata, pons, and midbrain constitute the **brainstem**. The brainstem connects the spinal cord to the remainder of the brain and is responsible for many essential functions.

The word **pons** means "bridge" in Latin, and true to its name, the pons contains bundles of axons traveling between the cerebellum and commands.

the rest of the CNS. In addition, the pons functions with the medulla oblongata to regulate breathing rate, and has reflex centers concerned with head movements in response to visual and auditory stimuli.

The **medulla oblongata** contains a number of reflex centers for regulating heartbeat, breathing, and blood pressure. It also contains the reflex centers for vomiting, coughing, sneezing, hiccuping, and swallowing. The medulla oblongata lies just superior to the spinal cord, and it contains tracts that ascend or descend between the spinal cord and higher brain centers.

White and gray matter

The brain and spinal cord contain gray matter and white matter. **Gray matter** is primarily made up of neuron cell bodies. **White matter** consists of bundled axons. In the spinal cord, white matter forms the outer layer, reflecting its role in linking the CNS to sensory and motor neurons of the PNS. In the brain, white matter is predominantly in the interior, where signaling between neurons functions in learning, feeling emotions, processing sensory information, and generating

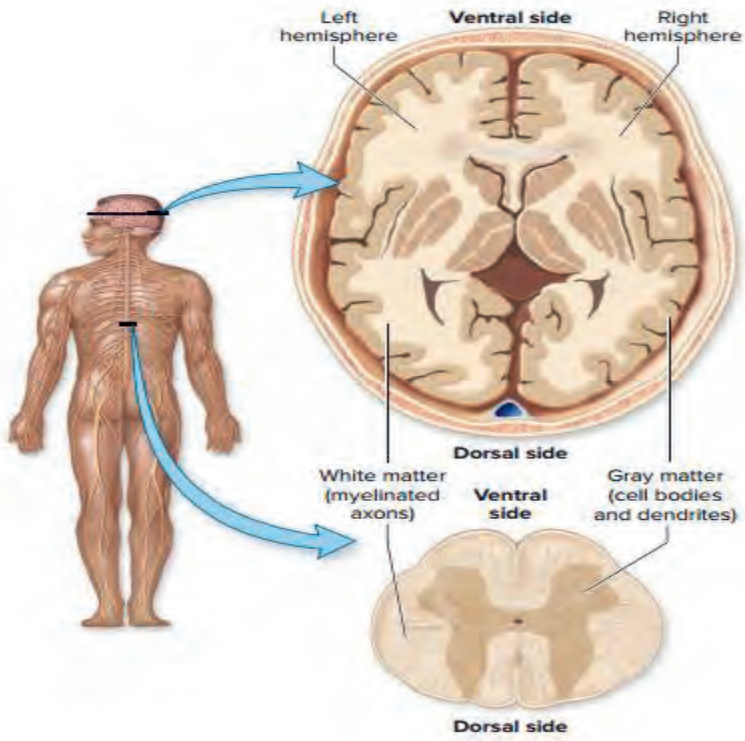


Figure 5.10 White and Gray matter of a brain

Keywords

Fore brain the large frontal area of the human brain

Grey matter areas in the brain and spinal cord that consist of unmyelinated nerve cells

White matter areas of the brain and spinal cord composed mainly of long-range myelinated axons

Affector (afferent) neuron that sends impulses from organs to the spinal cord and brain

Effector (efferent) neuron that sends impulses from the brain and spinal cord to

The spinal cord

The spinal cord has a much simpler structure than the brain. It is a tubular structure composed of the nervous tissue that extends from the brainstem and continues distally before tapering at the lower thoracic/upper lumbar region as the conus medullaris (the terminal end of the spinal cord). The spinal cord runs out from the brain down the body. The spinal cord is encased and protected by the vertebrae making up the spine. The majority of nerves come out of the spinal cord are known as the **spinal nerves**. They stretch to the arms, legs, trunk and to the rest of the body (Figure 5.4). In the spinal cord **the grey matter** is located in the middle whereas the **white matter** is found on the outside (Figure 5.5). At regular intervals along the spinal cord there are entrance points for **affector** nerves that bring information into the CNS and exit points for **effector** nerves carrying instructions from the CNS.

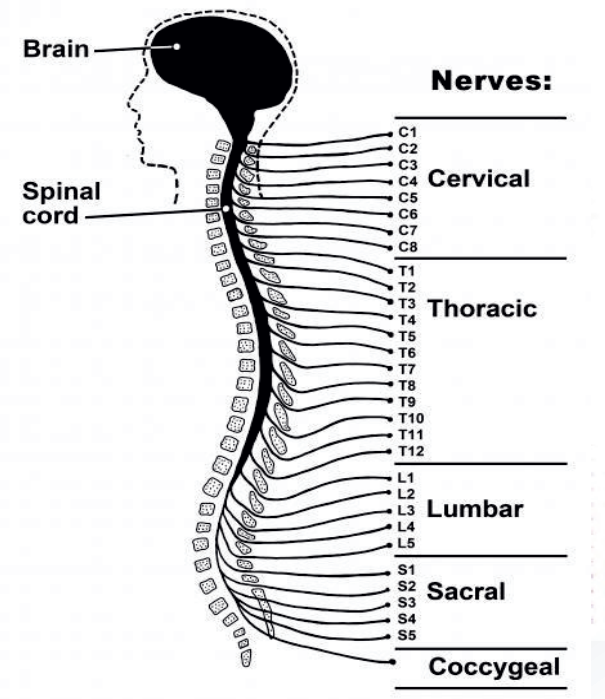


Figure 5.11 The spinal cord and spinal nerves

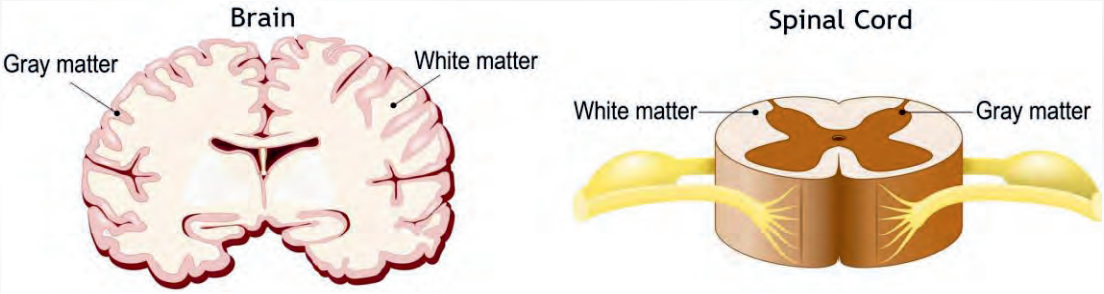


Figure 5.12 The cross-section of the CNS showing the position of grey and white matters.

Dura mater – This is the thick outermost covering (meninges) of the brain and spinal cord. It is the layers of connective tissue that make up the meninges of the brain (dura, arachnoid, and pia). It is the outermost layer of the three meninges that surround and protect the brain and spinal cord

Activity 5.7

Make a small group and read from different books and other sources about:

1. Cranial and spinal nerves; the names of the different types of the cranial and spinal nerves and the role of these nerves in the human nervous system.

2. The opposite sides of the brain (the right and left hemispheres of the brain), their basic difference in function and how they work and control your body. Report your work to the class.

Pia mater – This is the innermost covering of the spinal cord. Intimately adhered to its surface, the pia mater stabilizes the spinal cord through lateral extensions of the pia called the denticulate ligaments, extending between the ventral and dorsal roots unto the dura mater.

The Peripheral Nervous System

The Peripheral Nervous System (PNS) is the division of the nervous system that contain all the nerves that lie outside of the central nervous system. The peripheral nervous system is subdivided into the afferent division and the efferent division . The afferent division of the PNS carries signals to the CNS and includes all the neurons that transmit sensory information from their receptors. The efferent division of the PNS carries signals from the CNS to the muscles and glands, which act as effectors to bring about the desired response. In mammals, 31 pairs of spinal nerves carry signals between the spinal cord and the body trunk and limbs, and 12 pairs of cranial nerves connect the brain directly to the head, neck, and body trunk. The efferent division of the PNS is

further subdivided into the somatic nervous system and the autonomic nervous system.

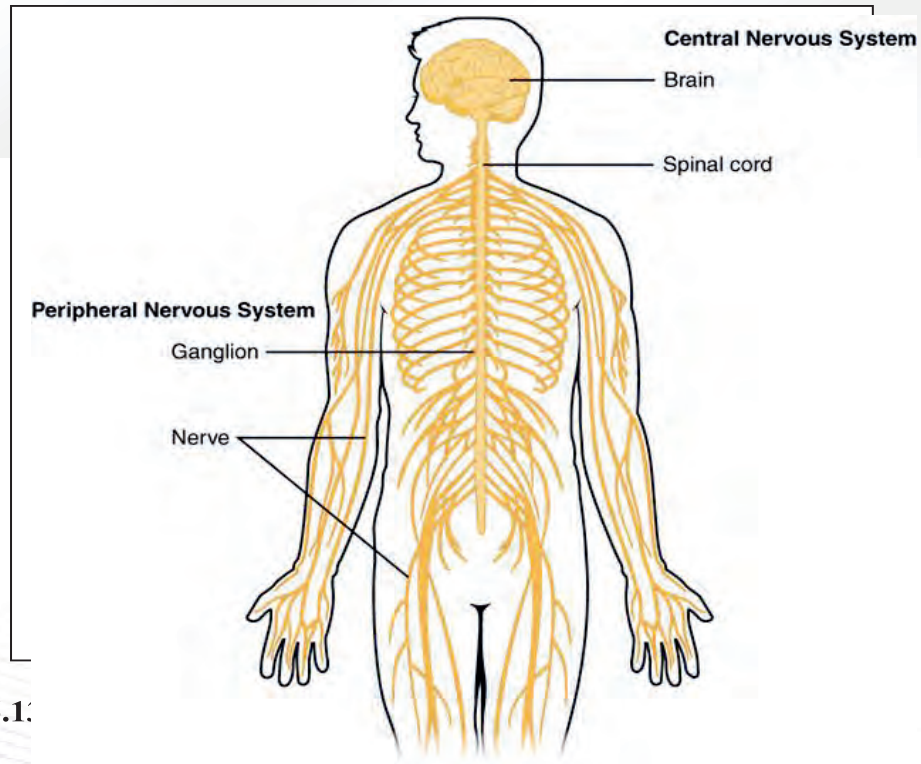


Figure 5.1

The Somatic Nervous System

The somatic system is the part of the PNS that is responsible for carrying out sensory and motor information to and from the central nervous system. The somatic nervous system derives its name from the Greek word *soma*, which means "body." The somatic system is responsible for transmitting sensory information as well as for voluntary movement.

The Autonomic Nervous System

The autonomic system is part of your PNS that is responsible for regulating involuntary body functions, such as blood flow, heartbeat, digestion, and breathing. In other words, it is the autonomic system that controls aspects of the body that are usually not under voluntary control. This system allows these functions to take place without the need to consciously think about what is happening. The autonomic system is further divided into two branches:

Sympathetic system: By regulating the 'flight-or-fight' response, the sympathetic system prepares the body to expend energy to respond to environmental threats. When action is needed, the sympathetic system triggers a response by accelerating heart rate and increasing breathing rate, boosting the blood flow to

muscles, activating sweat secretion, and dilating the pupils.

Parasympathetic system: This helps maintain the normal body functions and conserve physical resources. Once a threat is recognized, this system will slow the heart rate, slow breathing, reduce blood flow to muscles, and constrict the pupils. This allows us to return our bodies to a normal resting state.

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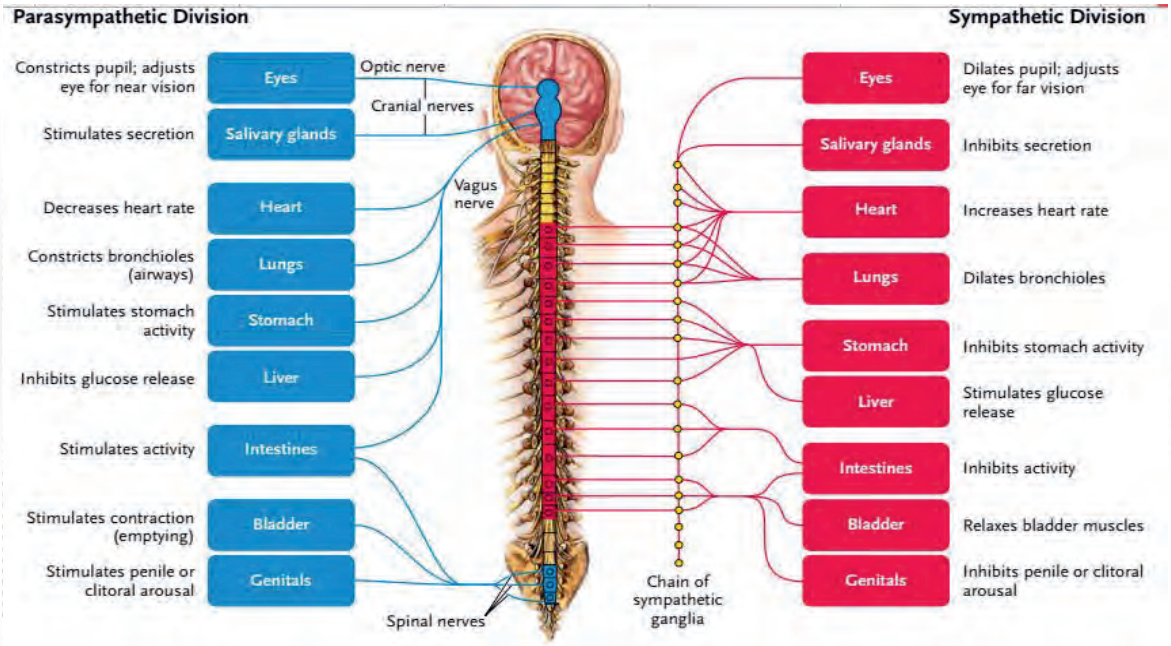


Figure 5.14 The Autonomic Nervous System

5.1.6. Reflex action

At the end of this section the learner will be able to:

- Define reflex action
- State the types of reflexes with examples.

A **reflex action** is a sudden, automatic and uncontrolled response of parts of the body or the whole body to the external stimuli. They are usually involved in helping us to avoid danger or damage. When the body is in danger, it can respond to the situation, without conscious, thought. This causes a faster response, preventing or minimizing damage to the body. This is known as reflex action. A reflex action is an automatic response to a stimulus.

Keywords

Reflex action- automatic, instinctive, unlearned reactions to stimuli

Voluntary- actions that involve conscious thought

Reflex arc neural path of a reflex

Reflex Arc

The parhway of neurons involved in a reflex action is known as reflex arc. Most reflexes follow the same steps between the stimuli and the response.

One of the simplest situations where impulses cross synapses to produce action is in the reflex arc. A reflex action is an automatic response to a stimulus. (A stimulus is a change in the external or internal environment of an organism.) It provides a means of rapidly integrating and co-coordinatinga stimulus with the response of an effector (a muscle or a gland) without the need for thought or a decision.

A reflex arc is the pathway that

Self-questioning

- Why reflexes can happen so quickly?
- Do you think that all reflexes are simple?
- How reflexes are so important for keeping us safe and helping us to learn
- What are the differences between voluntary and reflex action?

nerve impulses travel when a reflex is elicited, and there are five essential parts:

1. **Receptors**:- detect a change (the stimulus) and generate impulses.

2. **Sensory neurons**:-transmit impulses from receptors to the CNS.

3. **Central nervous system:-**contains one or more synapses (interneurons may be part of the pathway).

4. **Motor neurons:-**transmit impulses from the CNS to the effector.

5. **Effector**: -performs its characteristic action.

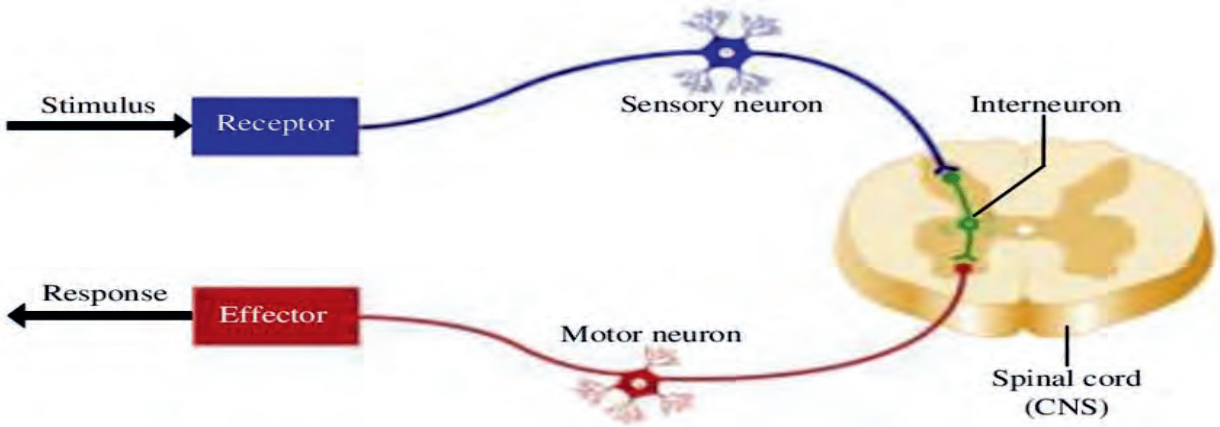


Figure 5.15 Path way for a nerve impulse in reflex action

- Let us now look at the reflex arc of a specific reflex, the patellar (or knee-jerk) reflex, with which you are probably familiar. In this reflex, a tap on the patellar tendon just below the kneecap causes extension of the lower leg. This is a stretch reflex, which means that a muscle that is stretched will automatically contract.

Keywords

Dorsal root- root at the back of the spinal cord

Ventral root- root at the front of the spinal cord

- As shown in the Figure 5.15:
- Impulses from a sensory receptor in the skin pass along an affector neuron to the central nervous system – the spinal cord.
- The neuron enters the spinal cord through the dorsal root.
- When an impulse from the affector neuron arrives in the synapse with a short relay neuron a transmitter is released, which causes an impulse to be sent along the relay neuron.

When the impulse reaches the synapse between the relay neuron and an effector

neuron returning to the arm again another transmitter chemical is released.

This starts impulses travelling along the effector neuron to the organ (effector), which brings about change. The effector neuron leaves the spinal cord by the ventral root. In this example the impulses arrive in the muscles of the upper arm, causing them to contract and move your hand upwards sharp.

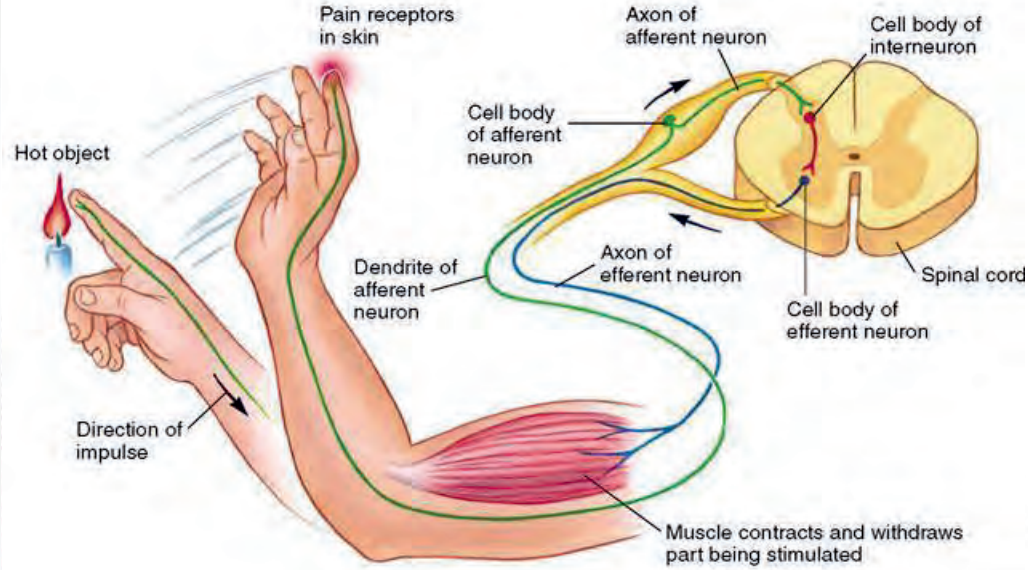


Figure 5.16 The reflex action

Some reflexes are important for avoiding injury, but the knee-jerk and ankle-jerk reflexes are important for normal physiological functions. For example, the knee-jerk reflex helps a person stand erect. If the knee begins to bend slightly when a person stands still, the quadriceps femoris is stretched, and the leg straightens.

The spinal cord has a much simpler structure than the brain. It is a tubular structure composed of the nervous tissue that extends from the brainstem and continues distally before tapering at the lower thoracic/upper lumbar region as the conus medullaris (the terminal end of the spinal cord). The spinal cord runs out from the brain down the body. The spinal cord is encased and protected by the vertebrae making up the spine. The nerves that originate from the spinal cord are known as the **spinal nerves**. They stretch to

the arms, legs, trunk and to the rest of the body (Figure 5.4). In the spinal cord **the grey matter** is located in the middle whereas the **white matter** is found on the outside (Figure 5.5). At regular intervals along the spinal cord there are entrance points for **affector** nerves that bring information into the CNS and exit points for **effector** nerves carrying instructions from the CNS.

Activity 5.8

Make a small group and read from different books and other sources about:

1. Cranial and spinal nerves; the names of the different types of the cranial and spinal nerves and the role of these nerves in the human nervous system.

2. The opposite sides of the brain (the right and left hemispheres of the brain), their basic difference in function and how they work and control your body. Report your work to the class.

Self-questioning

Distinguish voluntary and reflex action? Can you tell your friend one example to each actions?

Activity 5.9 Investigating reaction time

The length of time it takes you to recognize a stimulus and react to it is your reaction time. These are very important in many situations – for example, when you are driving, or at the start of a race. Some people react more quickly than others and you can train yourself to speed up. In this investigation you will be looking at reaction times by measuring how quickly your partner catches a meter ruler when you let it fall. If you collect all the data for the class, you can produce a graph to show the range of reaction times for your science group, and also do some statistical analysis to find the average, median and mean reaction times for your class.

You will need:

• A meter ruler (or a stick and a ruler)

Method

1. Work in pairs, with one partner holding the ruler and recording the distance and the other catching.
2. Hold the ruler so that your partner's hand is level with the 10 cm mark. They should be able to see the ruler and your hand.
3. Warn them that you will soon be dropping the ruler and, after a few seconds, let go.
4. Repeat this three times and calculate the average distance the ruler travelled before your partner caught it.
5. Swap roles.
6. Draw up a table and collect the results for the whole class.
7. Write up your investigation, including a graph to show the distribution of reaction rate across the class.
8. How could you develop or refine this investigation?

The key point about a reflex action is that the messages do not reach a conscious area of your brain before instructions are sent out to take action. Many reflexes involve the spinal cord, whereas others involve the brain. They involve three types of neuron affector neurons, relay neurons and effector neurons. Relay neurons (interneurons with short axons) connect the

affector and effector neurons directly in the CNS, without input from other areas. The receptors, neurons and effectors involved are referred to as a **reflex arc**. The brain and spinal cord act together as a coordinators that process the information coming from sensory receptors and neurons and instruct effector neurons and effectors to react.

Activity 5.10 The Knee jerk reflex

You can demonstrate a simple spinal reflex used by doctors.

1. Work in pairs.
2. One person sits down with one leg loosely crossed over the other.
3. The other hits the crossed leg just below the knee cap (patella) gently but firmly. Use the edge of the hand, a special hammer or the edge of a ruler to do this. It should not hurt!
4. This hits a ligament which stretches a muscle. This is picked up by stretch receptors setting up a reflex arc. The impulse travels to the spinal cord and back to the quadriceps muscles in the thigh. These contract, causing the lower leg to jerk upwards and outwards as the leg straightens. It drops straight back into its normal position.
5. Change places and repeat!

Activity 5.11

1. Make a small group and search from books and other materials (the internet) about Ivan Pavlov's experiment on dog. Report your finding to the class.

2. Take one example of a reflex action. Draw a diagram to show this reflex arc, label it carefully and discuss your work in the class

At the end of this section, the learner will be able to:

- Illustrate the effects of drug abuse on nervous systems with local and international examples.

Self-questioning

1. What is a drug?
2. Can you define
   a. what drug use mean?
   b. What drug abuse mean?

5.1.7. Drug abuse

Drugs are substances that change a person's mental or physical state. They can affect the way the brain works, how you feel and behave, your understanding and your senses. This makes people, especially young people develop unpredictable and dangerous behavior.

In every society there are certain drugs which are used for medicine and there are others which are used for pleasure. Usually some of these substances are socially acceptable and others are illegal. The status of a drug may be related to its effect on people, or it may be simply down to the history of its use. Most of the drugs being used for medicine affect our bodies. Some of the drugs used for pleasure tend to have a distinct effect on our minds. Drug (substance) use is when a substance is used to the extent that affects the brain and/or body function and mental activity.

Keywords

Addicted compulsively or physiologically dependent on something habit-forming

Withdrawal symptoms a set of unpleasant effects upon the body caused by a sudden stopping of using a drug

Legal drugs are used for the mild pleasure they bring, to be sociable and as a habit. People may start to use illegal drugs for the same reasons. **Drug (or substance) abuse** is when a person uses a substance to the point of excess and/or dependence. When drugs are taken in excess they cause a serious health risk and even death. Drug dependence is when a person use a drug again and again and become **addicted**. Addiction is marked by a change in behavior caused by the biochemical changes in the brain after continued substance abuse.

Activity 5.12

Read from different books and other resources about drug addiction, drug dependence and withdrawal symptoms. Reflect your finding to the class.

Substance use disorder (SUD) or drug addiction is a disease that negatively affects a person's brain and behavior. A person can become obsessed with any legal or illegal drugs. Some people can get addicted to certain medications. This addiction gradually starts developing when the individual continues to consume the drug despite the impairment it causes.

Nicotine, marijuana and alcohol are commonly misused drugs in today's world.

Drug abuse, of both legal and illegal substances, is becoming more of a public health problem. School surveys have shown that alcohol, khat and tobacco are substances widely used in Ethiopia. Alcohol, khat and tobacco are legal drugs in Ethiopia but cannabis is illegal. Drugs such as lysergic acid diethylamide (LSD), ecstasy, cocaine and heroin are illegal but rarely used in Ethiopia. Khat, alcohol and tobacco are linked to a wide range of health problems. The health issues linked to these drugs are mainly the result of their effect on the systems of the body.

Activity 5.13

Discuss in small group and list the three most commonly abused substances by young people in Ethiopia.

Smoking

The addictive drug in cigarette smoke is **nicotine**, which affects the brain and produces a sensation of calm, well-being and being able to cope with. However, it is very physically addictive. Unfortunately cigarette smoke also contains many very harmful chemicals, and these are linked to a number of very serious health problems.

What is dopamine?

Dopamine is a chemical released in the brain that makes you feel good. Having the right amount of dopamine is important both for body and brain. Dopamine helps nerve cells to send messages to each other. It's produced by a group of nerve cells in the middle of the brain and sends out messages to other parts of the brain. Within seconds of inhaling cigarette smoke, nicotine causes the release of dopamine in the brain, which gives people a good feeling. Over time, the brain begins to crave that feeling from nicotine and people need to use more and more tobacco to get that same good feeling.

How does nicotine use lead to addiction?

Nicotine releases a chemical called dopamine in the same regions of the brain as other addictive drugs. It causes mood-altering changes that make the person temporarily feel good. Inhaled smoke delivers nicotine to the brain within 20 seconds, which makes it very addictive.

Figure 5.17 Cigarette is an addictive drug

Alcohol

Alcohol is one of the drugs most commonly used in Ethiopia. Alcohol, like other drugs, has a powerful effect on the brain, producing pleasurable feelings and blunting negative feelings. These feelings can motivate some people to drink alcohol again and again, despite possible risks to their health and well-being. Alcohol addiction is a chronic relapsing disorder associated with compulsive alcohol drinking, the loss of control over intake, and the emergence of a negative emotional state when alcohol is no longer available. Alcohol use disorder (AUD) is a condition characterized by an impaired ability to stop or control alcohol use despite adverse social, occupational, or health consequences

Activity 5.14

Make a small group and search from books and other resources about some of the harmful chemicals found in smoking cigarettes and diseases that are caused as a result of smoking. Report your finding to the class.

Figure 5.18 Alcohol is addictive and poisonous

Activity 5.15

Read books and other sources about how alcohol can directly and/or indirectly affect the health of young people. Present your findings to the class.

Khat

Catha edulis (Khat) is a plant grown commonly in the horn of Africa. The leaves of khat are are chewed by people for it is stimulant. Its young buds and tender leaves are chewed to attain a state of euphoria and stimulation. Khat contain a drug that affects the brain. The plant has a stimulant effect similar to amphetamine, cocaine or very strong coffee depending on how much it is consumed. Many people can develop a tolerance to it over time, as well as dependence and addiction. Because of this, it is important to know what khat is, how it affects the body and how to treat a khat addiction. Khat contains a drug that affects the brain. Chewing the leaves of the evergreen shrub releases an amphetamine-like stimulant. The drug **cathinone** from the khat leaves is absorbed into the bloodstream through the membranes lining of the mouth and the stomach. It acts quickly, within 30 minutes of time, before it is broken down and removed by the liver. When addicted people cannot get the drug, feel depressed, tired and unable to concentrate.

Figure 5.19 A monoculture of Khat in Ethiopia

Cannabis (marijuana)

Cannabis is a plant that contains 400 known chemicals, 60 of which are the cannabinoids, unique to the plant. The most potent is delta-9-tetrahydrocannabinoid (THC). THC is known to affect the brain cells that are responsib

Keywords

Cannabis/marijuana drug made from the cannabis plant

Hallucinogens psychotoxic drugs that affect the mind in a way that produces distorted sensations abnormal in content

Activity 5.16

1. Search from different sources (books, the internet) about the economic importance of khat and its negative effect in damaging the economy. Report your findings to the class.

2. Can you list some of the relationship between unprotected sex, HIV/AIDS infection and Khat? How?

le for memory, emotion and motivation. Cannabis is usually smoked but it can also be eaten, whenit has a much stronger effect because your liver converts it into a much more powerful drug. It can make you feel a great sense of wellbeing and relaxation, happy and euphoric – and this is why people use it. It is a mild hallucinogenic drug. **Hallucinogens** are drugs that produce vivid waking dreams, where the user sees or hears things that are not really there, or has a distorted view of the world. However, many people find the effect of the drug a very unpleasant and disturbing experience. The effect of cannabis is very variable. It affects different people in different ways, and even the same person can react very differently depending on how it is used. Cannabis is illegal in Ethiopia.

Activity 5. 17

Make a small group and work on the following question:

What are the health and social effects of drug abuse on individuals, families and communities? Report your answers to the class.

5.2. Sense organs

At the end of this section, the learner will be able to:

- Discuss the structure and function of skin
- Draw taste sites on the tongue
- Indicate the different taste sites on a diagram of tongue correctly.
- Explain the structure and functions of nose
- Draw the structure of the human eye
- Label the structure of the human eye on a diagram correctly
- Explain the functions of the human eye
- Explain the way the eye sees things
- Identify common eye defects and their causes in humans
- Draw the structures of the human ear
- Indicate parts of human ear on a diagram
- Label the structures of the human ear
- Describe the functions of the human ear

5.2.1. Skin

Self-questioning

- What are the functions of human skin.
- What is the average weight of a human skin?

The skin is one of the largest organs in the human body in surface area and weight. It is a remarkably complex organ which carries out a number of important functions in human body. It gives you senses of touch, temperature and pain.

The skin:

- Contains a huge variety of sense organs (touch, temperature, pressure, pain).
- Forms a waterproof layer around the body tissues, which protects against the loss of water by evaporation and prevents gaining water by osmosis while swimming in the river or wash.
- It protects the body from the entry of bacteria and other pathogens.
- It protects the body from damage by UV light.
- It is an excretory organ (nitrogenous wastes are lost with the sweat).
- It is vital in controlling the body temperature.

Basic components of the human skin

Keywords

Hypodermis the lower fatty layer of skin which helps to

insulate the body against heat loss

Dermis the middle layer of skin, which is made up of blood vessels, lymph vessels, hair follicles, and sweat glands

The human skin is the largest organ of the body, with a total area of about 20 square feet. The skin protects us from microbes and elements, helps regulate body temperature, and permits the sensations of touch, heat, and cold. Skin has three main layers. These are **epidermis, dermis** and **hypodermis.**

Activity 5. 18

Form a small group. Each group needs to search for human skin model or picture and discuss the basic parts of the skin. Then present your small group discussion to the class.

- The **epidermis**, the outermost layer of skin, provides a waterproof barrier and creates our skin tone.
- The **dermis**, beneath the epidermis, contains tough connective tissue, hair follicles, and sweat glands.
- The deeper subcutaneous tissue (**hypodermis**) is made of fat and connective tissue.

The lower layer, the **hypodermis**, contains fatty tissue which is both an energy store that acts as an insulation layer, protecting against heat loss. The middle layer or **dermis** contains the blood vessels, the sweat glands, the sensory receptors and the hair follicles. This layer is closely involved in temperature control in homeostasis and in your sense of touch. The upper layer or **epidermis** is made up of dead cells. **Keratinocytes**, **melanocytes** and **langerhans cells** are some of the important cells found in the epidermis**.** Keratinocytes- produce the protein known as **keratin**, the main component of the epidermis. Keratin makes up hair, nails, and the surface layer of the skin. Keratin is what forms the rigidity of your skin and helps with the barrier protection that your skin offers. **Melanocytes** produce skin pigment, which is known as **melanin**. **Langerhans cells**, which prevent things from getting into your skin. Epidermal layer of the skin stop water loss and also protect against the entry of pathogens.

The **dermis** is a complex combination of blood vessels, hair follicles, and sebaceous (oil) glands. In the dermis, there are collagen and elastin proteins necessary for skin health because they offer support and elasticity (the skin's ability to go back to its original state after being stretched). **Collagen** is the most plentiful protein in the skin, making up 75-80% of the skin. Collagen and elastin are responsible for warding off wrinkles and fine lines. Over time, the environment and aging reduce the body's ability to produce collagen. It is the dermis which is particularly involved in the homeostatic mechanisms of the skin.

The thickness of the epidermis and dermis may vary. Due to variation in the thickness the skin can be categorized as **thick** and **thin**. Thin skin covers most of the body and can vary in thinness, with the thinnest skin covering the eyelids. Thick skin is present on the soles of the feet and palms of the hands. In addition to differing thicknesses, the skin also differs in what is available in the layers. For example, thick skin has no hair follicles or sebaceous glands, whereas thin skin does.

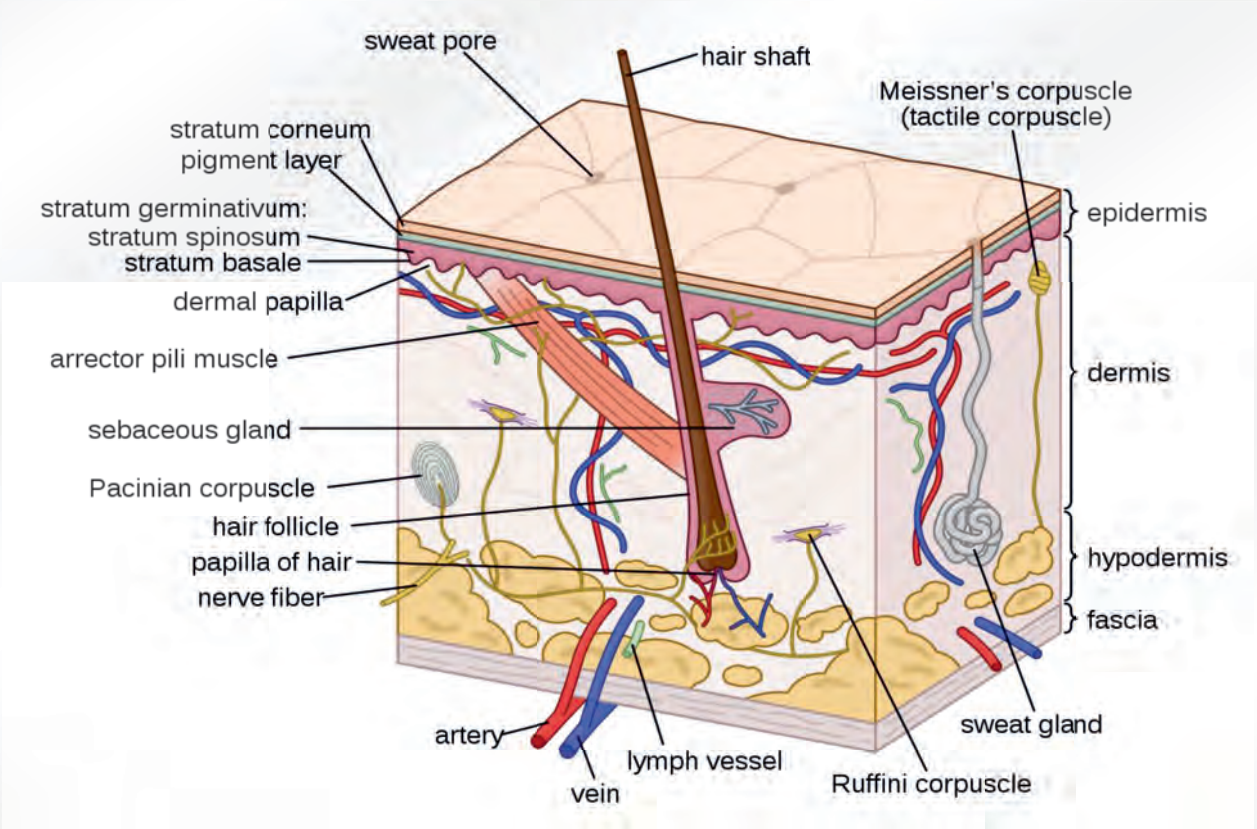


Figure 5.20 Part of the human skin

Activity 5.19

Read from different sources about human skin and report your finding to the class.

1. How a human skin color is influenced by a number of pigments.

2. List some of the skin disorders.

Activity 5.20

Investigating the sense of touch and temperature

1. Touch:

You are going to investigate the sense of touch in different areas of the skin.

You will need:

• a fine ballpoint or felt pen

• a bristle mounted on wooden holder or blunt seeker or fine piece of wire or very sharp pencil

2. Method

1. Work in pairs. Take turns to carry out the investigation.

2. With the pen, draw a grid of 25 squares on the back of your partner's hand. Each square should be 2 mm x 2 mm.

3. Draw an identical grid on paper and label it with the name of the subject and the area of the body.

4. The subject should close their eyes or look away – they must rely on the sense of touch alone. Ask them to say Y ES when they feel a touch.

5. Press the tip of the bristle against the skin in one of the squares until it just bends, or touch the skin with the blunt seeker as gently as possible. Touch each of the squares in turn, marking on your paper each one that gives a positive response.

6. Now try other areas of the skin that you might expect to be more or less sensitive,

e.g. the palm of the hand, the arm, the leg, the foot, etc.

7. Once you have tested three different areas, swap roles.

8. Are some parts of the skin more sensitive than others? W rite up your experiment along with the results and explain your observations as well as you can.

2. Temperature:

Is your sense of temperature absolute – or comparative? In other words, are your

5.2.2. The Tongue

The tongue is a muscular organ in the mouth. The tongue is covered with moist, pink tissue called mucosa. Tiny bumps called **papillae** give the tongue its rough texture. Thousands of taste buds cover the surfaces of the papillae. Taste buds are collections of nerve-like cells that connect to nerves running into the brain.

The tongue is anchored to the mouth by webs of tough tissue and mucosa. The tether holding down the front of the tongue is called the frenum. In the back of the mouth, the tongue is anchored into the hyoid bone. The tongue is vital for chewing and swallowing food, as well as for speech.

Activity 5.21

Identifying parts of human skin

You will need

- A light microscope
- A prepared slide of human skin

Method

1. Using the techniques for using a light microscope, examine your slide of the skin and identify as many features as possible.
2. Draw and label section through the skin.
3. Identify and draw higher magnification details of particular tissues such as the sweat gland, any sense organs, etc.

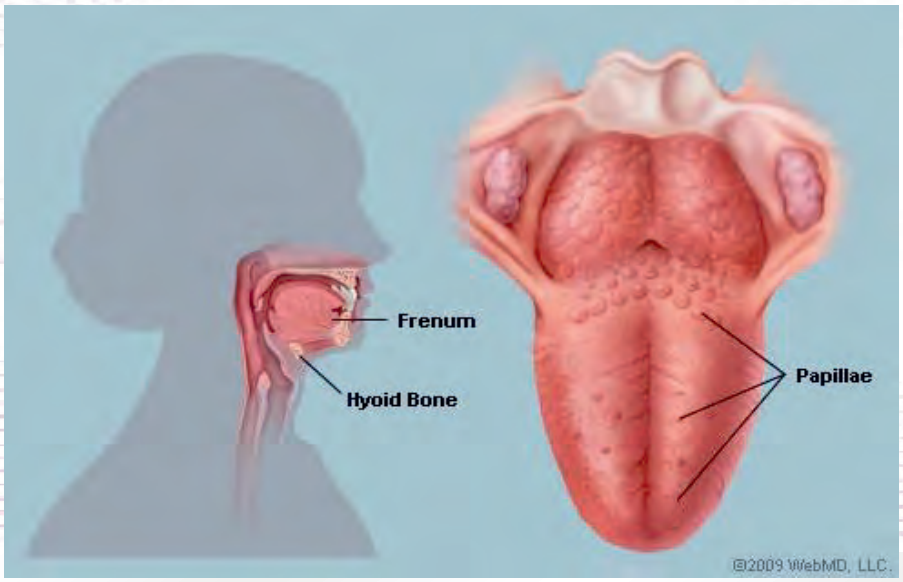


Figure 5.21 The human tongue

The sensory receptors of the tongue and those found in the nostrils are sensitive to solutions of certain chemical substances. The sensory receptors of taste are located on the upper surface of the tongue, and to a lesser extent on the surface of the throat. The receptors for smell are located in the upper parts of the nasal passages. There are five basic taste sensations. The first four are sweet, sour, bitter and salt. We have known too much about these for many years. Scientists have also discovered a fifth taste called umami (a very savoury flavour found in foods such as meat, cheese, broth and mushroom). For many years it was thought that the receptors for the four known senses had their areas of greatest concentration on different parts of the tongue. It has now been clearly shown that in fact all of the five different taste organs are spread out all over the tongue, although some of them may seem to be in a greater concentration in certain places.

Activity 5.22

Read from various sources about the tongue and reflect to the class the following questions.

1. Why do we see variation in some people in response to the same stimulus?

2. Explain why the map used in the Figure 5.19 below is wrong.

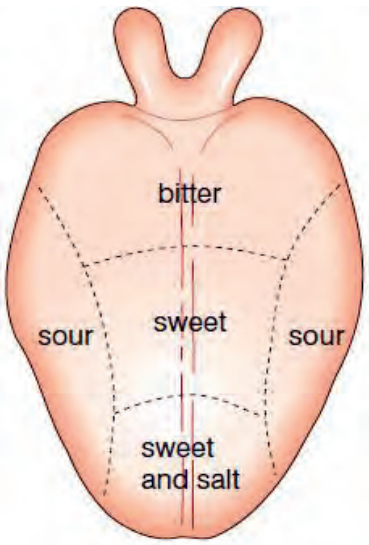


Figure 5.22 The main taste areas of the tongue that has been taught for many years.

5.2.3. The Nose

The human nose is a sense organ of smell. Another word for smell is **olfaction**. The olfactory receptors in the nose help to identify food, mates, predators, and provides both sensual pleasure in the odor of flowers and perfume, as well as warnings of danger. For example, spoiled food, fire or chemical dangers. For both humans and animals, it is one of the important means by which the environment communicates with us. Another function of the nose is the conditioning of inhaled air, which the nose makes it more humid and warmer. Hairs inside the nose prevent large particles from entering the lungs.

Anatomy and Physiology

- The nasal passages not only allow the passage of air for respiration through the nose but also warm and filter the air.
- The nasal concha or turbinate bone is a long, narrow and curled bone shelf that protrudes into the breathing passage of the nose. In humans, the turbinates divide the nasal airway into four groove like air passages, and are responsible for forcing inhaled air to flow in a steady, regular pattern around the largest possible surface of cilia and climate controlling tissue.

How do you smell?

Specialized receptor cells of the olfactory epithelium detect and recognize smells. The air passes through the nasal cavity and through a thick layer of mucus to the olfactory bulb. The olfactory bulb is situated in the forebrain. The smells are recognized here because each smell molecule fits into a nerve cell like a puzzle piece. The cells then send signals to the brain via the olfactory nerve. The brain then interprets those molecules as the sweet flowers, or the curdling milk that you have held up to your nose.

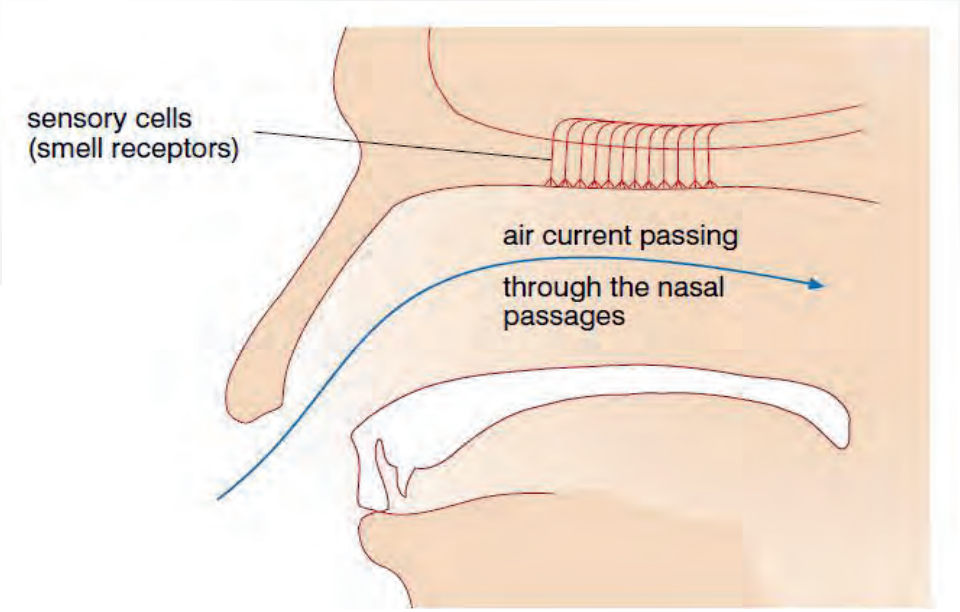


Figure 5.23 Internal structure of nose

Activity 5.23

Discuss in group why you cannot 'taste' foods well when catch a common cold.

For human being to be able to taste and smell, chemicals must go into solution in the form of liquid coating the membranes of receptor cells before they can be detected. The major functional difference between the two kinds of receptors is that smell receptors are more specialized for detecting vapors coming to the organism from distant sources. Taste receptors are specialized for detection of chemicals present in the mouth itself. Furthermore, smell receptors are much more sensitive than taste receptors.

5.2.4. The Eye

The eye structure, function, and defects

The eye transmits visual stimuli to the brain for interpretation and thereby functions as **sensory organ** of vision. The eyeball is located in the eye orbit, a round, bony hollow formed by several different bones of the skull. In the orbit, the eye is surrounded by a cushion of fat. The bony orbit and fat cushion protect the eyeball. To perform a thorough assessment of the eye, it is crucial to understand the external structures of the eye, the internal structures of the eye, the

Keyword

Sensory organ an organ that receives and relays information about the body's senses to the brain

visual fields and pathways, and the visual reflexes.

External structures of the eye

The parts of the eye that include the following are visible externally.

The **eyebrows** protect the eyes by preventing perspiration from running down the forehead and into the eyes, causing irritation. They also help shade the eyes from direct sunlight.

The **eyelids (upper and lower)** with their associated lashes are two movable structures composed of skin and two types of muscle: striated and smooth. Their purpose is to protect the eye from foreign bodies and limit the amount of light entering the eye. In addition, they serve to distribute tears that lubricate the surface of the eye (see Figure 5.24). The eyelids join at two points: the lateral (outer) canthus and medial (inner) canthus. The medial canthus contains the puncta, two small openings that allow drainage of tears into the lacrimal system, and the caruncle, a small, fleshy mass that contains sebaceous glands.

The white space between open eyelids is called the palpebral fissure. When closed, the eyelids should touch. When open, the upper lid position should be between the upper margin of the iris and the upper margin of the pupil. **Eyelashes** are projections of stiff hair curving outward along the margins of the eyelids that filter dust and dirt from air entering the eye.

The **conjunctiva is** a thin, transparent, continuous membrane that is divided into two portions: a palpebral and a bulbar portion. The palpebral conjunctiva lines the inside of the eyelids, and the bulbar conjunctiva covers most of the anterior eye, merging with the cornea at the limbus. This transparent membrane allows for inspection of underlying tissue and serves to protect the eye from foreign bodies.

The **lacrimal apparatus** consists of glands and ducts that serve to lubricate the eye (Figure 5.22). The lacrimal gland, that is located in the upper outer corner of the orbital cavity just above the eye, produces tears. The **extraocular muscles** are the six muscles attached to the outer surface of each eyeball (Figure 15.25). These muscles control six different directions of eye movement. Four rectus muscles are responsible for straight movement,

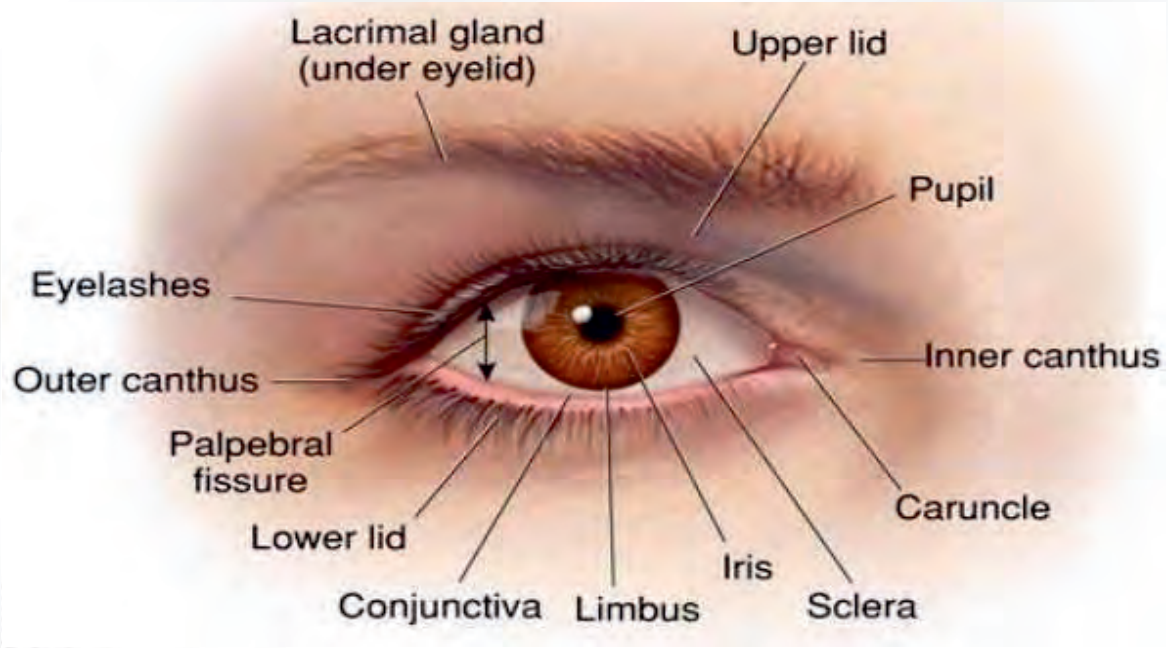


Figure 5.24 External structures of the eye

- and two oblique muscles are responsible for diagonal movement. The **cornea** is the transparent, anterior or front part of our eye, which covers the pupil and the iris. The main function is to refract the light along with the lens. The cornea permits the entrance of light, which passes through the lens to the retina. It is well supplied with nerve endings, making it responsive to pain and touch.
- **Iris:** It is the pigmented, colored portion of the eye, visible externally. The main function of the iris is to control the diameter of the pupil according to the light source.
- **Pupil:** It is the small aperture located in the centre of the Iris. It allows light to enter and focus on the retina.

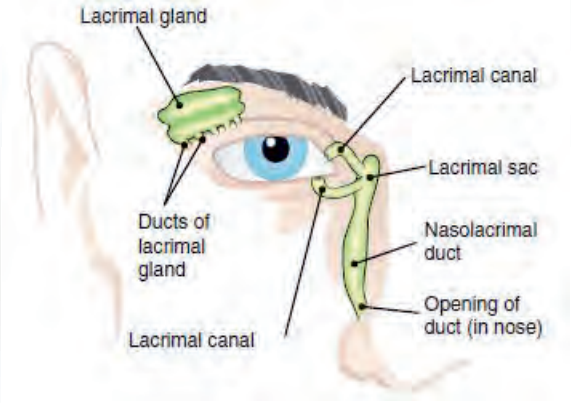


Figure 5.25 Lacrimal apparatus in the human eye

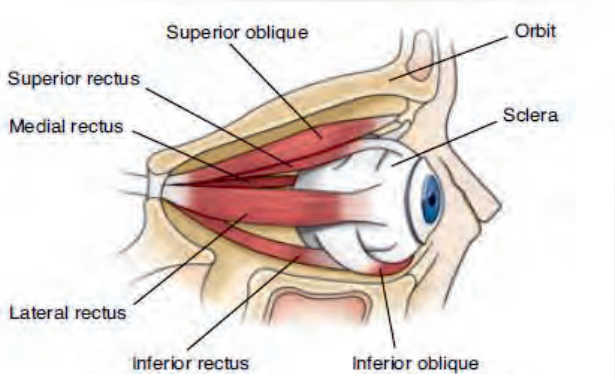


Figure 5.26 Extraocular muscles of the human eye.

Internal structure of the eye

The eyeball is composed of three separate coats or layers (Figure 5.27). The external layer consists of the **sclera** and **cornea**.

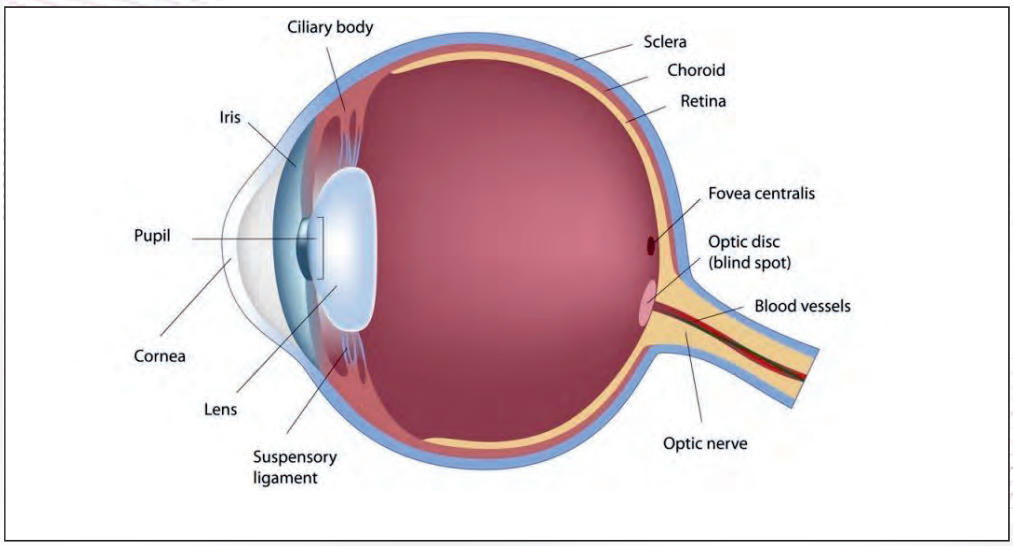


Figure 5.27 Anatomy of the human eye

- The **sclera** is a white visible portion. It is a dense, protective, white covering that physically supports the internal structures of the eye. It is continuous interiorly with the transparent cornea (the "window of the eye").

Activity 5. 24

Understanding the structure of the human eye

In this investigation you are going to look at eyes – your own and one from another animal

You will need:

• A mirror

• A white tile

• Sharp dissecting scissors

• Fresh eye of a sheep, cow or other mammal

Method

1. Look at your eyes in a mirror. How many of the features shown in Figure 5.16 can you see? Draw and label your eyes.
2. Move your eyes from side to side or up and down to see the whites of your eyes which part of the eye is this?
3. Look at the animal's eye provided by your teacher. Draw the external appearance of this eye and label it clearly. Can you see any of the muscles that are used to move the eyeball in the socket?
4. Using your scissors CAREFULLY cut a hole in the side of the eyeball. What comes out? How does this affect the eyeball?
5. Open the eyeball up more fully. Find the lens, the retina and as many other features. NB you won't be able to see them all.
6. Draw and label what you can see once you have opened the eyeball up fully.

The **lens** is a biconvex, transparent, avascular, encapsulated structure located immediately posterior to the iris. **Suspensory ligaments** attached to the **ciliary muscles** support the position of the lens. The lens functions to refract (bend) light rays onto the

Keywords

Lens a flexible disc that helps focus light on the retina

Suspensory ligaments elastic-like structures that suspend the lens and pull it into shape for focusing distant objects onto the retina

Ciliary muscles eye muscles that automatically contract or relax the shape of the lens of the eye to help focus light on the retina

Sclera the tough, opaque tissue that serves as the eye's protective outer layer

Cornea transparent structure over the front of the eye that allows light to enter. A cornea resembles a contact lens in size and appearance

Choroid the middle layer filled with blood vessels that nourish the retina

Pupil a hole in the centre of the iris that changes size in response to changes in lighting

Iris a membrane in the eye, responsible for controlling the amount of light reaching the retina

Dilated becomes wider

Constricted becomes smaller

retina. Adjustments must be made in refraction depending on the distance of the object being viewed. The refractive ability of the lens can be changed by a change in the shape of the lens (which is controlled by the ciliary body). The lens bulges to focus on close objects and flattens to focus on far objects. The **choroid layer** contains the vascularity necessary to provide nourishment to the inner part of the eye and prevents light from reflecting internally. Anteriorly, it is continuous with the ciliary body and the iris. Once the light has travelled through the cornea it has to pass through the **pupil** in the center of the **iris**. The iris is the colored part of the eye, but it is not there simply to look pretty. The iris is made up of muscles that contract or relax to control the size of the pupil and to control the amount of light reaching the retina. The circular muscles run around the iris, whereas the radial muscles run across it like the spokes of a bicycle wheel. When the light is relatively dim, the radial muscles contract and the circular muscles relax and the pupil is pulled open

wide (it **dilates**). When the pupil is dilated, lots of light can get into the eye that enable us to see even in relatively low light conditions. In bright light, however, the circular muscles of the iris contract and the radial muscles relax, which makes the pupil very small (it **constricts**). This reduces the amount of light that goes into the eye so that the delicate light-sensitive cells are not damaged by too much bright light (see Figure 5.28).

The innermost layer, the **retina,** extends only to the ciliary body anteriorly. It receives visual stimuli and sends it to the brain. The retina consists of numerous layers of nerve cells, including the cells commonly called **rods** and **cones**. These specialized nerve cells are often referred to as "**photoreceptors**" because they are responsive to light. The rods are highly sensitive to light, regulate black and white vision, and function in dim light. The cones function in bright light and are sensitive to color. The **optic disc** is a cream-colored, circular area located on the retina toward the medial or nasal side of the eye. It is where the optic nerve enters the eyeball. The optic disc can be seen with

Keyword

Retina a light-sensitive tissue lining the inner surface of the eye

Rods cells in the retina that perceive light and movement and work well in dim light

Cones cells in the retina that perceive light and movement and only work in bright light

Activity 5.25

You can observe the way the size of the pupil changes in response to light in two simple ways – either work with a partner or use a mirror and observe your own eyes.

1. Observe the size of your partner's/your own pupils in the normal working light of your classroom.

2. Cover your eyes with your hands, or with a piece of cloth, for a minute or so – but keep your eyes open normally (you can blink).

3. Remove the cover from the eyes and observe the pupils closely. Note down what you see both immediately after removing the cover and as the eye adjusts to normal light levels (diagrams may help).

4. Now increase the light intensity – move outside into the sunlight or just move nearer to the windows. Again watch and record what happens to the pupils in the brighter light – and when you return to normal classroom levels of light.

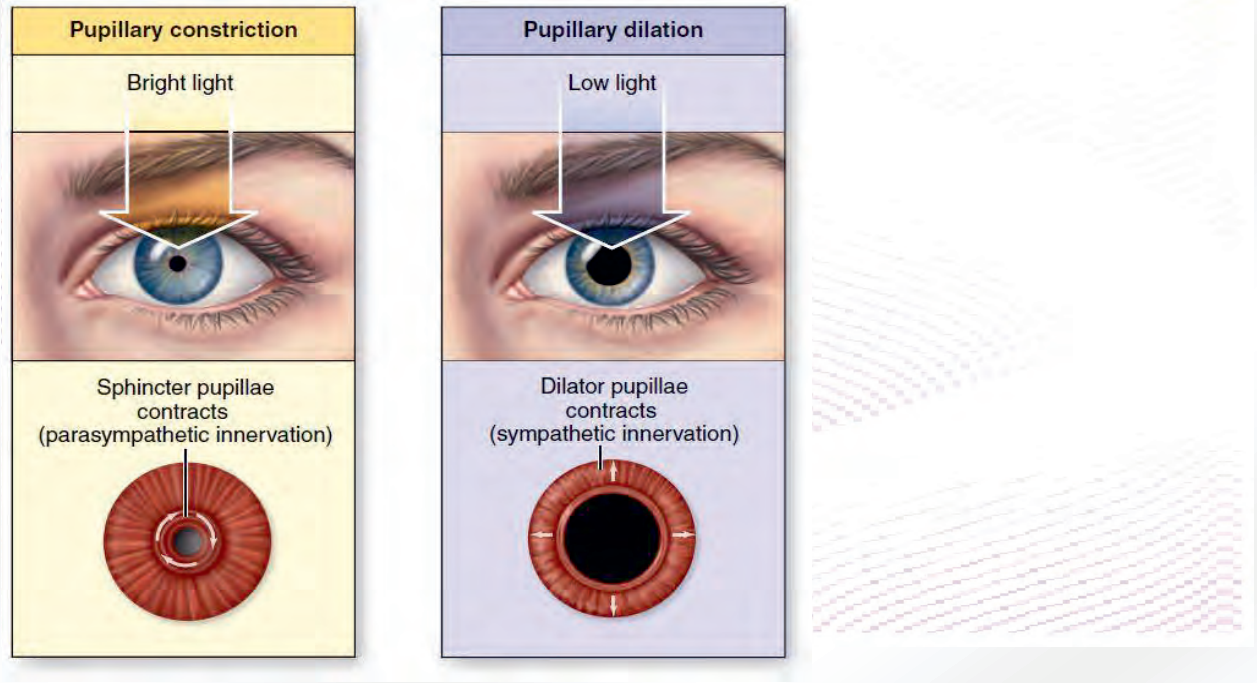


Figure 5.28 Figure showing the constriction and relaxation of pupil

When an image is produced on the retina, the light-sensitive cells are stimulated. They send impulses to the brain along affector (sensory) neurons in the optic nerve. When the brain

receives these messages it interprets the information and enable the person to see. At the point where your optic nerve leaves the eye there is no retina but there is a **blind spot**.

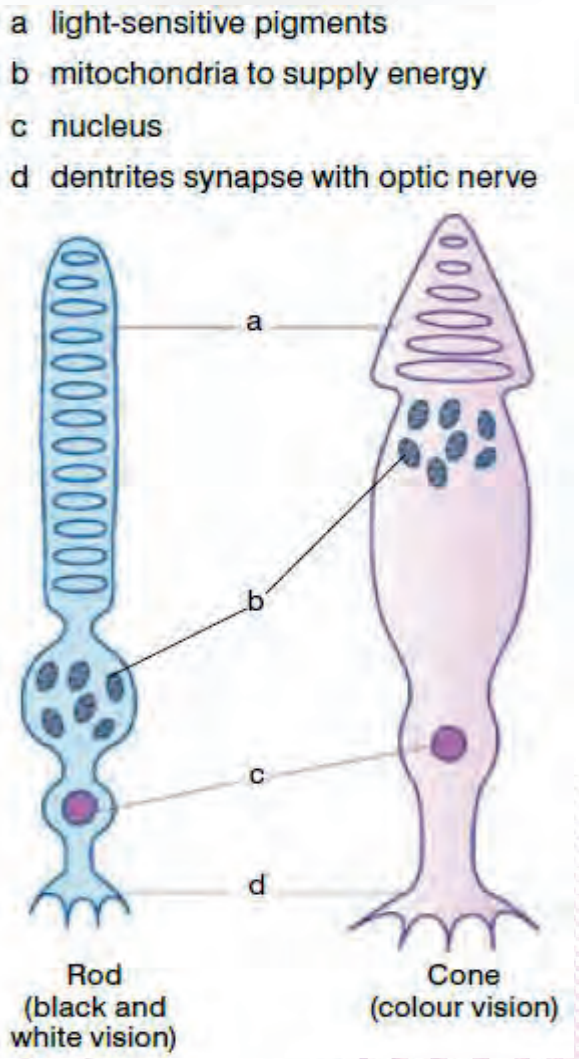


Figure 5.29 The rods and the cones in the retina of human eye

Activity 5.26

1. Look at the picture below. Hold the book so it is 30 cm from your eyes, with the pictures below level with your eyes.

2. Close your left eye completely and focus on the person with your right eye.

3. SLOW LY bring the book towards your face, keeping your right eye focused on the person all the time.

4. What happens to the lion as you move the picture towards you? How do you explain what happens?

5. What happens if you repeat the investigation with both eyes open? Explain any differences you observe.

A retinal depression known as the **fovea centralis** (see Figure 5.29) is located adjacent to the optic disc (blind spot) in the temporal section of the fundus. This area is surrounded by the macula, which appears darker than the rest of the fundus. The fovea centralis and macular area are highly concentrated with cones and form the area of highest visual resolution and color vision. The eyeball contains several chambers that serve to

maintain structure, protect against injury, and transmit light rays. The **anterior chamber** is located between the cornea and iris, and the **posterior chamber** is the area between the iris and the lens (see Figure 5.29). These chambers are filled with aqueous humour, a clear liquid substance produced by the ciliary body. **Aqueous humour** helps to cleanse and nourish the cornea and lens as well as maintain intraocular pressure. The aqueous humour filters out of the eye from the posterior to the anterior chamber then into the *canal of Schlemm* through a filtering site called the *trabecular meshwork.*

Another chamber, the **vitreous chamber,** is located in the area behind the lens to the retina. It is the largest of the chambers and is filled with a clear and gelatinous vitreous humour.

Activity 5.27

Discuss in group and reflect your answer for the following questions.

1. What retina is and how it works.

2. What is color blindness? What do you think is the cause of color blindness?

Activity 5.28 Bending light rays

For this investigation you will need to raid the stores of the physics department.

You will need:

• Light boxes or small, bright light sources

• Lenses

Method

Switch on the light box and observe the light emerging. Try putting the different lenses in the light and make drawings to show how they affect the light rays.

Activity 5. 29 studying refraction

This is a very simple exercise to remind you of the way light can be refracted as it changes medium.

You will need:

• A beaker or glass of water

• A glass rod, ruler or your finger

Method

Look carefully at your rod, ruler or finger. Then dip it into the water in the beaker or glass and observe what appears to happen. Any changes you see are due to the rays of light bending as they pass from water to air.

5.2.5. The Ear

The structure, function, and defects of the ear

Ears are specialized organs for hearing. They are also concerned with the balance and position of the body. The ear is divided into three regions: **the outer ear, middle ear and inner ear**. The outer ear consists of a flap called a **pinna (auricle)**. Leading from the pinna is a tube, the **ear canal**. In a human being, this is about 2 cm long. The pinna helps to trap and funnel sound into the ear. This is particularly important in animals, which can move the pinna to pick up sounds with longer ears than humans. At the end of the ear canal that closes the tube is a sheet of very thin membrane called the **eardrum** or **tympanum**

Activity 5.30

Search from different books and other sources about the different types of human eye defects and how these defects can be resolved? Report your finding to the class.

Activity 5. 31

Make a small group and read from different sources about **cataract** and **glaucoma**. Then, interview health professionals in your locality to obtain additional information about how serious the problem is in your area. Write a report of your finding and present to the class**.**

Figure 5.30 The pinna of human ear

Keywords

Outer ear the part of the ear visible externally

Middle ear the main cavity of the ear, between the eardrum and the inner ear

Inner ear a complex system of interconnecting cavities, concerned with hearing and equilibrium

Eardrum the membrane in the ear that vibrates to sound

The pinna, ear canal and the eardrum form the outer part of the human ear. At the entrance of the ear canal are a number of small hairs. These filter out dust particles from the air entering the ear canal. The cells lining the ear canal produce waxy material which traps dust and germs, and lubricates the eardrum.

Behind the eardrum is a cavity filled with air. This cavity contains three tiny bones and forms the middle ear. The three tiny bones– called the **malleus** (**hammer**), the **incus** (**anvil**) and the **stapes** (**stirrup**) (see Figure 5.31) are the smallest bones in the human body. They form joints with one another, with the malleus attached to the eardrum and the stapes to the oval window. The cavity of the middle ear is connected to the throat by a tube called the Eustachian tube. This is usually closed but when the pressure in the middle ear increases, the tube opens until the air pressure in the middle ear is equal to that in the throat and therefore to the atmosphere. At one end of the middle ear, opposite to the eardrum, there are two openings: one of them is oval in shape and hence it is called the oval window. The other is round and is called the round window. The openings are covered by very thin membranes.

The inner ear consists of a cavity filled with a fluid, two sac-like structures called the sacculus and utriculus, three semicircular canals and a coiled tube called the cochlea. The sacculus, utriculus, semicircular canals and the cochlea are filled with a liquid. A cross section of the cochlea reveals that it is made up of three tubes in one (Figure 5.31). The floor of the middle tube is lined with sensory cells linked to affector neurons. These nerve fibers join to form the auditory nerve which leads to the brain.

Keywords

Malleus **(hammer)** *is the ossicle attached to the eardrum*

Incus (anvil) is the ossicle between the malleus and the stapes

Stapes (stirrup) is the stirrup-shaped ossicle that transmits sound from the incus to the cochlea

Ampullae is the e swelling at the base of each semicircular canal, containing sensory cells which detect movement of the fluid within the canals

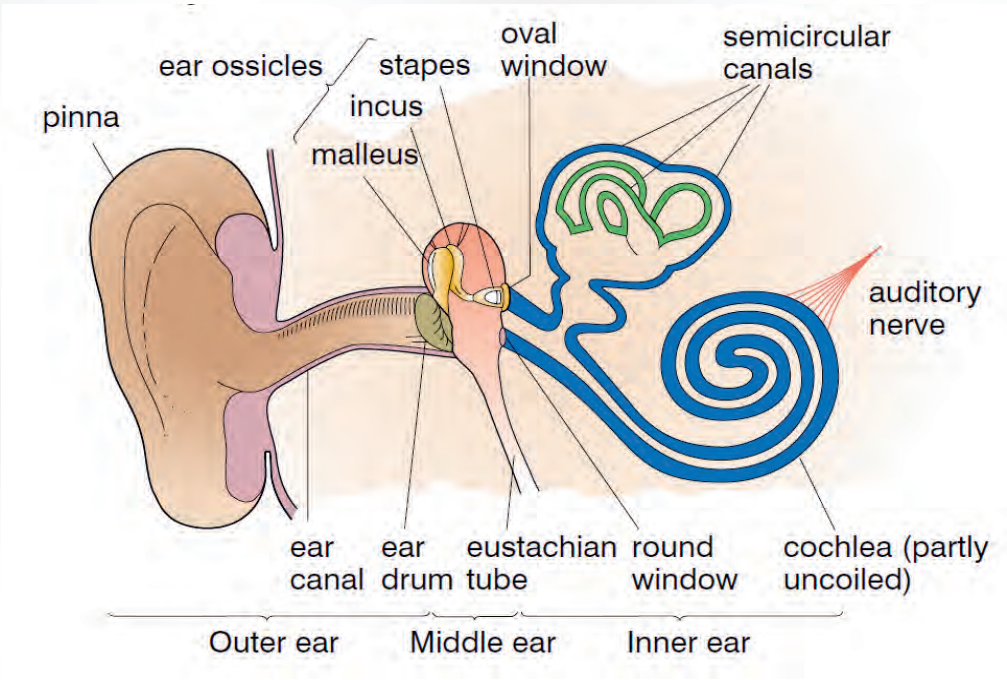


Figure 5.31 Structure of human ear

Activity 5.32

Examination of the human ear

You will need:

- A model of a mammalian ear.

Method

1. Examine the external part of the ear of your classmate. Note its shape and texture.
2. Examine a model of a mammalian ear. Note the shape of the various parts (external and internal parts).
3. Make a drawing of the model.
4. Compare your diagram with that in Figure 5.31

The mechanism of hearing

The pinna collects sound waves and directs them to the eardrum through the ear canal.

When sound waves hit the eardrum, it vibrates. This magnifies the vibrations which are then transmitted through the ear ossicles (the small bones) to the oval window. The ear ossicles also amplify the vibrations (make them bigger).

The vibrations of the stapes make the membrane at the Vibrations of the fluid cause the hair-like sensory cells to move. These movements in turn cause production of nerve impulses in the affector nerve fibres. These impulses are transmitted to the brain for interpretation.

The human ear is sensitive to vibrations ranging from those of a very low note of about 20 vibrations per second, to a very high note of about 30 000 vibrations per second. High notes are detected in the first part of the cochlea and low notes are recorded in the last part of the cochlea.

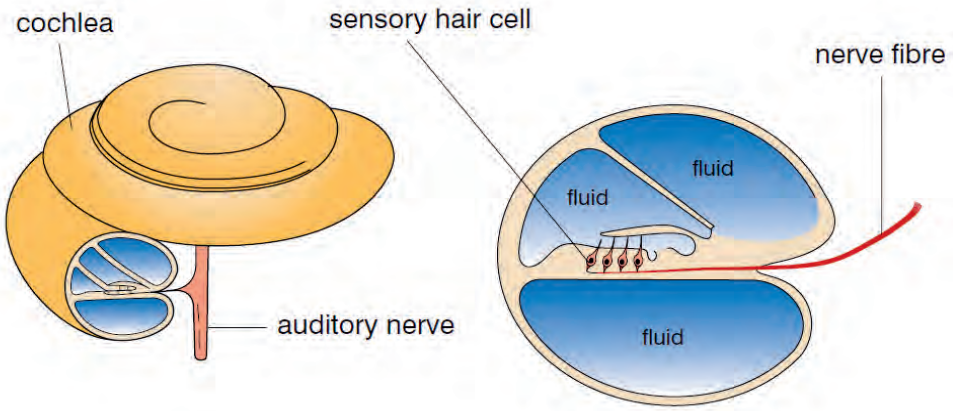


Figure 5.32 The cross section of a cochlea

The sense of balance and movement

The semicircular canals in the inner ear are concerned with the detection of motion. The swellings on each of the semicircular canals (the ampullae) contain sensory cells attached to the sensory nerve endings. The **sensory cells** have hairs which are enclosed in a core of jelly substance called a cupula (Figure 5.32). Whenever the body or the head moves, the semicircular canals move with the head.

The fluid in the semicircular canals also starts to move but it lags behind in its motion, so it apparently moves in the opposite direction. The moving fluid causes the cupula to tilt, thus pressing the hairs of the sensory cells. The pressing of the sensory hairs creates nerve impulses in the sensory nerve endings. The nerve impulses are transmitted to the brain. The brain then interprets the direction and speed of the motion of the body or head.

The semicircular canals are all at right angles to each other, so each one is sensitive to movement in a different plane. One canal responds to nodding, one to shaking and one to head tilting. Fast spinning of the body followed by instant interruption causes dizziness. This is because the fluid in the semicircular canals keeps on moving after the spinning has stopped

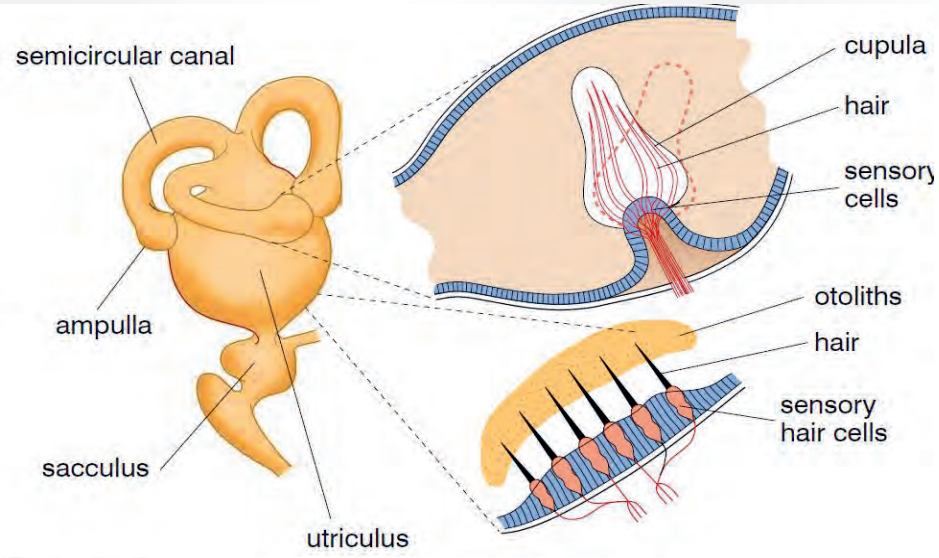


Figure 5.33 The balance organs of the ear

Activity 5.33

Organize yourself to form a small group and search for different materials (books, internet) to understand the possible causes of deafness (hearing loss). The, reflect your finding to the class.

5.2.6. The endocrine system

By the end of this section, the learner will be able to:

- Define glands and hormones
- Identify the location and function of principal endocrine glands, including pituitary thyroid, parathyroid, adrenal, and pancreas
- Compare and contrast exocrine and endocrine glands
- Explain the function of glands and hormones

The endocrine glands

The endocrine system is a diverse collection of cells, tissues, and organs including specialized endocrine glands that produce and secrete hormones, chemical messengers that regulate many physiological processes. The endocrine system has two main components: glands and hormones. An **endocrine gland** consists of cells that produce and secrete hormones into the bloodstream, which carries the secretions throughout the body. A hormone is a regulatory chemical that is secreted into the blood by an endocrine gland or an organ of the body exhibiting an endocrine function. The blood carries the hormone to every cell in the body, but only the target cells for a given hormone can respond to it

The endocrine system interacts with the nervous system to coordinate and integrate body activities by means of **hormones**. Endocrine tissues and organs secrete **hormone** into the body fluids (mainly blood and lymph) directly using diffusion. **Hormones** act as chemical messages that are produced in one part of the body, but they have an effect somewhere entirely different. Glands are structures which produce hormones and other useful substances.

Self- questioning

What are the communication methods used by the endocrine system?

What are the functions of hormones?

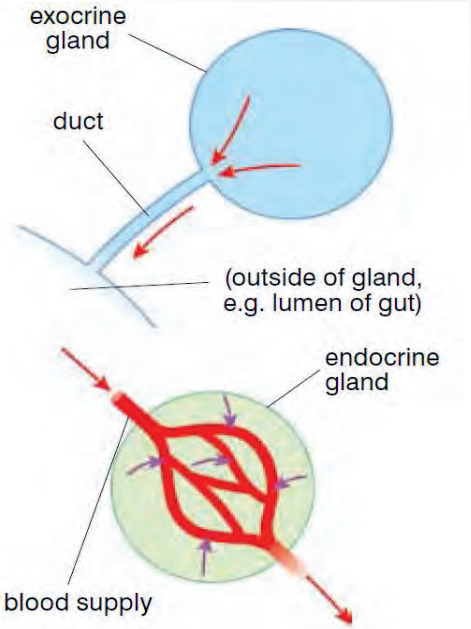
What are the difference between exocrine and endocrine glands?

The **endocrine glands** that produce hormones have no ducts, so they are sometimes known as ductless glands. They secrete hormones directly into the blood, and the chemicals are carried from glands to the body parts throughout the bloodstream.

Most hormones only affect certain tissues or organs – their target organ and the hormone is picked up from the blood by receptors in the cell membranes. They can act very rapidly, but often their effects are slower and longer lasting than the results of nervous control.

In human body there are also glands that produce other substances (not hormones)

released through ducts. These glands are known as **exocrine glands**. **Exocrine glands** are glands which release their cellular secretions through a duct or tube. These include certain sweat glands, salivary and pancreatic glands, sebaceous, and mammary glands. These glands are not considered as a part of the endocrine system.



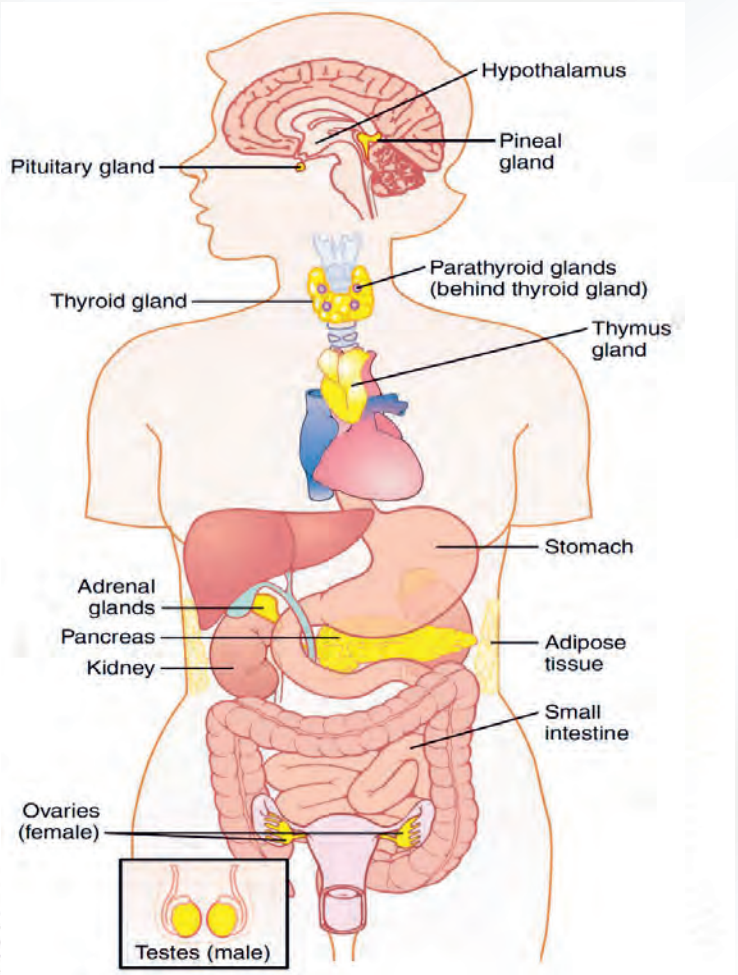


Figure 5.34 Figure showing differences between exocrine gland and endocrine gland, and their secretions.

The major endocrine glands

Figure 5.35 The location of endocrine glands in human body.

Hypothalamus

The hypothalamus connects the nervous system to the endocrine system. It receives and processes signals from other brain regions and pathways where the hypothalamus translates them into hormones that are the chemical messengers of the endocrine system. These hormones flow to the pituitary gland, by the infundibulum. Some hormones are stored in the pituitary stores for later release; others spur it to secrete its own hormones. The hormones are released

by the pituitary gland whereas the hypothalamus controls the other endocrine glands and regulate all the major internal functions.

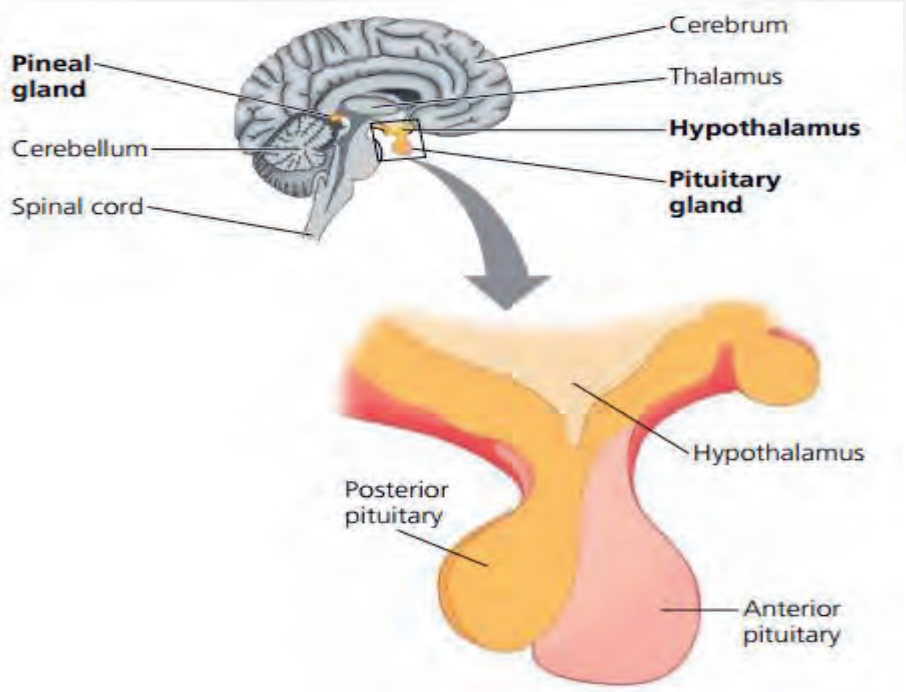


Figure 5.36 Hypothalamus and pituitary gland

The pituitary gland

The pituitary gland, found in the brain and about the size of a pea, is sometimes described as the controller of the endocrine orchestra (master gland). The hormones made in this tiny gland control the secretion of many other hormones. Because of its position in the brain, it is also involved in the co-ordination between the nervous and hormonal systems of control. It is divided into anterior lobe (adenohypophysis) and posterior lobe (neurohypophysis). Anterior lobe is about 3 times larger than the posterior lobe.

A). Anterior pituitary gland

The anterior pituitary makes peptide hormones whose secretion is regulated by the hypothalamus. Most hypothalamic hormones that target cells of the anterior pituitary are releasing hormones, which encourage secretion of hormones. The hypothalamus also makes inhibiting hormones that slow secretion of anterior pituitary hormones. Six major peptide hormones plus several other hormones of lesser importance are secreted by the anterior pituitary. The hormones of the anterior pituitary play major roles in the

control of metabolic functions throughout the body.

• **Growth hormone (GH)**:- promotes growth of the entire body by affecting protein formation, cell multiplication, and cell differentiation. Growth hormone is produced by the anterior pituitary. The quantity is greatest during childhood and adolescence, when most body growth is occurring. If too little GH is produced during childhood, the individual has pituitary **dwarfism**, characterized by perfect proportions but small stature. If too much GH is secreted, gigantism may result. Individuals with gigantism often have additional health problems, primarily because GH has a secondary effect on the blood sugar level, promoting an illness called diabetes mellitus. On occasion, GH is overproduced in the adult and a condition called acromegaly results. Long bone growth is no longer possible in adults, so only the feet, hands, and face (particularly the chin, nose, and eyebrow ridges) can respond, and these portions of the body become overly large.

• **Adrenocorticotropin (corticotropin):-** controls the secretion of some of the adrenocortical hormones, which affect metabolism of glucose, proteins, and fats.

• **Thyroid-stimulating hormone (thyrotropin):-** controls the secretion rate of thyroxine and triiodothyronine by the thyroid gland, and these hormones control the rates of most intracellular chemical reactions in the body.

• **Prolactin**:-is an anterior pituitary hormone that stimulates milk production in a woman's breasts after she gives birth. Shortly after a baby begins to suckle, prolactin binds to receptors on the milk-producing cells of the breasts. The receptors' affinity for prolactin depends on whether the breast is full or empty, enabling the mother's body to adjust the rate of milk production according to the baby's needs.

• Two separate gonadotropic hormones, follicle stimulating hormone and luteinizing hormone, control growth of the ovaries and testes, as well as their hormonal and reproductive activities.

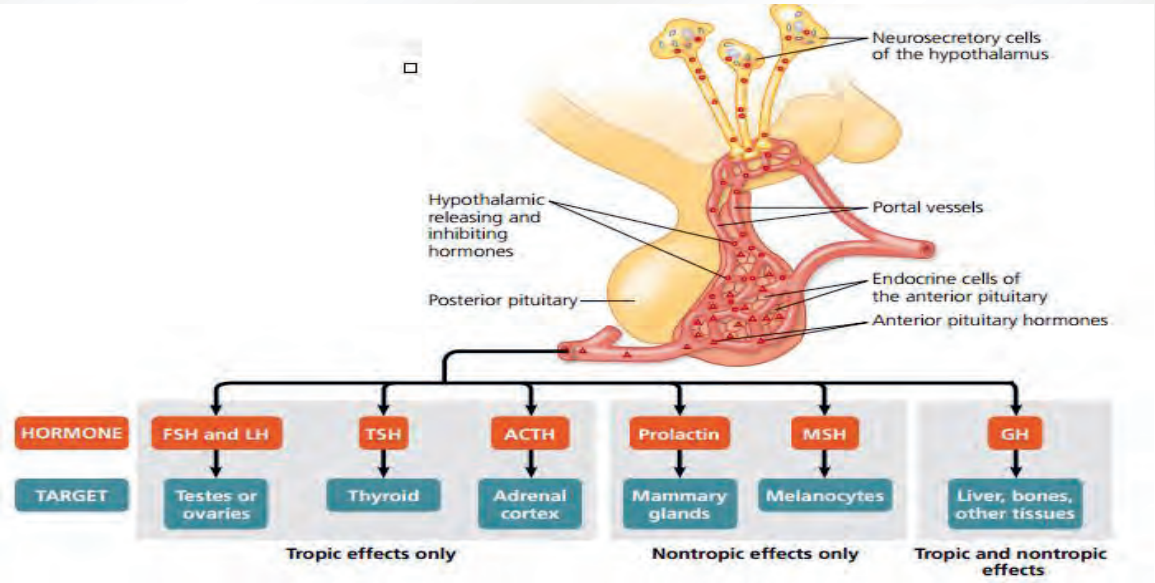


Figure 5.37 hormones secreted by anterior pituitary gland

B). Posterior pituitary gland

The posterior pituitary gland stores and releases hormones when needed that are actually produced by the hypothalamus. These two hormones of the posterior pituitary gland are actually produced by the hypothalamus and simply stored in the posterior pituitary until needed.

These hormones are:

• **Antidiuretic hormone** (also called vasopressin) controls the rate of water excretion into the urine, thus helping to control the concentration of water in the body fluids.
• **Oxytocin** helps express milk from the glands of the breast to the nipples during suckling and helps in the delivery of the baby at the end of gestation.

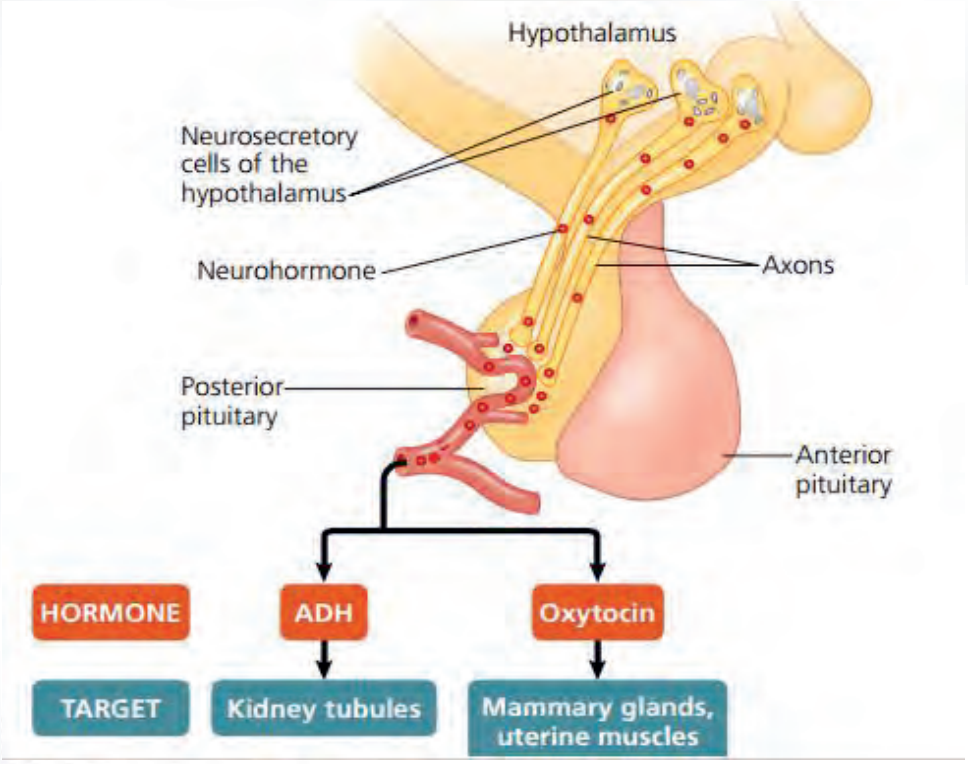


Figure 5.38 Hormones secreted by hypothalamus and stored in the posterior pituitary gland

Table 5.2 Hormones of the posterior pituitary gland

Hormone	Functions	Regulation of secretion
Antidiuretic hormone (ADH or vasopressin)	Increases water reabsorption by the kidney tubules (water returns to the blood) • Decreases sweating • Causes vasoconstriction (in large amounts)	Decreased water content in the body (alcohol inhibits secretion)
Oxytocin	Promotes contraction of myometrium of uterus (labor) • Promotes release of milk from mammary glands	Nerve impulses from hypothalamus, the result of stretching of cervix or stimulation of nipple Secretion from placenta at end of gestation—stimulus unknown

Activity 5.34

Search from different books and other sources about the different types of hormones produced (released) by the anterior pituitary and posterior pituitary gland. List down some of the problems associated with these hormones. Report your finding to the class.

Thyroid gland

The thyroid gland (Greek thyros ,"shield") is a small, butterfly-shaped gland located inferior to the larynx (voice box) and attached to the trachea. It is divided into two lateral lobes. Thyroid follicles utilize iodine and synthesize **thyroglobulin (TGB)** to be stored in the colloids. Upon stimulation of thyroid-stimulating hormone (TSH), TGB is converted into two hormones: **Triiodothyronine (T3)** and **Thyroxine (T4)** to promote normal metabolism. Hypo secretion causes hypothyroidism (goiter, cretinism and myxedema) and hyper secretion causes hyperthyroidism that result in Graves' disease. Thyroid gland also secretes Calcitonin to lower blood calcium and phosphate levels and regulate digestive hormones. Both hypo secretion and hyper secretion would affect the normal balances of calcium and phosphate.

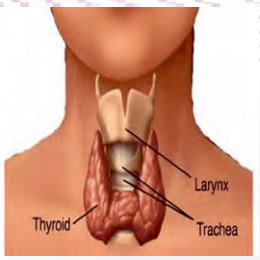


Figure 5.39 The thyroid gland

Activity 5.35

Search from different sources (books and the internet) and rea about:

1. The function of thyroxine
2. The current situation of goiter cases in Ethiopia?
3. Why women and children are more affected by iodine deficiency than men?
4. If there is a solution for most of the problems of iodine deficiency disorders? Reflect your findings to the class.

Figure 5. 40 A woman with goiter

Malfunction of the thyroid gland leads to specific disorders

Extreme hypothyroidism during infancy and childhood results in low metabolic rate and can lead to **cretinism,** a condition characterized by retarded mental and physical development. When diagnosed early enough, hypothyroidism can be treated with thyroid hormones, and cretinism can be prevented. An adult who feels like sleeping all the time, has little energy, and is mentally slow or confused may be suffering from hypothyroidism.

High levels of TSH stimulate the thyroid gland, whose cells enlarge in a futile attempt to manufacture more thyroxine. Without iodine, the thyroid gland keeps getting bigger and bigger - a condition known as a **goiter**. In most countries, goiter is prevented through the addition of iodine to table salt.

Parathyroid glands

Four oval-shaped glands are embedded in the posterior surface (back) of the thyroid gland. Each parathyroid gland is normally about the size of a grain of rice (about 3-5 millimeters in diameter and 30 - 60 milligrams in weight). Parathyroid glands release **parathyroid hormone (PTH)** which controls the calcium levels in the blood stream. Other areas of the body, especially the bones, kidneys and small intestine, respond to PTH by increasing the calcium levels in the blood. Calcium is very important for our bodies, especially for muscle and nerve function. Hypo secretion causes tetany, and hyper secretion causes osteitis fibrosa cystica.

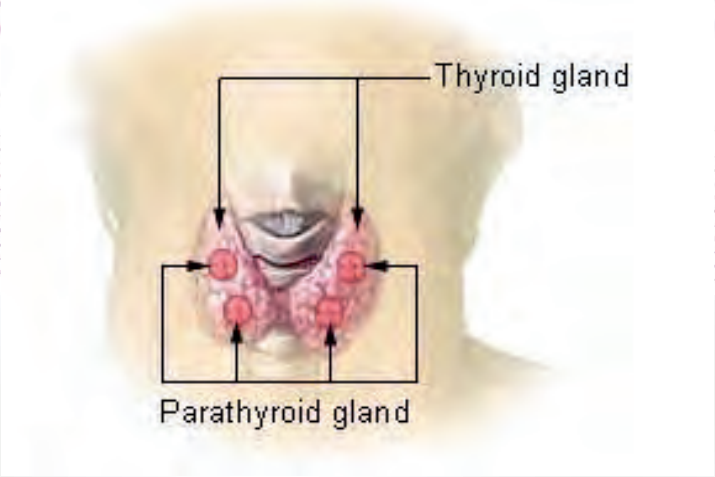


Figure 5.41 Parathyroid gland

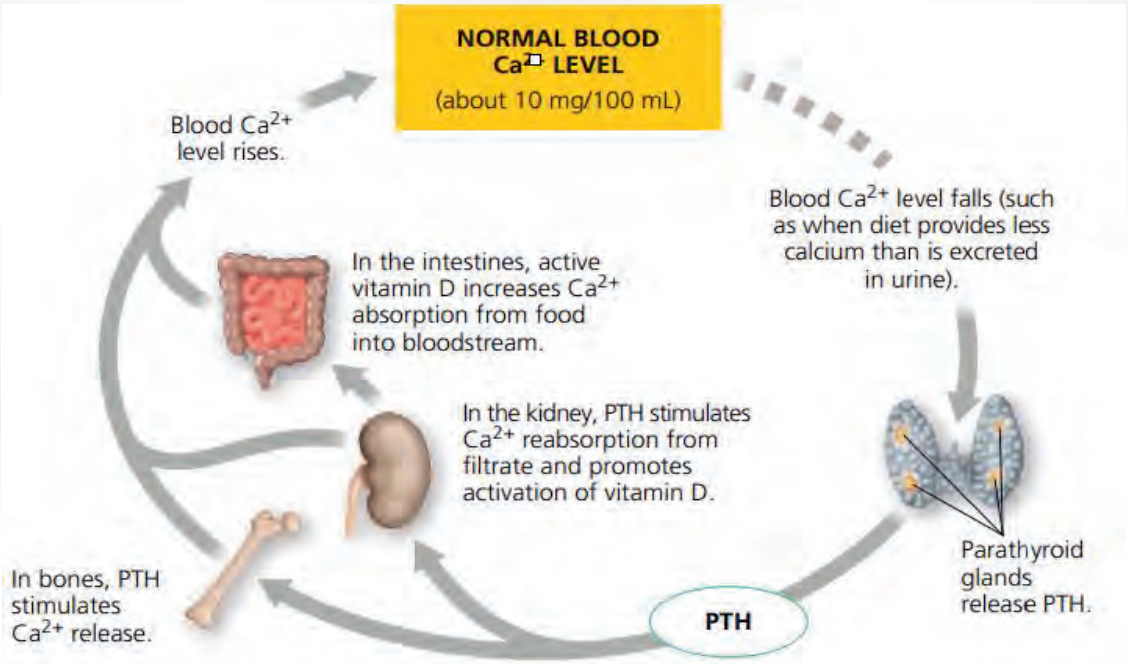


Figure. 5.42 The function of PTH in the regulation of blood calcium levels in mammals

Adrenal glands

Adrenal glands, also known as suprarenal glands, are small, triangular-shaped glands located on the top of both kidneys. Adrenal glands are composed of two parts. The cortex and the medulla-which are responsible for the production of different hormones.

A. Adrenal Cortex: is the outer portion of the adrenal gland which is attached to the superior surface of the kidney. It is divided into 3 regions, from outside to inside: **Zona glomerulosa**, **Zona fasciculate** and **Zona reticularis**. Adrenal cortex secretes over 30 steroid-based substances and several steroid hormones, all crucial for normal homeostasis.

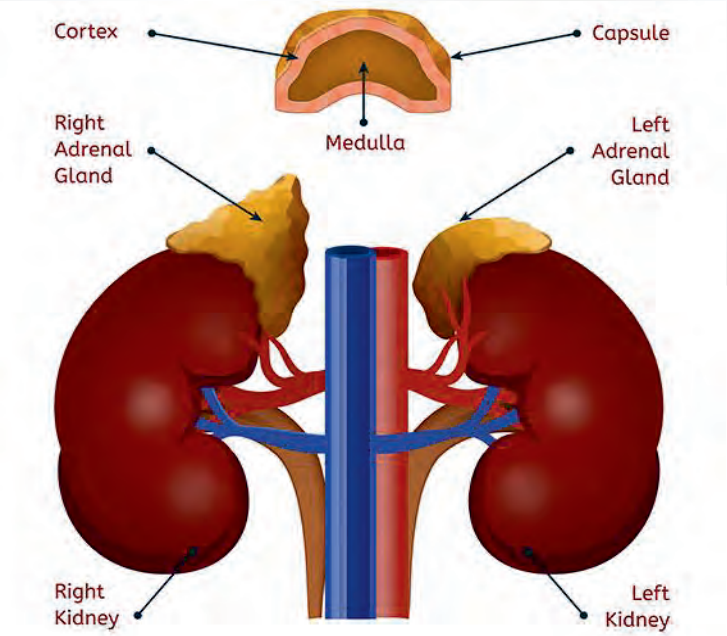


Figure 5.43 Adrenal glands

Activity 5.36

Search from different sources and read about the types of hormones secreted by Zona glomerulosa, Zona fasciculate and Zona reticularis and the role of each hormone. Also, find some of the problems associated with hyper secretion or hypo secretion of each hormone. Discuss your finding in the class.

B. Adrenal Medulla: This is the inner portion of the adrenal gland. It is made of modified nerve tissue that is under direct regulation of sympathetic nerves of the autonomic nervous system. Contains glandular cells called chromaffin cells which secrete 2 closely related hormones- **Epinephrine (adrenaline)** and **Norepinephrine (noradrenaline).**

Adrenaline is a well-known hormone produced by your adrenal glands. It is the hormone of 'fight or flight'. If you are stressed, angry, excited or frightened your adrenal glands will secrete lots of adrenalin. Carried rapidly round in your blood, adrenalin affects many different organs from the pupils of your eyes (it dilates them) to the *beating of your heart (it speeds it up).* Adrenalin basically prepares

Activity 5.37

Discuss in group the role of adrenaline.

your body for action, so that you can run fast to escape or fight successfully if you need to.

Table 5.3 Hormones of adrenal glands

Hormone	**Functions**	**Regulation of secretions**
Norepinephrine	Causes vasoconstriction in skin, viscera, and skeletal muscles	
Epinephrine/ Adrenalin	• Increases heart rate and force of contraction • Dilates bronchioles • Decreases peristalsis • Increases conversion of glycogen to glucose in the liver • Causes vasodilation in skeletal muscles • Causes vasoconstriction in skin and viscera • Increases use of fats for energy • Increases the rate of cell respiration	Sympathetic impulses from the hypothalamus in stress situations
Aldosterone	Increases reabsorption of Na ions by the kidneys to the blood • Increases excretion of K ions by the kidneys in urine	Low blood Na+ level Low blood volume or blood pressure High blood K level
Glucocorticoid hormones (cortisol (hydrocortisone), cortisone and corticosterone)	The glucocorticoid hormones have several effects: • they influence the metabolism of most body cells; • they promote glycogen storage in the liver; • during fasting they stimulate the generation of glucose; • they increase blood glucose levels; • they are involved in providing resistance to	ACTH (anterior pituitary) during physiological stress

Hormone	Functions	Regulation of secretions
	stressors; • they decrease the permeability of vascular endothelium; • they promote the repair of damaged tissues by promoting the breakdown of stored protein to create amino acids; • they suppress the immune system; • they suppress inflammatory processes.	

Pancreas

Pancreas is a small pink organ found below the stomach. It is both exocrine and endocrine in physiology. In its exocrine aspect, 99% of its mass is composed of cells called acini which secrete digestive enzymes and fluids into the small intestine through the pancreatic ducts. In its endocrine aspect, 1% of its mass is little groups of cells called **islets of langerhans (or pancreatic islets)** secrete hormones to regulate blood glucose level. In each pancreatic islet, **alpha cells (α cells)** secrete **glucagons** to raise blood glucose level, **beta cells (β cells)** secrete **insulin** to lower blood glucose level. Hypo secretion causes **diabetes mellitus** and hyper secretion causes **hyperinsulinism**. **Delta cells (δ cells)** secrete **somatostation** or **growth hormone inhibiting hormone (GHIH)** which helps regulate carbohydrate metabolism by inhibiting the secretion of glucagons.

Keywords

Pancreas is a gland that produces digestive enzymes and manufactures hormones, including insulin and glucagon

Insulin is a hormone that lowers the blood glucose level

Glucagon is a hormone that raises the blood glucose level

Glycogen is a form of carbohydrate stored primarily in the liver and

Activity 5.38

What are the basic differences between Type I and Type II diabetes? Read from different sources and discuss in the class.

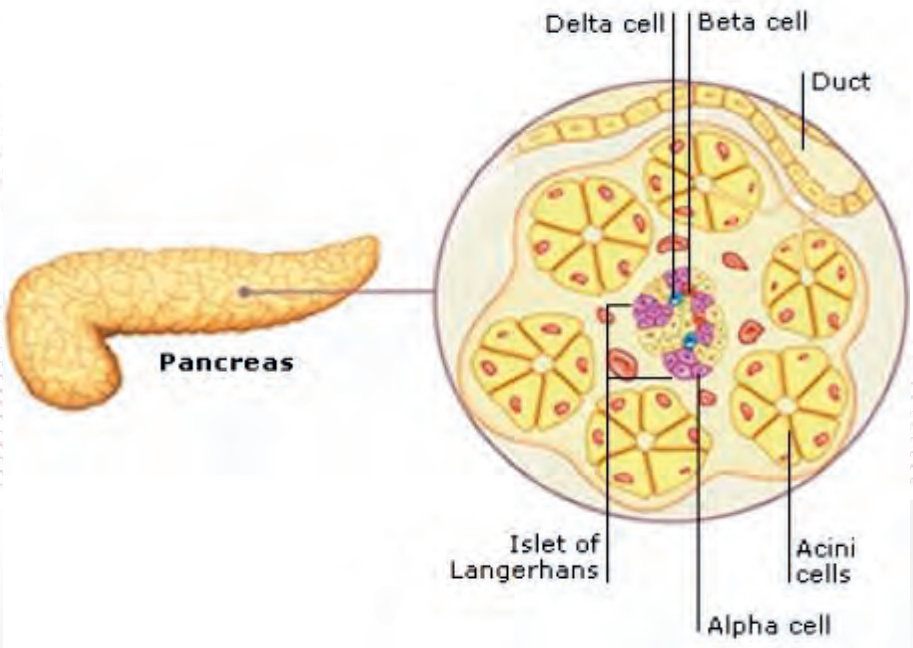


Figure 5.44 Islets of Langerhans in pancreas

A). Insulin

Insulin is well known for its effect in reducing the blood glucose levels. It does this by:

A. It increases conversion of glucose into glycogen and deposition of it in liver and muscles.

B. It increases the rate of oxidation of glucose in the tissues.

C. It increases the rate of conversion of glucose into fat and facilitates its storage in adipose tissue.

D. It also regulates the rate at which amino acids are catabolized into water and CO2.

Types of diabetes

Type 1 diabetes:- in which the pancreas can no longer manufacture insulin, may be the result of an autoimmune reaction that specifically targets the beta cells. Antibodies to specific components of beta cells have been found in the pancreases of Type 1 diabetes patients. Research suggests that hereditary factors may also play a role in the onset of this disease.

Type II diabetes:- is due to insulin resistance. The insulin target cells do not respond normally to the circulating insulin. This may result due to obesity, over-eating and lack of exercise. The insulin hypo responsiveness can be corrected if the person reduces his or her caloric intake. Thus dietary control without any other therapy is frequently sufficient to eliminate the elevated blood glucose level of type II diabetics. An exercise programmed is also useful, since it will help to increase the number of insulin receptors.

B). Glucagon

Glucagon has an important role in maintaining normal blood glucose levels, especially as the brain and neurons can only use glucose as a fuel. Glucagon has the opposite effect on blood glucose levels to insulin it:

- stimulates the breakdown of glycogen stored in the liver;
- activates hepatic gluconeogenesis (the creation of glucose from substrates such as amino acids);
- has a minor effect enhancing triglyceride breakdown in adipose tissue – providing fatty acid fuel for most cells, and thus conserving glucose for the brain and neurons.

The production and secretion of glucagon are stimulated in response to a reduction in blood glucose concentrations and elevated blood levels of amino acids (for instance,

after a protein-rich meal). It has also been found that glucagon levels in the blood rise in response to exercise, but it is unclear whether this is a response to the exercise itself or a response to the reduced blood glucose levels that exercise creates.

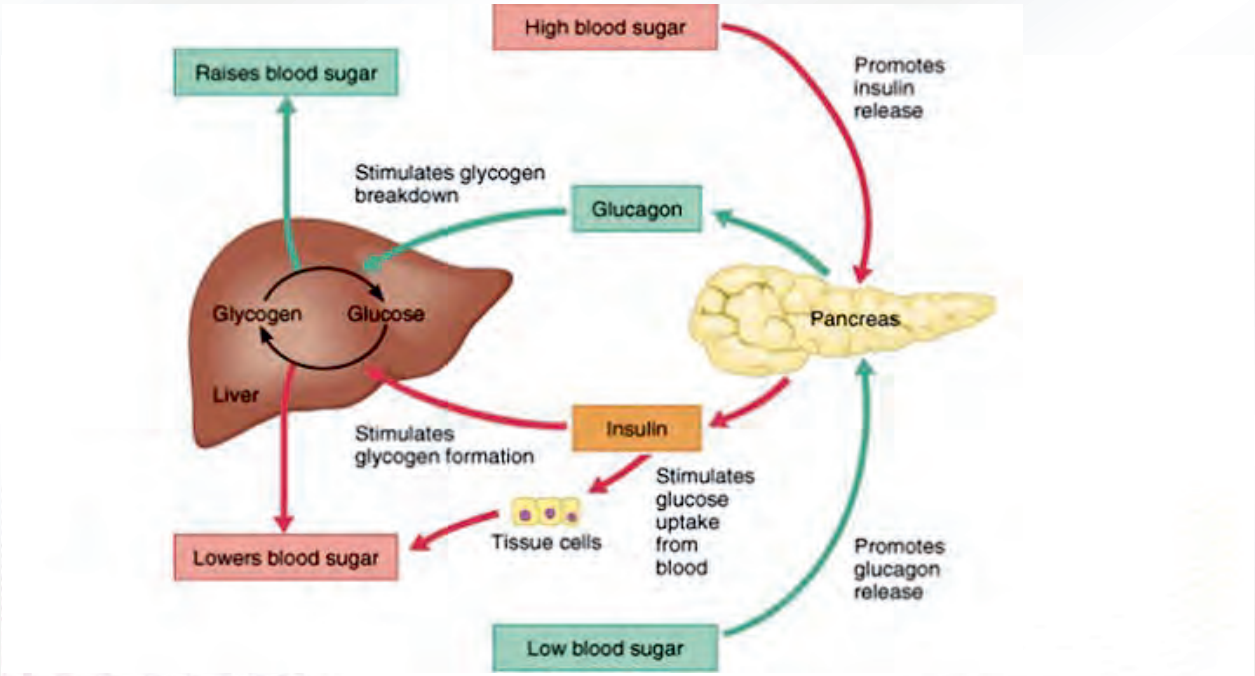


Figure 5. 45 Regulation of blood sugar level

The gonads: Ovary and testis

Self-questioning

Can you explain what causes secondary sexual characteristic in boys and girls?

The **gonads** are the endocrine glands which produce some of the sex hormones. These are the testes in boys and ovaries in girls who become active at the time of puberty. When the big physical changes take place, boys and girls look very different which the body takes its adult form. The changes come about in response to hormones released by the brain and by the gonads themselves.

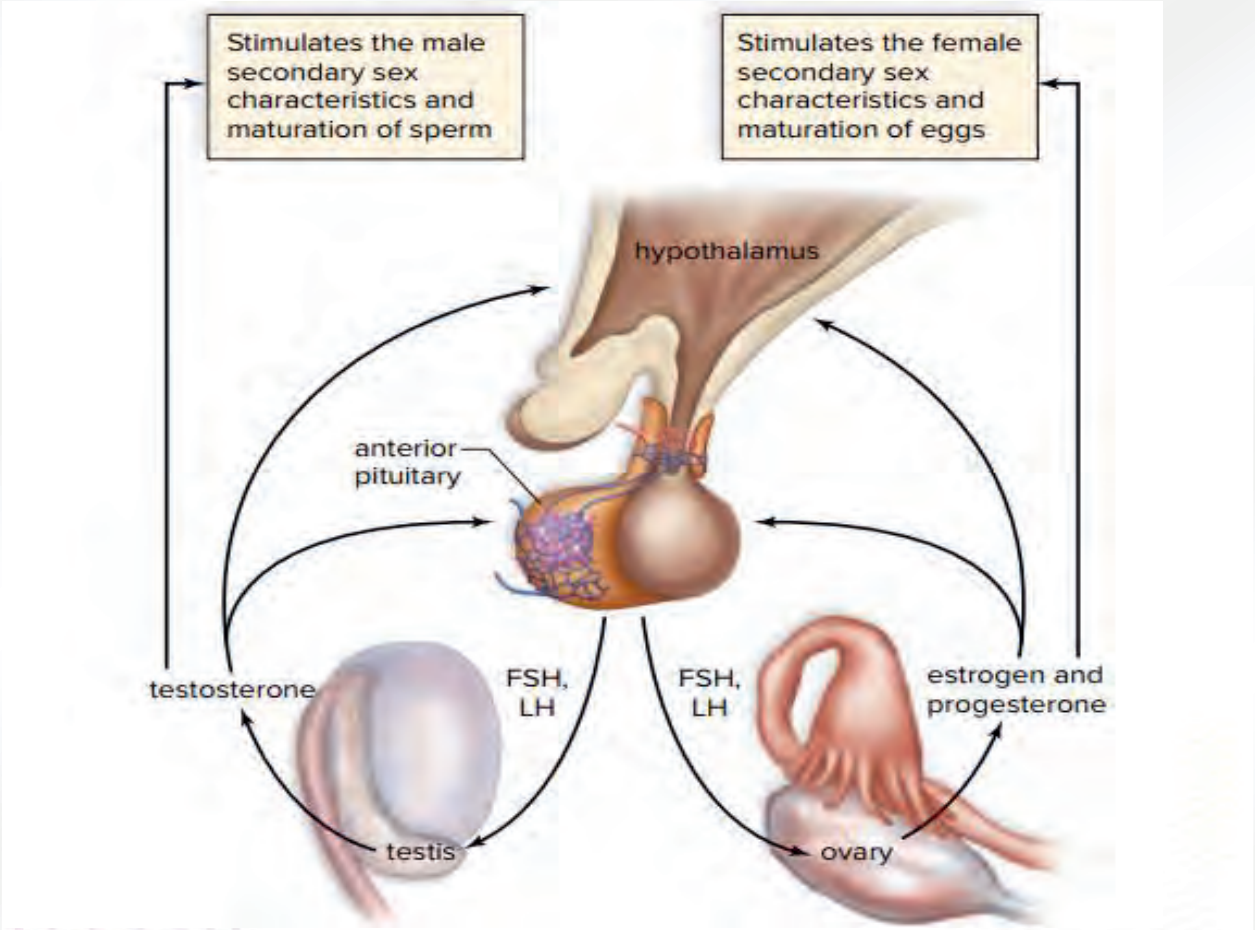


Figure 5.46 The hormones produced by the testes and the ovaries

The role of the ovaries

The female gonads are the ovaries, two walnut-sized organs found low in the abdomen in either side of the uterus. Ovaries produce eggs and hormones. Girls often go into puberty slightly earlier than boys. Between the ages of 8–14 most girls begin the changes which will take their bodies into sexual maturity. As with boys, the time and speed of puberty varies greatly from one person to another. Although it is different for everyone, and everyone ends up a slightly different shape and size – the basic changes which take place are the same.

Puberty in girls is controlled by hormones

Activity 5.39

Search from different books and other sources about the hormones that have effect on the female reproductive system, control of the menstrual cycle and female fertility. Present your work to the class.

from the pituitary gland in the brain and from the gonads themselves – in this case the ovaries. Follicle stimulating hormone (FSH) from the brain stimulates the ovaries to become active and start producing hormones. The hormones produced by the ovaries are the steroids estrogen and progesterone.

Estrogen

Estrogen is secreted by the follicle cells of the ovary; secretion is stimulated by **FSH** from the anterior pituitary gland. Estrogen promotes the maturation of the ovum in the ovarian follicle and stimulates the growth of blood vessels in the endometrium (lining) of the uterus in preparation for a possible fertilized egg.

The **secondary sex characteristics** in women also develop in response to estrogen. These include growth of the duct system of the mammary glands, growth of the uterus, and the deposition of fat subcutaneously in the hips and thighs.

Progesterone

When a mature ovarian follicle releases an ovum, the follicle becomes the **corpus luteum** and begins to secrete **progesterone** in addition to estrogen. This is stimulated by **LH** from the anterior pituitary gland. Progesterone promotes the storage of glycogen and the further growth of blood vessels in the endometrium, which thus becomes a potential placenta. The secretory cells of the mammary glands also develop under the influence of progesterone.

The role of the testes

Testis is the male sex organ that also serves as an endocrine gland. It contains interstital cells (or leydig's cells) that secrete **testosterone** to develop male secondary sexual characteristics. Puberty in boys usually begins somewhere between the ages of 9 and 15 years old. It may happen very rapidly, or it may take place much more slowly over a number of years. Two people do not experience puberty in exactly the same way.

The chemical changes which trigger puberty are unseen, which is another important example of hormonal co-ordination and control. The pituitary gland in the brain starts to produce increasing amounts of **FSH**. This in turn stimulates the male gonads or testes to begin developing and producing the male sex hormone **testosterone**. The rising levels of testosterone trigger the many changes which affect the body during puberty, causing the development of the secondary sexual characteristics.

Source	Ovaries		Testes
Hormone	Progesterone	Estrogen	Testosterone
Type	Steroid	Steroid	Steroid
Target cells	Uterine lining, hypothalamus, pituitary, other tissues	Uterine lining, hypothalamus, pituitary, other tissues	Sperm-producing cells, hypothalamus, pituitary, other tissues
Major responses	Regulates menstrual cycle, prepares body for pregnancy	Regulates menstrual cycle, maintains secondary sex characteristics in females	Promotes sperm development, maintains secondary sex characteristics in males

Figure 5.47 hormones secreted by ovaries and testes

Pineal gland

Pineal gland is pine cone shaped located deep in the cerebrum. It secrets **melatonin** to regulate **circadian rhythms** which are necessary to keep track of day or night cycles, sleep/wake rhythm, menstrual and ovarian cycles.

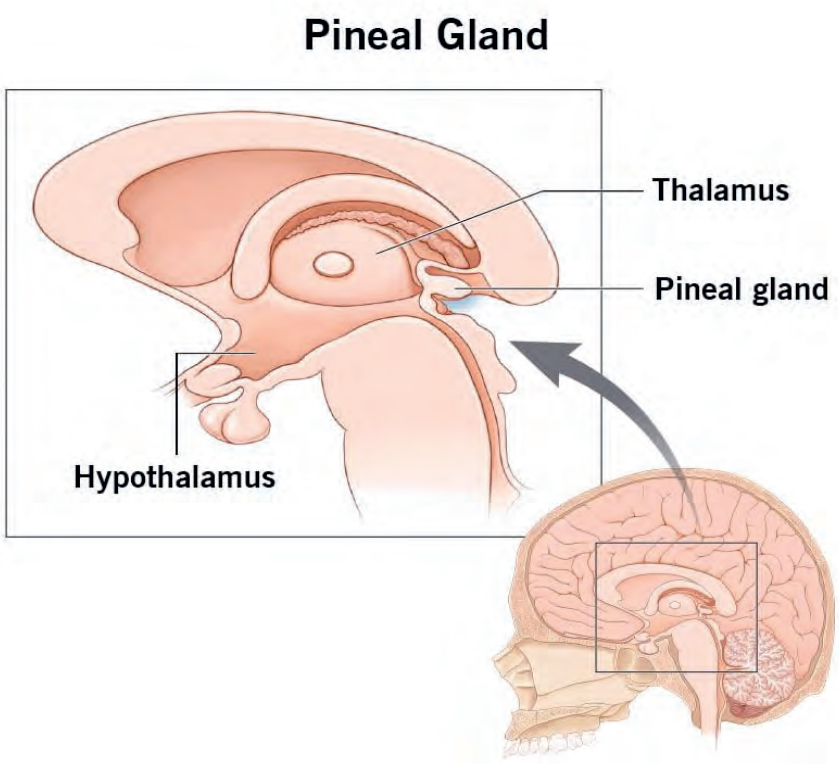


Figure 5.48 Pineal gland

Circadian Rhythms

Melatonin secretion by the pineal gland of the brain plays a pivotal role in the control of circadian rhythms

- Circadian rhythms are the body's physiological responses to the 24 hour day-night cycle
- Circadian rhythms are driven by an internal (endogenous) circadian clock, although they can be modulated by external factors

Melatonin is the hormone responsible for synchronising circadian rhythms and regulates the body's sleep schedule

- Melatonin secretion is suppressed by bright light (principally blue wavelengths) and hence levels increase during the night
- Over a prolonged period, melatonin secretion becomes entrained to anticipate the onset of darkness and the approach of day
- Melatonin functions to promote activity in nocturnal animals and conversely promotes sleep in diurnal animals (like humans)
- During sleep, necessary physiological changes occur in body temperature, brain wave activity and hormonal production

Melatonin levels naturally decrease with age, leading to changes in sleeping patterns in the elderly .

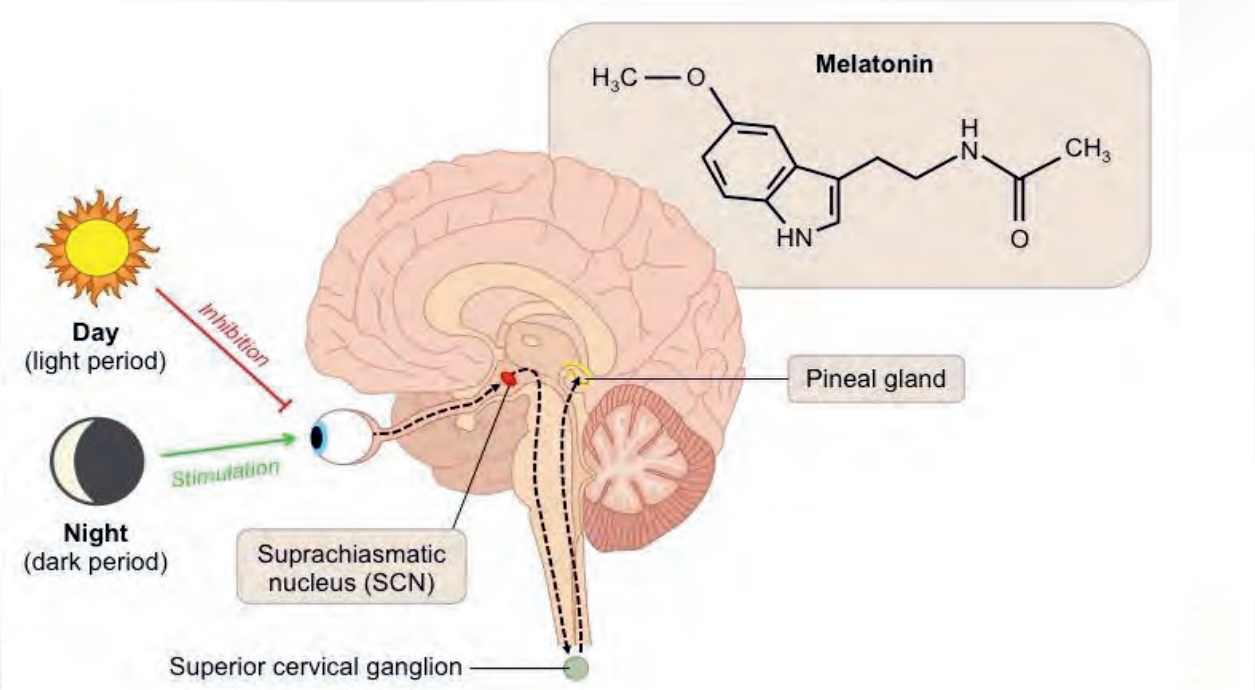


Figure 5.49 Pineal gland

Thymus gland

A thymus gland is a diminishing gland (over time) located between the lungs. It secretes a group of hormones, such as **thymosin**, to affect the production and maturation of lymphocytes in body defenses.

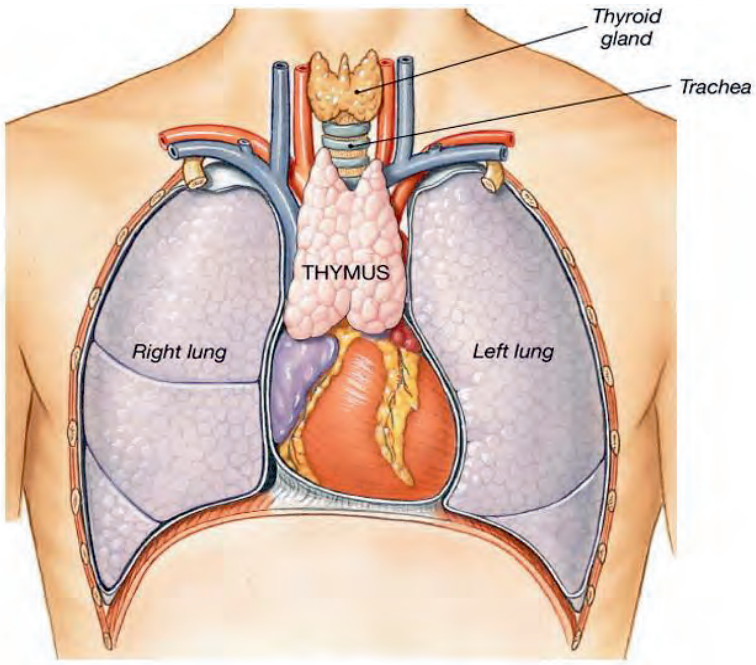


Figure 5.50 Thymus gland

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Activity 5.40

Discuss in group the difference between endocrine and exocrine glands and reflect your views to the class.

5.3. Homeostasis in the human body

At the end of this section, the learner will be able to:

- Define homeostasis
- Explain the regulatory mechanisms of body temperature, sugar level, water balance
- Discuss the structures and functions of the human kidney
- Investigate traditional mechanisms in your locality used to regulate body homeostasis

5.3.1. The structure and function of the human kidney

The kidneys are a pair of bean-shaped organs just above the waist. They are important organs with many functions in your body, including producing hormones, absorbing minerals, and filtering blood and producing urine. Internally, the kidney has three regions: an outer cortex, a medulla in the middle, and the renal pelvis, which is the expanded end of the ureter. The renal cortex contains the nephrons, which is the functional unit of the kidney. The renal pelvis collects the urine and leads to the ureter on the outside of the kidney. The ureters are urine-bearing tubes that exit the kidney and empty into the urinary bladder.

Self-questioning

How do kidneys control the levels of water and ions in the human body?

Blood flows into the kidney along the renal artery. The blood is filtered, so fluid containing water, salt, urea, glucose and many other substances is forced out into the kidney tubules. Then everything the body needs is taken back (reabsorbed), including all of the sugar and the mineral ions needed by the body.

The amount of water reabsorbed depends on the needs of the body. The waste product urea and excess ions and unwanted water of the body are released as urine. Each kidney has a very rich blood supply and is made up of millions of tiny microscopic tubules (nephrons) where all the filtering and reabsorption takes place.

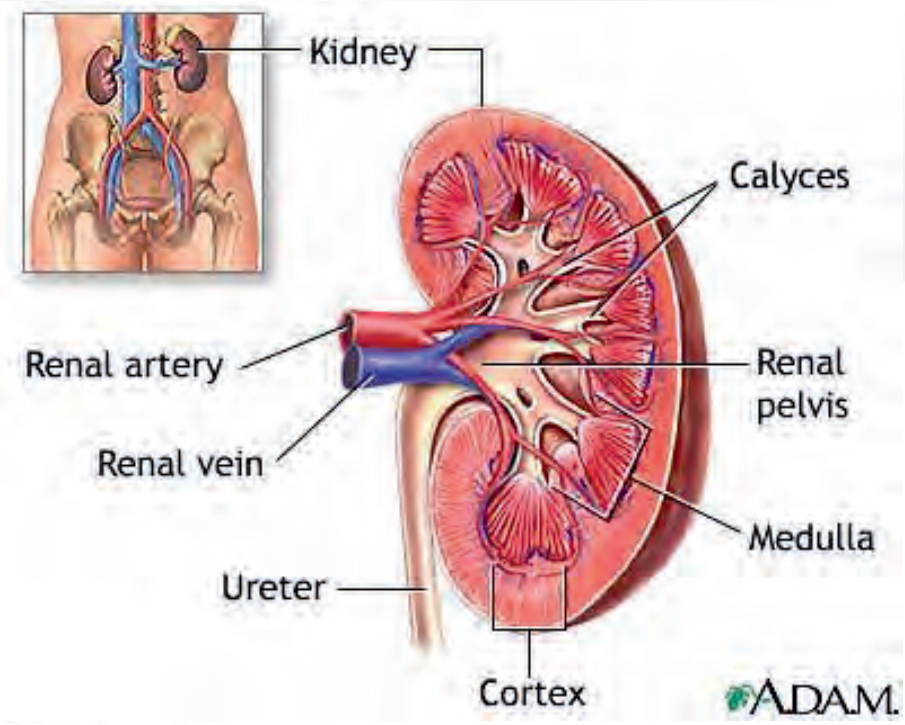

Figure 5.51 The anatomy of human kidneys

The roles of the different areas of a single kidney tubule in the production of urine are described below:

- **Bowman's capsule** is the site of the ultrafiltration of the blood. The blood vessel feeding into the capsule is wider than the vessel leaving the capsule, which means the blood in the capillaries is under a lot of pressure. Several layers of cells, the wall of the blood capillaries and the wall of the capsule act as a filter and the blood cells and the large blood proteins cannot leave the blood vessels as they are too big to fit through the gaps. However, water, salt, glucose, urea and many other substances are forced out into the start of the tubule. In fact, the concentration of substances in the liquid in the capsule is the same as that in the blood itself. This process is known as ultrafiltration – filtration on a very small scale.

- **Glomerulus:** This is the knot of blood vessels in the Bowman's capsule where the pressure builds up so that ultrafiltration occurs. The volume of the blood leaving the glomerulus is about 15% less than the blood coming in which is a measure of the liquid which has moved into the capsule as a result of ultrafiltration.

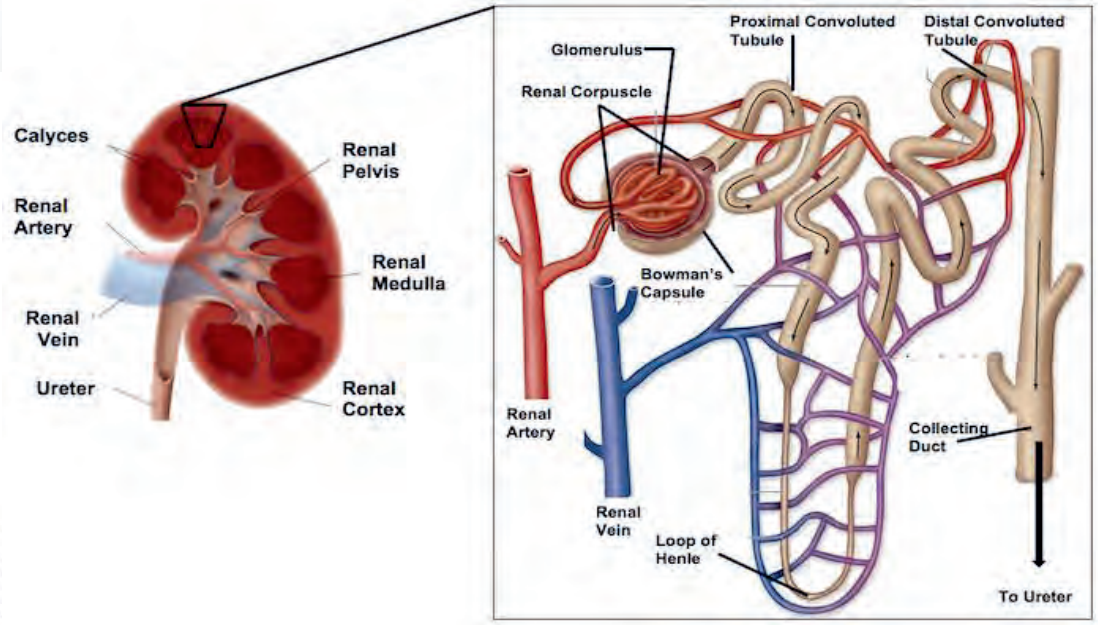

Figure 5.52 The kidney filters the blood and removes waste materials.

- **First coiled (convoluted) tubule:** This is the liquid which enters this first tubule is known as the glomerular filtrate. The first tubule is where much of the reabsorption takes place. All of the glucose is actively taken back into the blood along with around 67% of the sodium ions and around 80% of the water. It has many microvilli to increase the surface area for absorption.
- **Loop of Henlé:** is part of a kidney where the urine is concentrated and more water is conserved.

Second coiled (convoluted) tubule: is part of a kidney where the main water balancing is done. If the body is short of water, more is reabsorbed into the blood in this tubule under the influence of the anti-diuretic hormone or ADH. (Diuresis means passing urine, so anti-diuresis means preventing or reducing urine flow.) Also ammonium ions and some drugs (if they have been taken into the body) are secreted from the blood into this tubule to get rid of them. By the end of this second coiled tubule all of the salt which is needed by your

body has been reabsorbed, leaving the excess in the filtrate along with most of the urea.

- **Collecting duct:** is also part of a kidney where the liquid (essentially urine) is collected. It contains about 1% of the original water, with no glucose at all. The level of salt in the urine will depend on the amount of salt in your diet and the water content of the urine. There is also a much higher concentration of urea (about 60 times more) in the urine than in the blood. But, if your body badly needs more water, more may be reabsorbed along the collecting duct again under the influence of ADH – until the urine passes into the pyramid of the kidney and on into your bladder.

Urine is formed constantly in your kidneys as it drips down to collect in your bladder. The bladder is a muscular sac which can hold between 600 and 800 cm3 urine, although we usually empty it when it contains only 150–300 cm3.

The amount of water lost from the kidney in the urine is controlled by a sensitive feedback mechanism involving the hormone ADH. If the water content of the blood is too low (so the salt concentration of blood increases) special sense organs known as osmoreceptors in the brain detect this. They stimulate the pituitary gland in the brain to release ADH into the blood. This hormone affects the second coiled tubules of the kidneys, making them more permeable so more water is reabsorbed back into the blood.

Activity 5.41

By dissecting a kidney you can see the way the different tissues are arranged. Remember to indicate the magnification of your drawing each time – if it is life size, it is × 1.

You will need:

• Fresh kidney (lamb or sheep) from a butcher house, preferably with the fat surrounding the kidney in place

• Dissecting instruments – a scalpel, forceps and a seeker

• Ddissecting board

Method

1. Observe the outer appearance of the kidney with the fat on, if possible. Draw and label about what you observe.

Activity 5.43

Investigating kidney structure

2. Carefully remove the fat, clearing the tubes leading into and away from the kidney carefully. Again draw and label what you see.

3. Slice the kidney in half longitudinally (along its length) and open it out to see the internal structure. Again draw and label the regions carefully – use Figure 5.43 to help you identify them.

4. You may have the opportunity to look at prepared slides of kidney tissue under the microscope – if so, keep the drawings you make with these drawings from a fresh kidney to build up a record of the whole organ from your own observations.

This means less water is left in the kidney tubules and so a more concentrated urine is formed. At the same time the amount of water in the blood increases so that the concentration of salts in the blood returns to normal.

If the water content of the blood is too high, the pituitary gland releases much less ADH into the blood. The kidney then reabsorbs less water back into the blood, and thereby producing a large volume of dilute urine. Water is effectively lost from

the blood and concentration of salts returns to normal.

This system of osmoregulation is an example of negative feedback. As the water concentration of the blood falls, the level of ADH produced rises. Then as the water concentration of the blood rises again, the level of ADH released falls. On an average day the kidneys will produce around 180 l (that's about 50 gallons) of liquid filtered out of the blood in the glomerulus (glomerular filtrate) – but only about 1.5 l (just over 2.5 pints) of urine. So more than 99% of the liquid filtered out of the blood is eventually returned to it.

Activity 5.42

Discuss in small groups about homeostasis in action by considering the type of urine produced when someone drinks a lot of water and small volume of water?

Thermoregulation

It is vitally important that wherever we go and whatever we do our body temperature is maintained at the temperature (around 37 °C) at which our enzymes work best. It is not the temperature at the surface of an organism which matters as the skin temperature can vary enormously without causing harm. It is the temperature deep inside the body, known as the internal or core body temperature, which must be kept stable. Human beings are good examples of homeotherms. The body temperature is controlled by a number of physiological mechanisms which work together to allow gain or lose heat you need to.

Activity 5. 43

Search for different types of books and read about the different types of physiological and behavioral methods of temperature regulation in the human body. Present your work to the class.

Self-questioning

What types of mechanism do humans do have to regulate their body temperature?

1. Osmoregulation

If the concentration of the body fluids changes, water will move into or out of the cells by osmosis and they could be damaged or destroyed. Yet some days you may drink several liters of water of liquid and other days much less. How is the balance maintained?

We gain water when we drink and eat. We lose water constantly from the lungs when we breathe out, when water evaporates into the air in the lungs and is breathed out. This water loss is constant.

Whenever we exercise or get hot we sweat and lose more water. The water balance is maintained by the kidneys. They remove any excess water which leaves the body as urine. If we are short of water we produce very little urine and most water is saved for use in the body. If we have too much water then our kidneys produce lots of urine to get rid of the excess. The ion concentration of the body – particularly ordinary salt – is also important. We take in mineral ions with our food. Some are lost via our skin when we sweat. Again the kidney is most important organ to keep an ion balance. Excess mineral ions are removed by the kidneys and lost in the urine. The balance of water and salts in the body is very important because of the osmotic impact of the cells. If the balance is wrong controlling this balance is known as **osmoregulation**. The kidneys as discussed above are vitally important in two aspects of homeostasis, both in excretion and in osmoregulation.

2. Chemical regulation

Human liver plays a vital role in maintaining a constant internal environment. It is the largest individual organ in the body that makes up around 5% of the body mass. The liver cells are very active in carrying out a wide range of functions, many of which help to maintain a constant internal environment. The liver has a very special blood supply in addition to the usual artery and vein (hepatic artery and vein) there is another blood vessel which comes to the liver directly from the gut. This is the hepatic portal vein and it brings the products of digestion to the liver to be dealt with.

Activity 5. 44

Discuss in small groups if there are traditional mechanisms used to regulate body homeostasis in your locality.

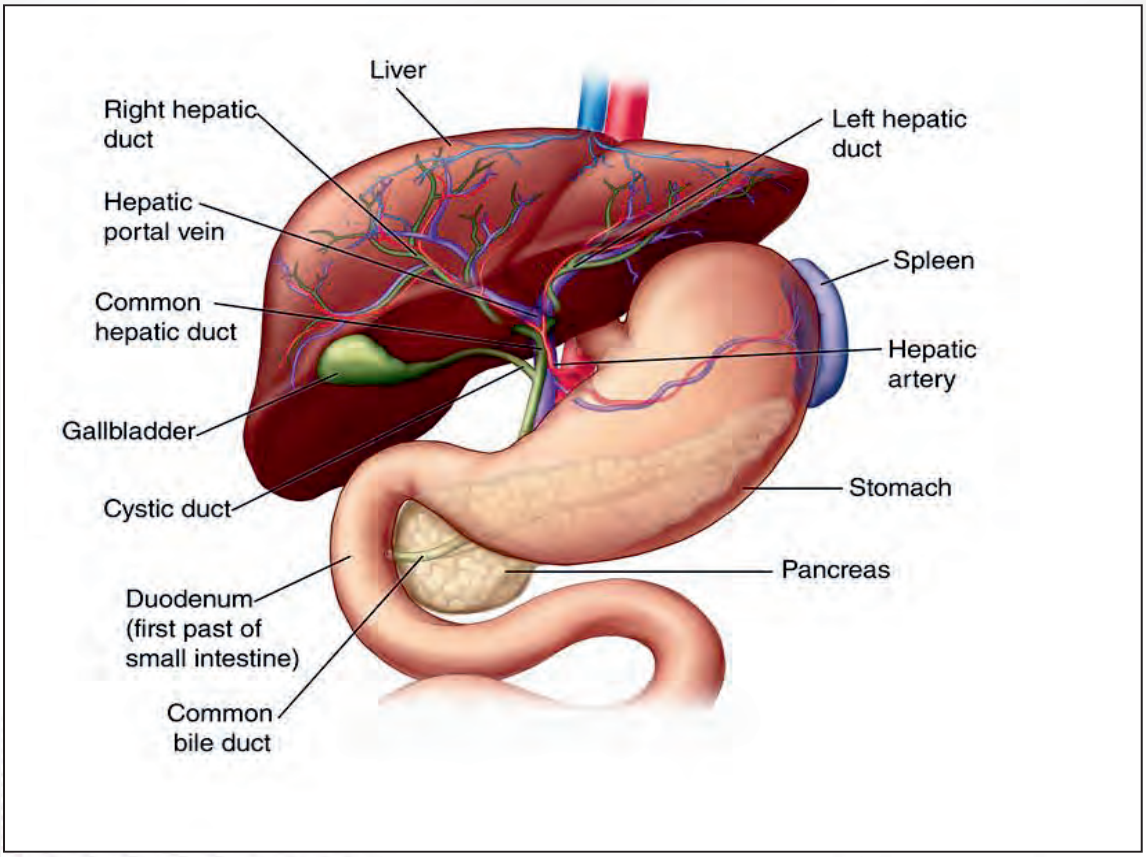

Figure 5.53 The liver is one of the most active organs in the body

A large number of reactions take place in the liver. Many of them are involved in homeostasis in one way or another. It plays a part in all of the following functions:

It controls the sugar levels in the body (through stored glycogen in the liver itself).

It controls and balances the fats that you eat and the cholesterol levels in the blood.

It is an important organ where protein metabolism takes place. The liver breaks down excess amino acids and forms urea. If you eat more carbohydrate or fat than you need in the diet the body simply stores the excess energy as fat. If you eat too much protein, it isn't so easy. The body cannot store the excess amino acids or simply convert protein to fat. Instead the amino acids which make up the protein are broken down in the liver. The amino (nitrogen containing) part of the amino acid molecule is removed and converted into ammonia and then urea in the liver. The rest of the amino acid can be used in cellular respiration or converted to fat for storage. The process of removing the amino group from excess amino acids is known as **deamination.** This it is a very important function of the liver.

It carries out the breakdown of worn-out red blood cells – in particular the red pigment hemoglobin.

It is vital organ for the formation of bile which is made in the liver and stored in the gall bladder before it is released into the gut to emulsify fats and help in their digestion.

It controls toxins. The liver breaks down most of the poisons you take into the body, including alcohol. This is why the liver is so often damaged when people drink heavily.

It is used to control temperature. Around 500 different reactions take place in the liver at any time. For many years it has been believed that as a result of all these reactions the liver generates a lot of heat which is then spread around the body by the bloodstream.

The regulation of tissue oxygenation is another typical example for chemical regulation in the body. The respiratory chemoreceptors work by sensing the pH levels of their environment through the concentration of hydrogen ions. Because most carbon dioxide is converted to carbonic acid (and bicarbonate) in the bloodstream, chemoreceptors are able to use blood pH as a way to measure the carbon dioxide levels of the bloodstream.

The main chemoreceptors involved in respiratory feedback are:

1. Central chemoreceptors: These are located on the ventrolateral surface of medulla oblongata and detect changes in the pH of spinal fluid. They can be desensitized over time from chronic hypoxia (oxygen deficiency) and increased carbon dioxide.
2. Peripheral chemoreceptors: These include the aortic body, which detects changes in blood oxygen and carbon dioxide, but not in the pH, and the carotid body which detects all three. They do not desensitize, but they have less impact on the respiratory rate compared to the central chemoreceptors.

Activity 5.45

Search from different books and materials about other types of chemical regulation that takes place in the human body. Report your finding to the class.

The need for different levels of respiration varies with the physiologic state of the organism (e.g., sleep, excitement, exercise). The respiratory system must try to maintain constant levels of O_2, CO_2 and H^+ in the arterial blood which then ensures relatively constant levels of these important substances in the interstitial fluid. For O_2, one needs an adequate supply to meet cellular metabolic requirements. For CO_2 and H^+, one needs to maintain the acid–base status of the body's cells. The respiratory system provides a

rapid, but usually incomplete, compensation for acid–base disturbances through altered partial pressure of CO2. (PCO2). Changes in the levels of O2, CO2 and H+ in the blood cause compensatory changes in the level of ventilation.

a. The side effects of drugs on the normal functioning of nervous and endocrine systems

Activity 5.46

In this chapter you have learned about the drug abuse and their effect on your body. Now it can be easier for you to further strength your understanding by doing research in this topic.

Make a small group as per the instruction of your teacher and research about the effects of drugs (smoking, alcohol use, chewing khat and cannabis) on the normal functioning of the nervous system and endocrine system. You are advised to prepare a questionnaire and an interview for drug users and health professionals in your locality. Report your findings to the class.

Unit Summary

In this unit you have learnt human beings have both nervous and hormonal co-ordination and control systems. The nervous system is the most rapid. A nerve cell or neuron consists of a cell body, dendrites and an axon. Sensory neurons carry information from the sense organs to the central nervous system (CNS). Motor neurons carry instructions from the CNS to the effector organs (muscles and glands). Neurons carry electrical impulses known as the action potential.

In any pathway the junctions between neurons are called synapses. When an impulse arrives in one neuron chemicals are released in the synapse to trigger an impulse in the next neuron. Nerve contains many neurons. There are sensory nerves, motor nerves and mixed nerves. The spinal cord carries information from all over the body to and from the brain. Cranial nerves come from the brain, while spinal nerves are from the spinal cord.

Reflex actions avoid danger without conscious thought. Reflex actions involve: stimulus → receptor → co-ordinator →effector → response but the co-ordinator is the relay neuron in the spinal cord and there is no conscious thought involved. The knee jerk reflex is a common example of a reflex. Drug abuse is when someone use a substance to the point of excess and/or dependence. Drug dependence is when an individual use a drug again and again and become addicted.

Drugs change the chemical processes in human body then become addicted to them. Alcohol, tobacco and khat are the widely used substances in Ethiopia.

Sense organs detect changes in the internal or external environment. The human eye includes: sclera, cornea, iris, pupil, lens, ciliary muscle, suspensory ligament, retina and optic nerve. The light-sensitive cells the rods and the cones are found in the retina. The iris controls the amount of light entering the eye. The cornea bends the light into the eye. The lens controls the fine focus of the image onto the retina. Short sight, long sight and astigmatism are three common defects of the eye. The ear is an organ of hearing and of balance. Hearing involves the outer, middle and inner ear. Balance involves the semicircular canals.

Chemical co-ordination and control of the body is brought about by hormones secreted by special endocrine glands. The hormones are secreted directly into the blood and are carried around the body in the blood. They may affect a single target organ or a range of organs and tissues. They have their effect through special receptor molecules on the cell membranes of the target organs and tissues. Hormonal

control may be rapid but is often relatively slow and long term. Important endocrine organs include the pituitary gland, the thyroid gland, the adrenal glands, the pancreas, the ovaries and the testes. Insulin produced by the pancreas controls the blood sugar levels Hormones from the pituitary and the ovary control the menstrual cycle.

Unit review questions

Choose the correct answer for the following questions.

1. The two types of main components of macromolecules in myelin are;
   a. carbohydrates and lipids
   b. proteins and nucleic acids
   c. lipids and proteins
   d. carbohydrates and nucleic acids
2. If a thermoreceptor is sensitive to temperature sensations, what would a chemoreceptor be sensitive to?
   a. Light
   b. Sound
   c. Molecules
   d. vibration
3. What ion enters a neuron causing depolarization of the cell membrane?
   a. Sodium
   b. Chloride
   c. Potassium
   d. phosphate
4. Which of the following response is not part of the fight-or-flight response?
   a. pupil dilation
   b. increased oxygen supply to the lungs
   c. suppressed digestion
   d. reduced mental activity
5. Which one of the following correctly traces the transmission of sound from the external environment to the nerves that carry the signal to the brain to be interpreted?
   a. Cochlea, tympanic membrane, ossicles, pinna, external auditory meatus.
   b. Pinna, external auditory meatus, tympanic membrane, ossicles, cochlea
   c. Tympanic membrane, Cochlea, ossicles, pinna, external auditory meatus
   d. External auditory meatus, Pinna, tympanic membrane, ossicles, cochlea

6. Which structure detect rotational acceleration of the head and body?

a. Ossicles

b. Cochlea

c. Tympanic membrane

d. Semicircular canals

7. Which section of the ear contains the malleus, incus, and stapes bones?

a. Outer ear

b. Middle ear

c. Inner ear

d. Pinna

8. Which of the following is not an example of homeostasis?

a. control of the blood sugar levels

b. control of the body temperature

c. control of the water content of the blood

d. control of the length of the limbs

9. Which of the following areas is NOT part of the nephron (kidney tubule)?

a. Bowman's capsule

b. urinary bladder

c. loop of Henle

d. first coiled tubule

10. Which of the following is not part of the eye?

a. lens

b. retina

c. pinna

d. eyelid

11. Imagine you have been out on the beach looking at some friends in the sea. You walk into the shade of a palm tree and begin to read a book. What changes would take place in your eyes?

a. Your pupils would constrict and your lens would become flatter and less convex.

a. Your pupils would constrict and your lens would become rounder and more convex.

b. Your pupils would dilate and your lens would become flatter and less convex.

c. Your pupils would dilate and your lens would become rounder and more convex.

12. Which is the correct order of the bones in the middle ear, from the eardrum inwards?

a. hammer, anvil, stirrup

b. anvil, hammer, stirrup

c. stirrup, anvil, hammer

d. none of these

13. Which is the most recently discovered sense of taste?

a. sweet

b. bitter

c. sour

d. umami

14. Which of the following endocrine glands secretes a hormone that directly affects the metabolic rate of the body?

a. pituitary gland b. ovary

c. thyroid d. pancreas

14. Which of the following reproductive hormones is produced by the pituitary gland?

a. oestrogen
b. testosterone
c. follicle stimulating hormone
d. progesterone

15. Which of the following is not an example of homeostasis?

a. control of the blood sugar levels

b. control of the body temperature

c. control of the water content of the blood

d. control of the length of the limbs

16. Which of the following areas is NOT part of the nephron (kidney tubule)?

a. Bowman's capsule
b. urinary bladder
c. loop of Henle
d. first coiled tubule

b. Which of the following statements is true about ADH?

17. ADH is a hormone produced in the brain which affects the second coiled tubules of the kidneys, making them more permeable so more water is reabsorbed back into the blood and little, concentrated urine is formed.

a. ADH is a hormone produced in the brain which affects the first coiled tubules of the kidneys, making them more permeable so more water is reabsorbed back into the blood.

b. ADH is a hormone produced in the kidney which affects the coiled tubules of the kidneys, making them more permeable so more water is reabsorbed back into the blood.

c. ADH is a hormone produced in the brain which affects the second coiled tubules of the kidneys, making them less permeable so less water is reabsorbed back

Provide correct answer for the following questions

1) Explain the role of neurotransmitters.

2) Compare and contrast endocrine and exocrine glands

3) Briefly explain the coordination of the body by means of hormones and electrical impulses.

4) Describe the role of negative feedback in the function of the parathyroid gland.

5) Describe the antagonistic actions of insulin and glucagon

6) Describe the role of some of the pigments that are found in your skin

7) What are the three regions of the adrenal cortex and what hormones do they produce?

8) List down the hormones produced by anterior pituitary gland.

9) Describe the role of negative feedback in the function of the thyroid gland.

10) Explain some of the eye disorders with their possible corrections.

11) Briefly explain how you can use your ears to hear a sound

12) Why is the pituitary gland considered as a master gland?

13) How does the hypothalamus interact with the posterior and anterior pituitary glands?

14) Describe the functions of the hormones released by the posterior and anterior pituitary glands.

15) Compare the hormones of the adrenal cortex and medulla